

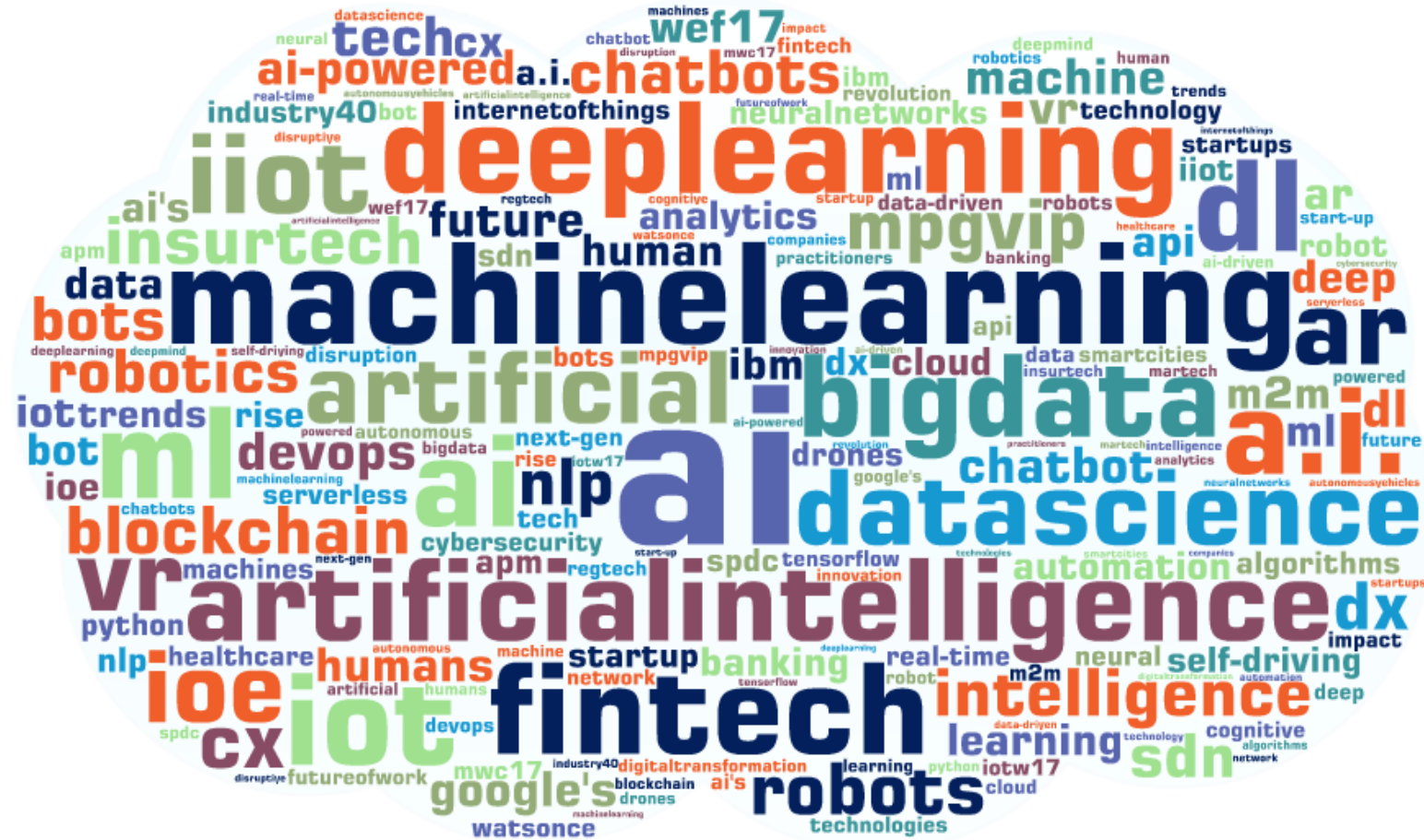
Introduction

Generative AI

What is Generative AI?



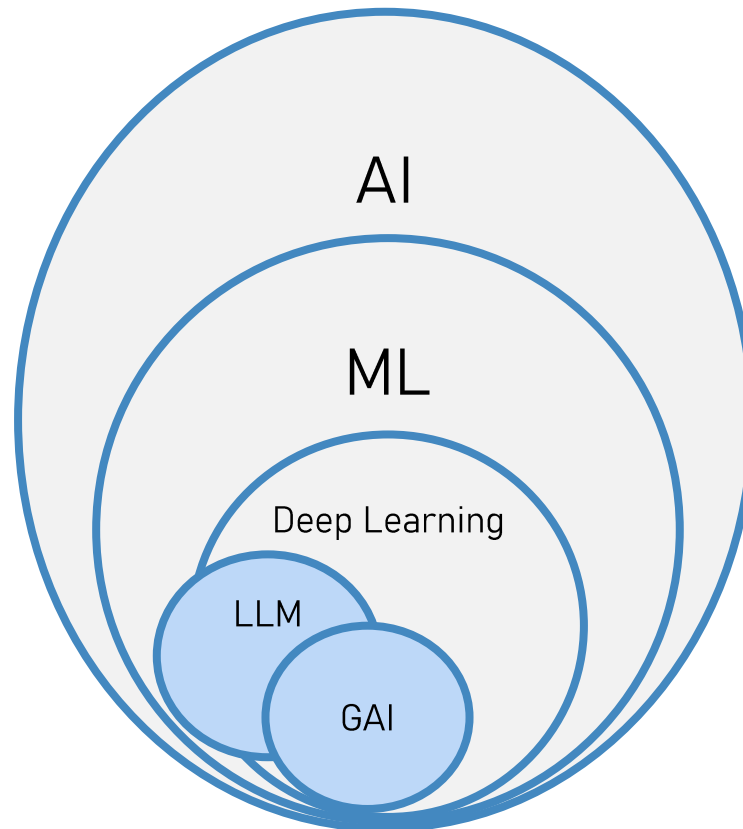
- Gen AI is a type of Artificial Intelligence that creates new content based on what it has learned from existing content.
- Generative AI was introduced in the 1960s in chatbots.
- In 2014 on introduction of GANs a type of machine learning algorithm that generative AI could create convincingly authentic images, videos and audio of real people.
- In 2017 Googles Brain team came up with Transformers as another ML algorithm with Deep Learning capabilities using LLM's.



What is Generative AI



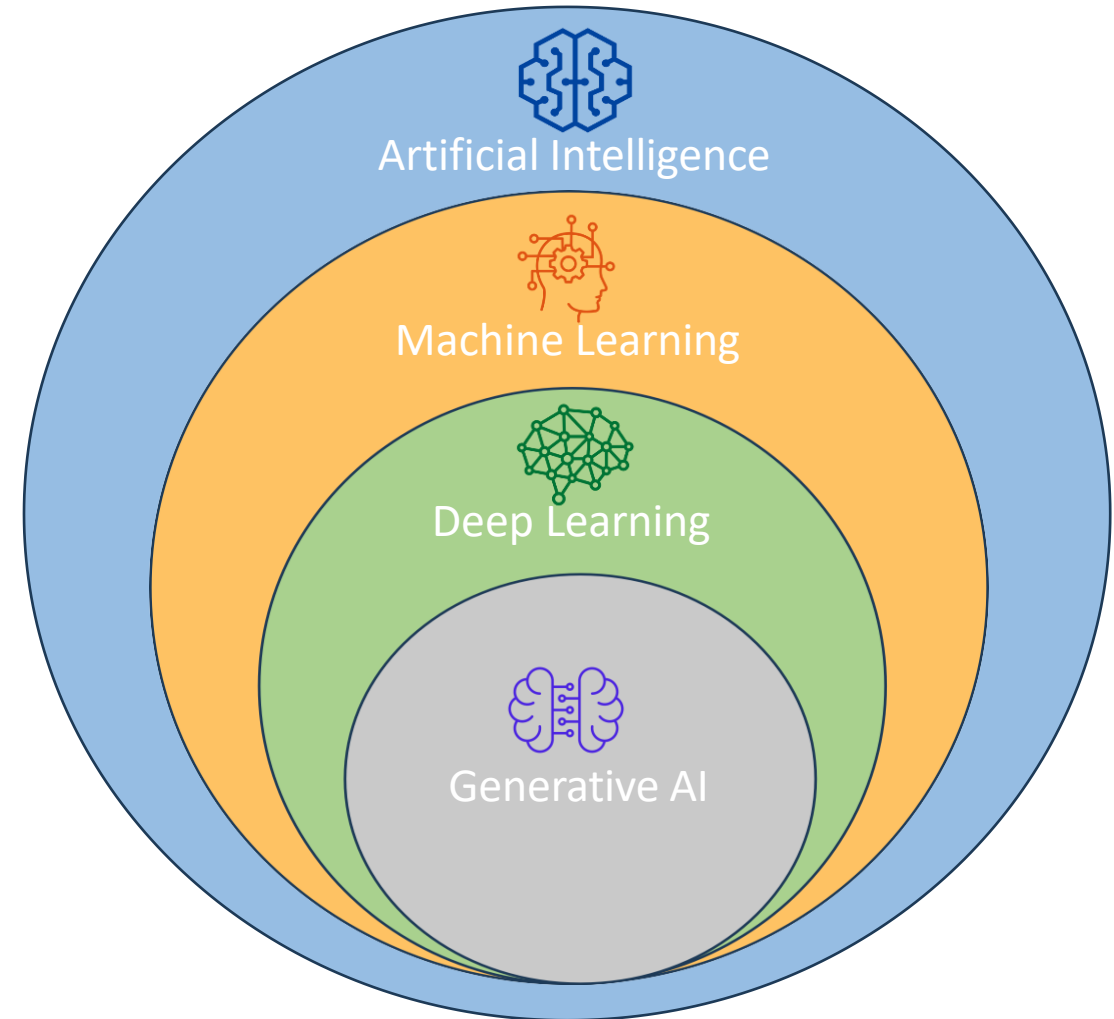
Generative AI is a type of artificial intelligence technology that can produce various types of content, including text, imagery, audio and synthetic data.



What is Artificial Intelligence (AI)?



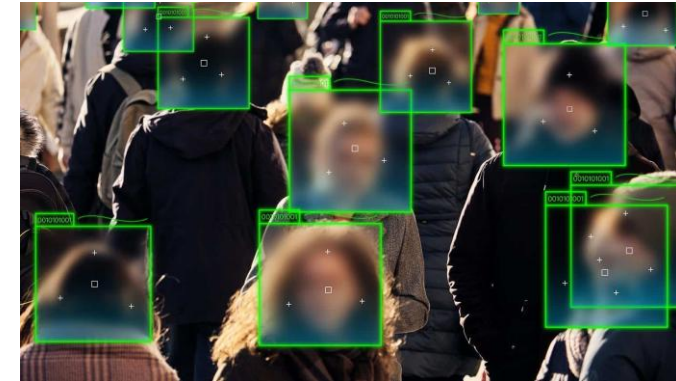
- AI is a broad field for the development of intelligent systems capable of performing tasks that typically require human intelligence:
 - Perception
 - Reasoning
 - Learning
 - Problem solving
 - Decision-making
 - Umbrella-term for various techniques



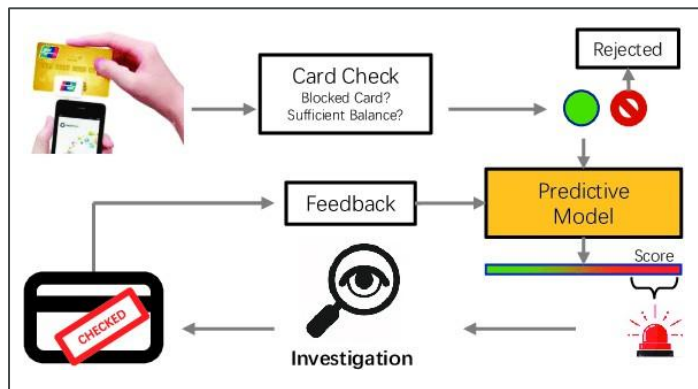
Artificial Intelligence – Use Cases



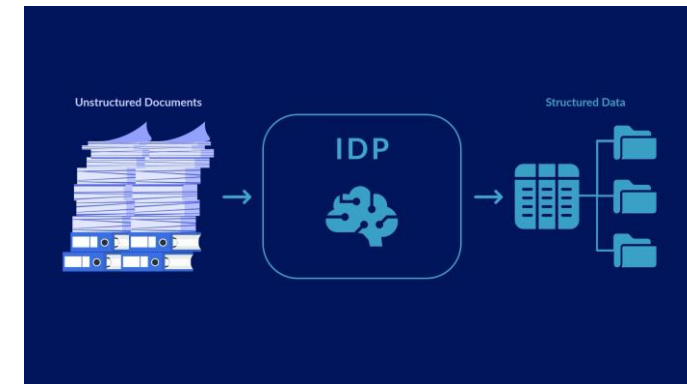
Computer Vision



Facial Recognition



Fraud Detection

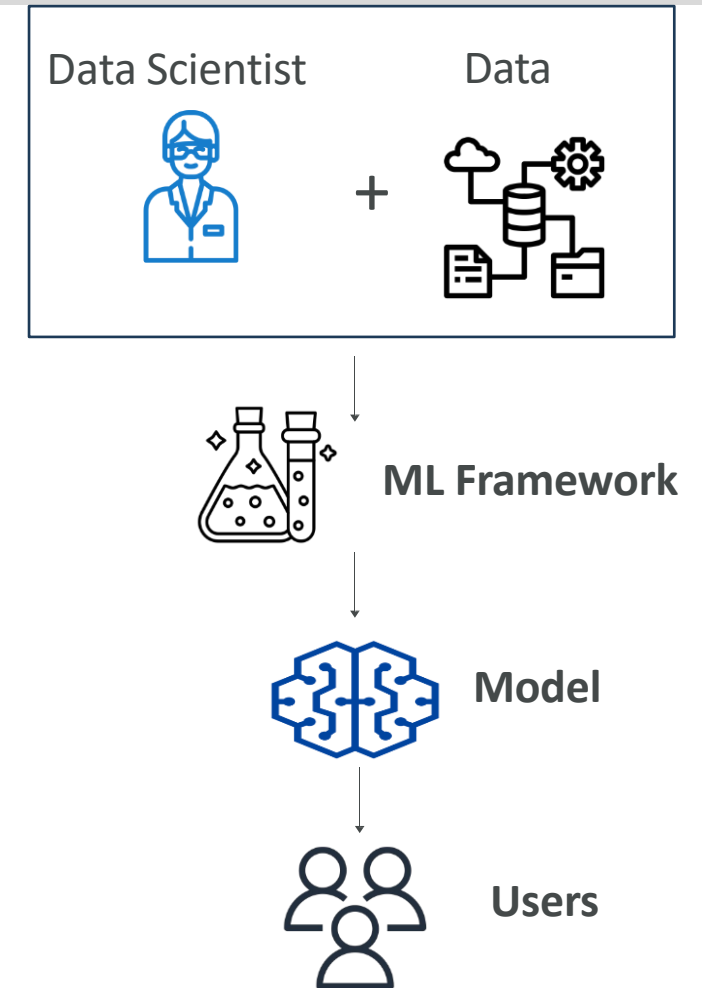


Intelligent Document Processing (IDP)

AI Components



- Data Layer – collect vast amount of data
- ML Framework and Algorithm Layer – data scientists and engineer work together to understand use cases, requirements, and frameworks that can solve them
- Model Layer – implement a model and train it, we have the structure, the parameters and functions, optimizer function
- Application Layer – how to serve the model, and its capabilities for your users

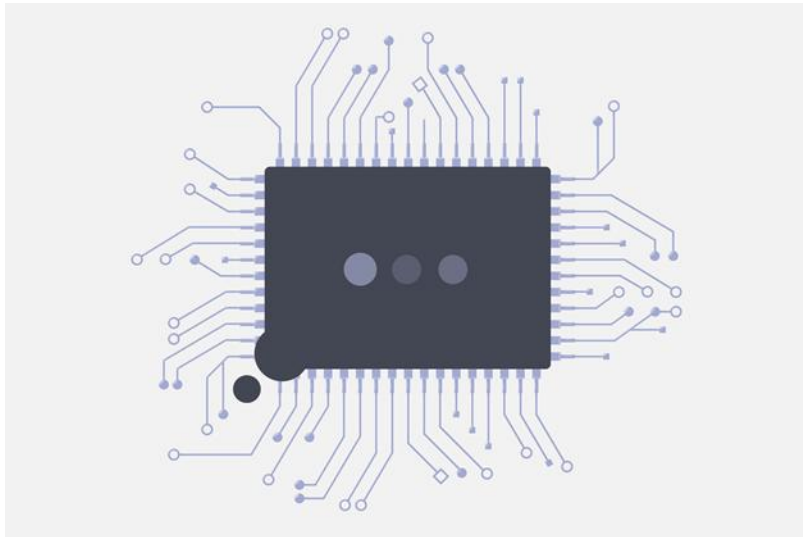


What is Machine Learning?



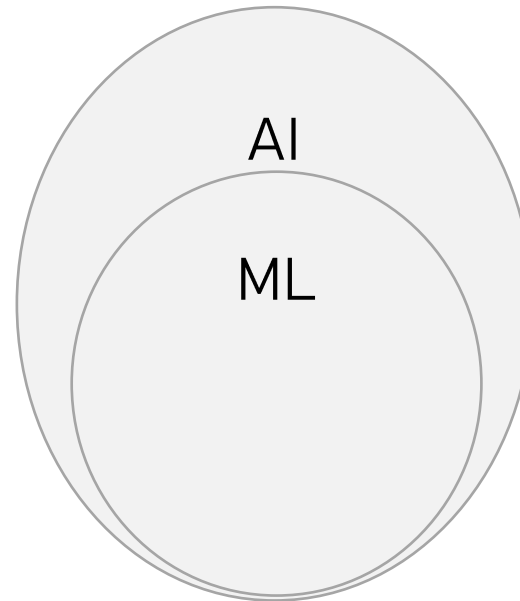
Machine learning is a branch of **artificial intelligence (AI)** and computer science which focuses on the use of **data** and **algorithms** to imitate the way that humans learn, gradually improving its accuracy.

How AI and ML Related?



Artificial Intelligence

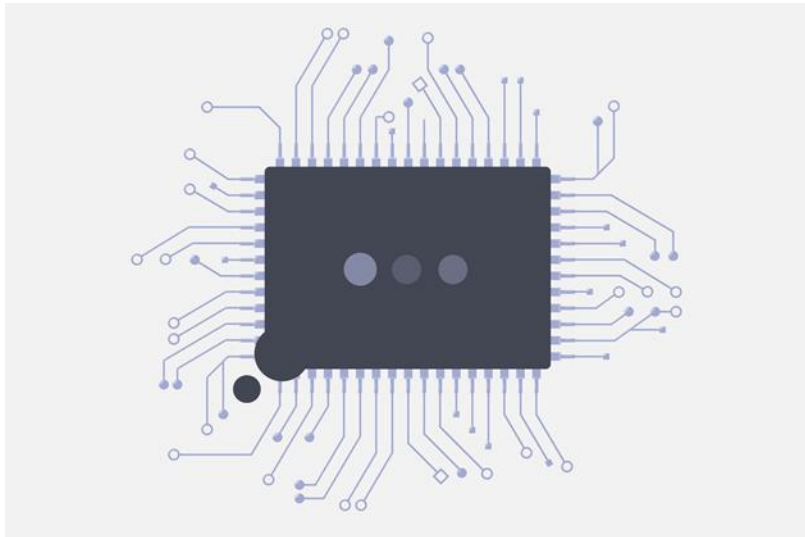
is a discipline



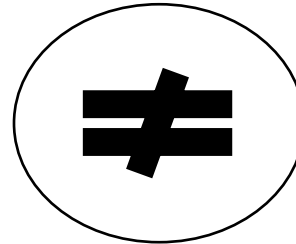
Machine Learning

is a subfield

AI != ML

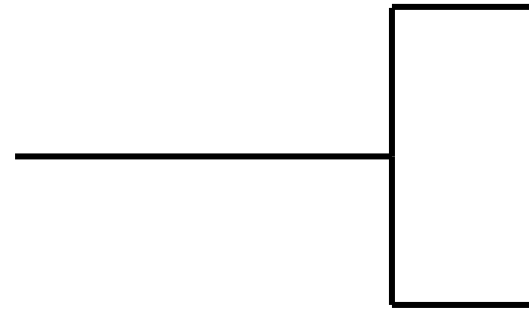


Artificial Intelligence



Machine Learning

ML Models

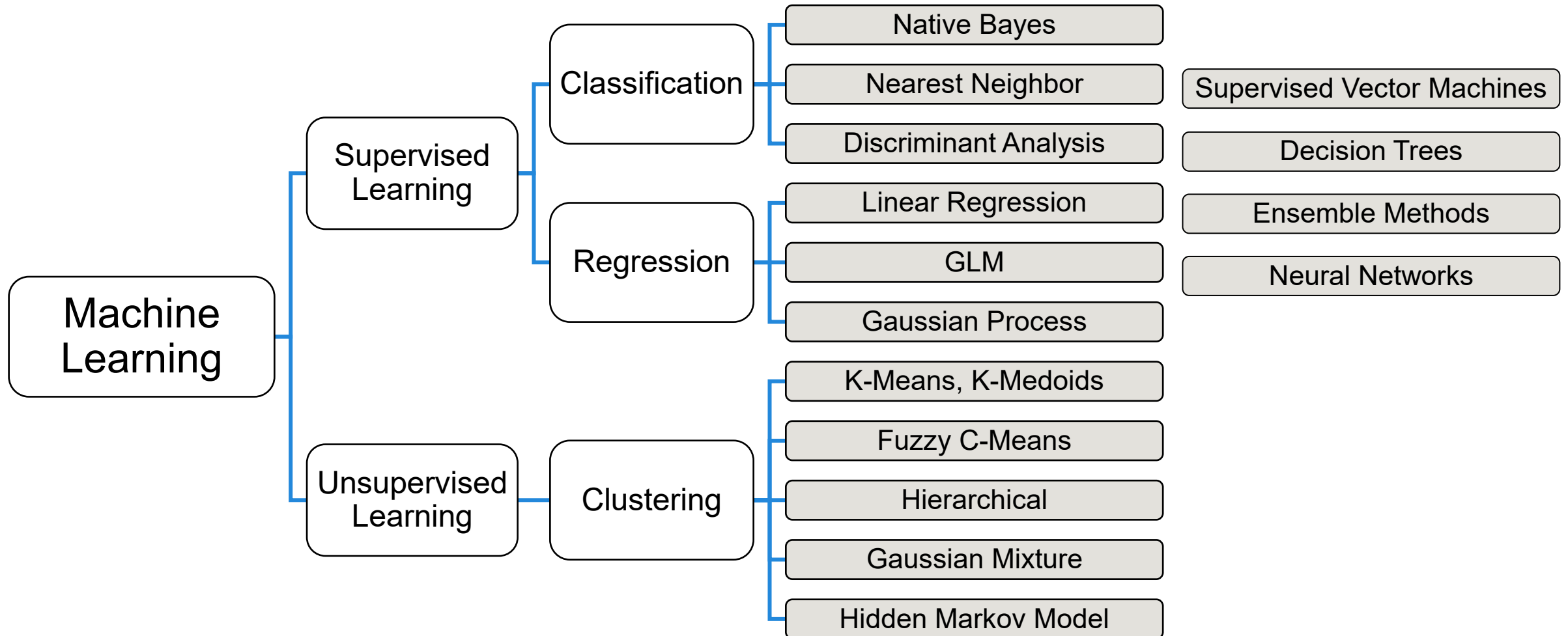


Unsupervised ML models

supervised ML models

ML Models

ML Models



Supervised ML



- Supervised learning implies the data is already labeled
- In supervised learning we are learning from past examples to predict future values.

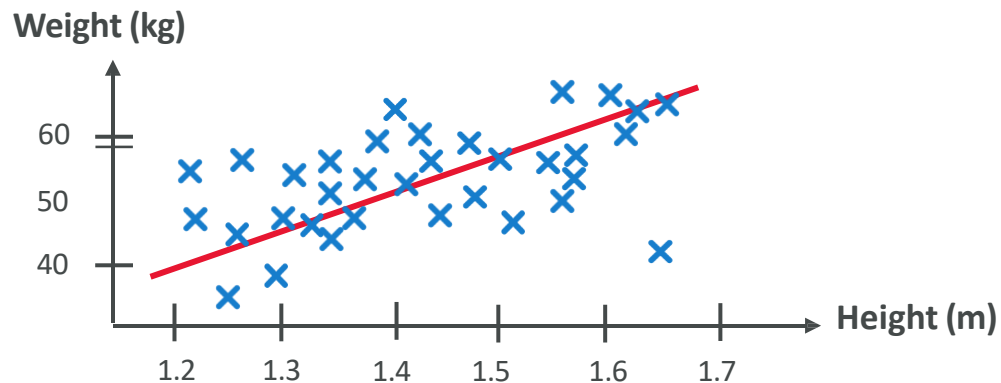


Types - Supervised Learning



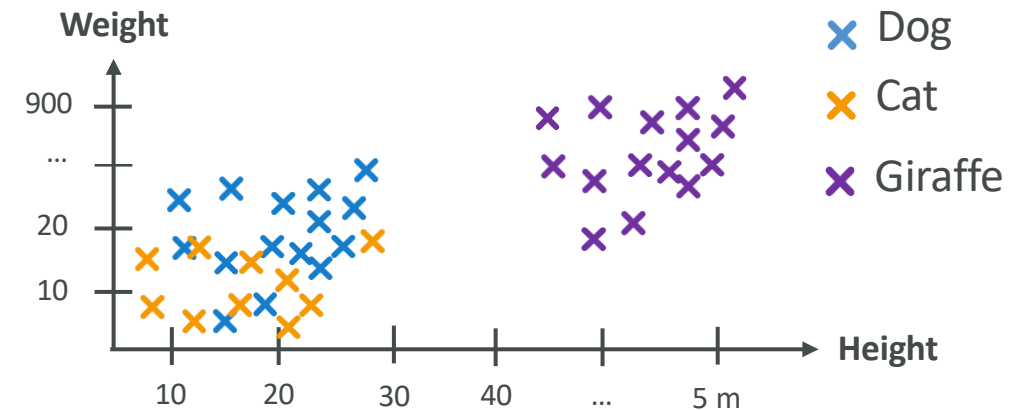
- Learn a mapping function that can predict the output for new unseen input data
- Needs labeled data: very powerful, but difficult to perform on millions of datapoints

Regression



What's the weight of a person which is 1.6m tall ?
=> Based on linear regression: 60kg

Classification



Which animal is this?
Height: 4.5m
Weight: 800kg

Classification
Model

Giraffe

Supervised Learning – Regression



- Used to predict a numeric value based on input data
- The output variable is continuous, meaning it can take any value within a range
- Use cases: used when the goal is to predict a quantity or a real value
- Examples:
 - Predicting House Prices – based on features like size, location, and number of bedrooms
 - Stock Price Prediction – predicting the future price of a stock based on historical data and other features
 - Weather Forecasting – predicting temperatures based on historical weather data

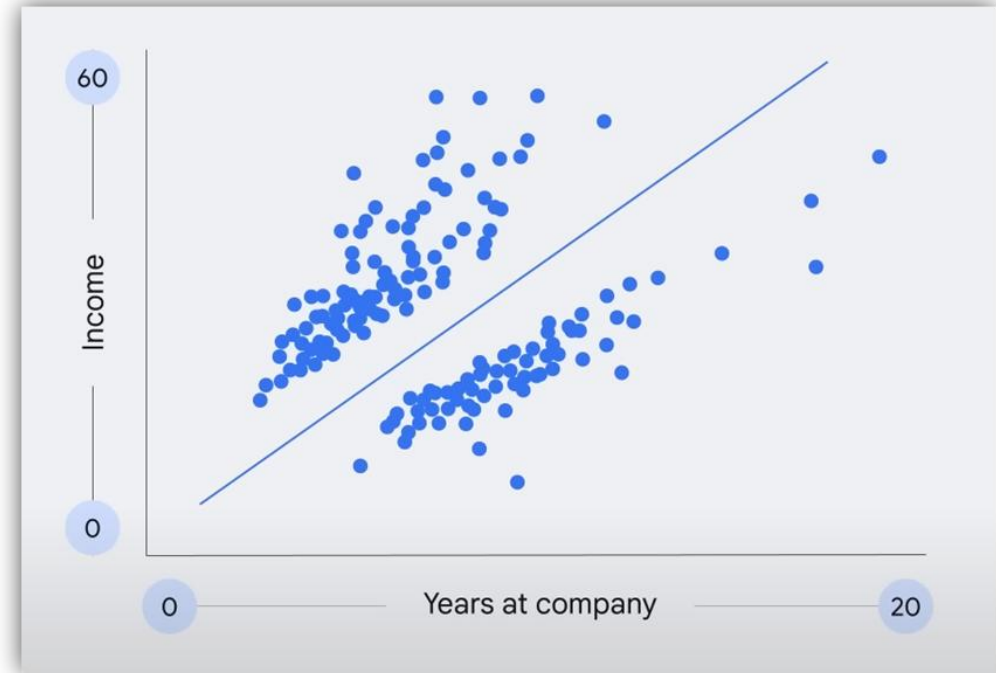


Unsupervised ML



- Unsupervised learning implies the data is not labeled
- Unsupervised problems are all about looking at the raw data, and seeing if it naturally falls into groups

Income vs Job tenure



Example Model: Clustering

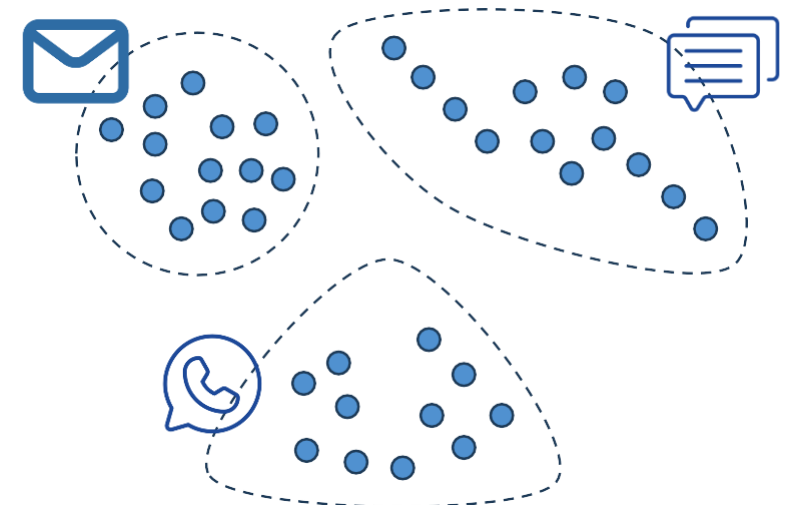
Is this employee on the "fast-track" or not?

ML Algorithms – Unsupervised Learning



- The goal is to discover inherent patterns, structures or relationships within the input data
- The machine must uncover and create the groups itself, but humans still put labels on the output groups
- Common techniques include **Clustering**, **Association Rule Learning**, and **Anomaly Detection**
- Clustering use cases: customer segmentation, targeted marketing, recommender systems
- Feature Engineering can help improve the quality of the training

Clustering Customers

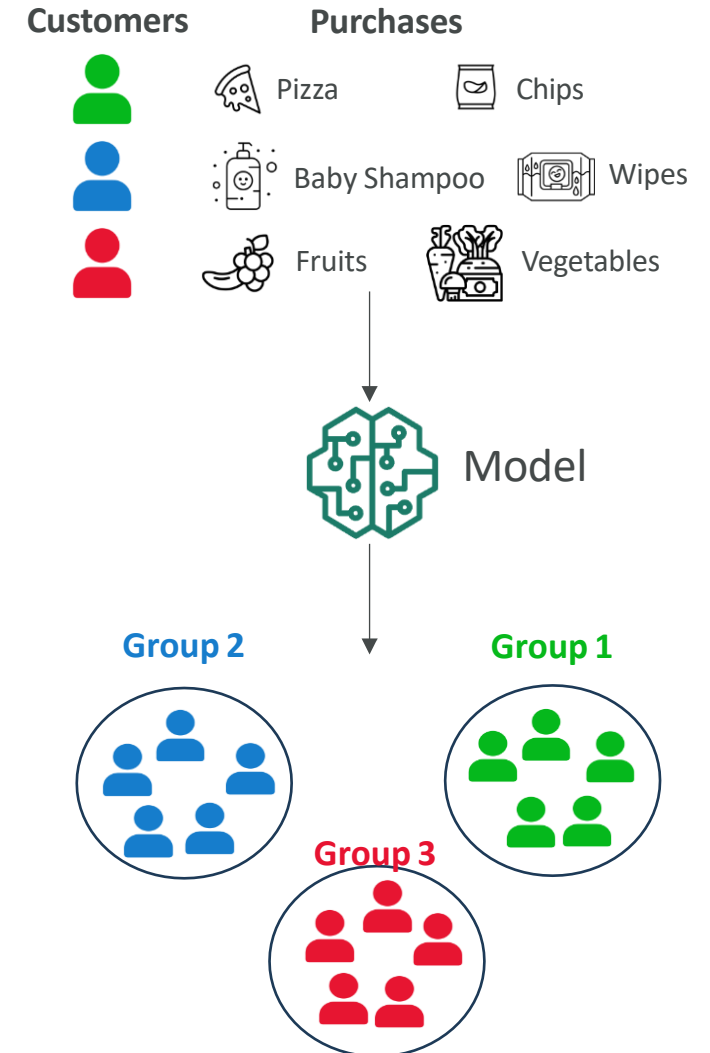


Unsupervised Learning
for Targeted Marketing

Unsupervised Learning: Clustering Technique



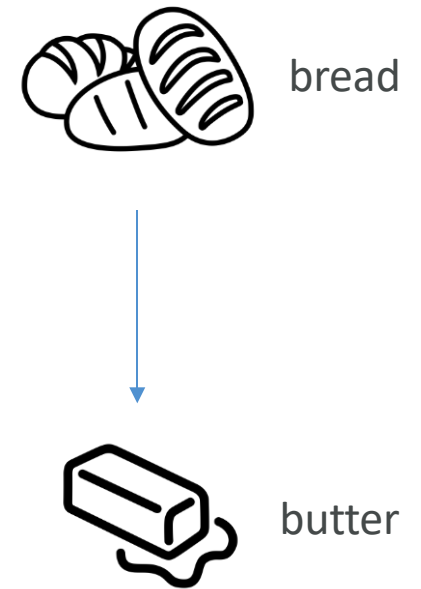
- Used to group similar data points together into clusters based on their features
- **Example:** Customer Segmentation
 - Scenario: e-commerce company wants to segment its customers to understand different purchasing behaviors
 - Data: A dataset containing customer purchase history (e.g., purchase frequency, average order value)
 - Goal: Identify distinct groups of customers based on their purchasing behavior
 - Technique: K-means Clustering
- **Outcome:** The company can target each segment with tailored marketing strategies



Unsupervised Learning: Association Rule Learning Technique



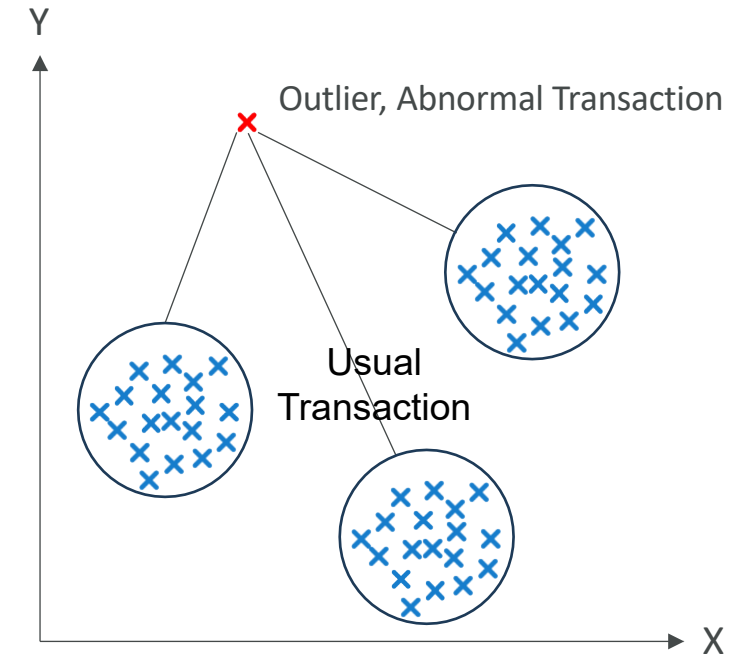
- **Example:** Market Basket Analysis
 - Scenario: supermarket wants to understand which products are frequently bought together
 - Data: transaction records from customer purchases
 - Goal: Identify associations between products to optimize product placement and promotions
 - Technique: Apriori algorithm
- **Outcome:** the supermarket can place associated products together to boost sales



Unsupervised Learning: Anomaly Detection Technique



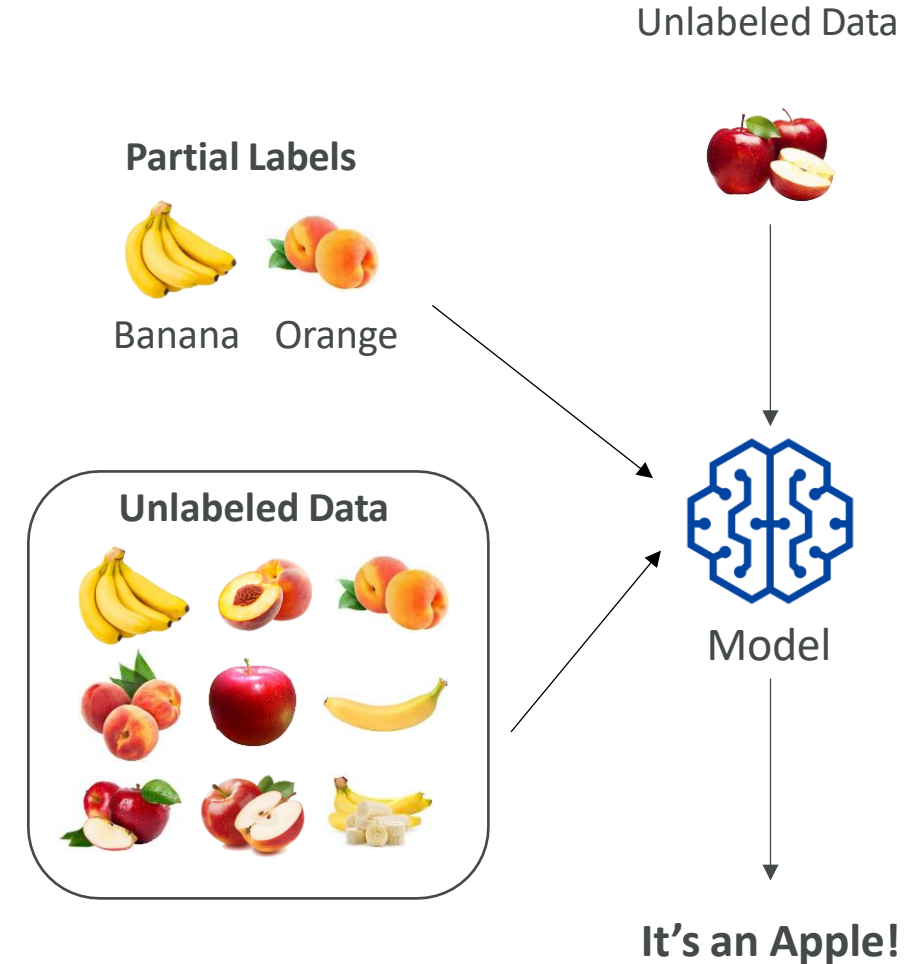
- **Example:** Fraud Detection
 - Scenario: detect fraudulent credit card transactions
 - Data: transaction data, including amount, location, and time
 - Goal: identify transactions that deviate significantly from typical Behavior
 - Technique: Isolation Forest
- **Outcome:** the system flags potentially fraudulent transactions for further investigation



Semi-supervised Learning



- Use a small amount of labeled data and a large amount of unlabeled data to train systems
- After that, the partially trained algorithm itself labels the unlabeled data
- This is called pseudo-labeling
- The model is then re-trained on the resulting data mix without being explicitly programmed



Self-Supervised Learning



- Have a model generate pseudo-labels for its own data without having humans label any data first
- Then, using the pseudo labels, solve problems traditionally solved by Supervised Learning
- Widely used in NLP (to create the BERT and GPT models for example) and in image recognition tasks



Huge amount of text data

Amazon Web Services, Inc. (AWS) is a subsidiary of Amazon that provides on-demand cloud computing platforms and APIs to individuals, companies, and governments, on a metered, pay-as-you-go basis. Clients will often use this in combination with autoscaling (a process that allows a client to use more computing in times of high application usage, and then scale down to reduce costs when there is less traffic). These cloud computing web services provide various services related to networking, compute, storage, middleware, IoT and other processing capacity, as well as software tools via AWS server farms. This frees clients from managing, scaling, and patching hardware and operating systems. One of the foundational services is Amazon Elastic Compute Cloud (EC2), which allows users to have at their disposal a virtual cluster of computers, with extremely high availability, which can be interacted with over the internet via REST APIs, a CLI or the AWS console. AWS's virtual computers emulate most of the attributes of a real computer, including hardware central processing units (CPUs) and graphics processing units (GPUs) for processing; local/RAM memory; hard-disk (HDD)/SSD storage; a choice of operating systems; networking; and pre-loaded application software such as web servers, databases, and customer relationship management (CRM)...

Self-supervised learning



Learn the English language, grammar, meaning of words, and relationships between words

Self-Supervised Learning: Intuitive example



- Create “pre-text tasks” to have the model solve simple tasks and learn patterns in the dataset.
- Pretext tasks are not “useful” as such, but will teach our model to create a “representation” of our dataset
 - Predict any part of the input from any other part
 - Predict the future from the past
 - Predict the masked from the visible
 - Predict any occluded part from all available parts
- After solving the pre-text tasks, we have a model trained that can solve our end goal: “downstream tasks”

Unlabeled Data

Amazon Web Services, Inc. (AWS) is a subsidiary of Amazon that provides on-demand cloud computing platforms and APIs to individuals, companies, and governments, on a metered, pay-as-you-go basis



Pretext Task

X	Y
Amazon Web	Services
that provides on-demand cloud	computing
APIs to individuals, _____, and governments, on a metered, pay-as-you-go	companies

What is Reinforcement Learning (RL)?



- A type of Machine Learning where an agent learns to make decisions by performing actions in an environment to maximize cumulative rewards
- Key Concepts
 - Agent – the learner or decision-maker
 - Environment – the external system the agent interacts with
 - Action – the choices made by the agent
 - Reward – the feedback from the environment based on the agent's actions
 - State – the current situation of the environment
 - Policy – the strategy the agent uses to determine actions based on the state

EXIT

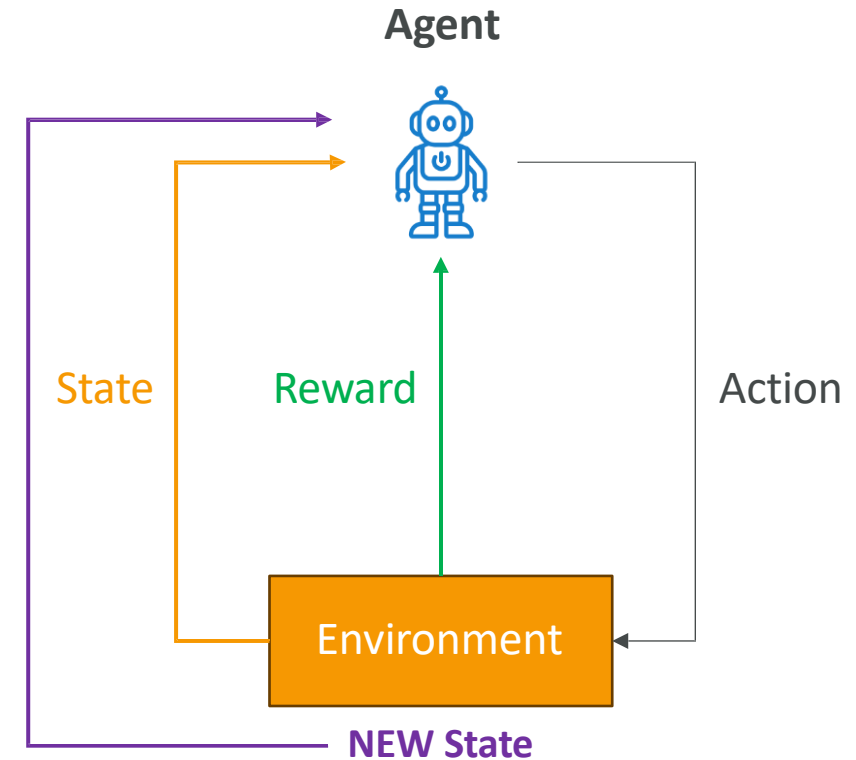
-1	-1	-1	-1	+100
-10	-10	-1	-10	-10
-1	-10	-1	-1	-1
-1	-10	-1	-10	-1
	-1	-1	-10	-1

Simulate many times
Learn from mistakes
Learn from successes

How Does Reinforcement Learning Work?



- Learning Process
 - The Agent observes the current State of the Environment
 - It selects an Action based on its Policy
 - The environment transitions to a new State and provides a Reward
 - The Agent updates its Policy to improve future decisions
- Goal: Maximize cumulative reward over time



Example: Reinforcement Learning in Action



- Scenario: training a robot to navigate a maze
- Steps: robot (Agent) observes its position (State)
 - Chooses a direction to move (Action)
 - Receives a reward (-1 for taking a step, -10 for hitting a wall, +100 for going to the exit)
 - Updates its Policy based on the Reward and new position
- Outcome: the robot learns to navigate the maze efficiently over time

EXIT

-1	-1	-1	-1	+100
-10	-10	-1	-10	-10
-1	-10	-1	-1	-1
-1	-10	-1	-10	-1
	-1	-1	-10	-1

Applications of Reinforcement Learning



- Gaming – teaching AI to play complex games (e.g., Chess)
- Robotics – navigating and manipulating objects in dynamic environments
- Finance – portfolio management and trading strategies
- Healthcare – optimizing treatment plans
- Autonomous Vehicles – path planning and decision-making



What is RLHF?



- RLHF = Reinforcement Learning from Human Feedback
- Use human feedback to help ML models to self-learn more efficiently
- In Reinforcement Learning there's a reward function
- RLHF incorporates human feedback in the reward function, to be more aligned with human goals, wants and needs
 - **What is RLHF?** assess the quality of the model's responses
- RLHF is used throughout GenAI applications including LLM Models
- RLHF significantly enhances the model performance
- Example: grading text translations from “technically correct” to “human”

RLHF Example: Internal company knowledge chatbot

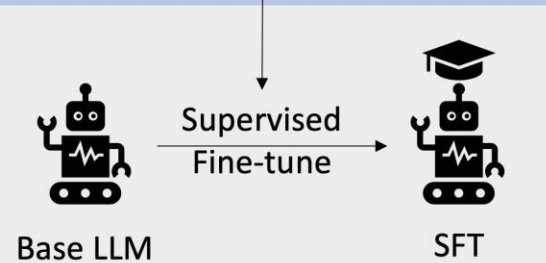


- Data collection
 - Set of human-generated prompts and responses are created
 - “Where is the location of the HR department in Boston?”
- Supervised fine-tuning of a language model
 - Fine-tune an existing model with internal knowledge
 - Then the model creates responses for the human-generated prompts
 - Responses are mathematically compared to human-generated answers
- Build a separate reward model
 - Humans can indicate which response they prefer from the same prompt
 - The reward model can now estimate how a human would prefer a prompt response
- Optimize the language model with the reward-based model
 - Use the reward model as a reward function for RL
 - This part can be fully automated

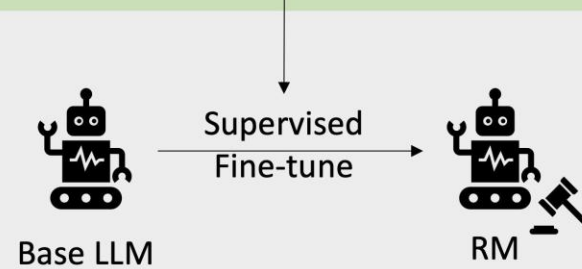
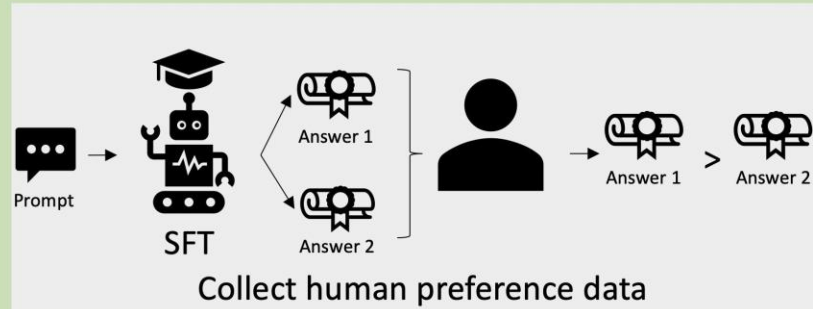
RLHF Process



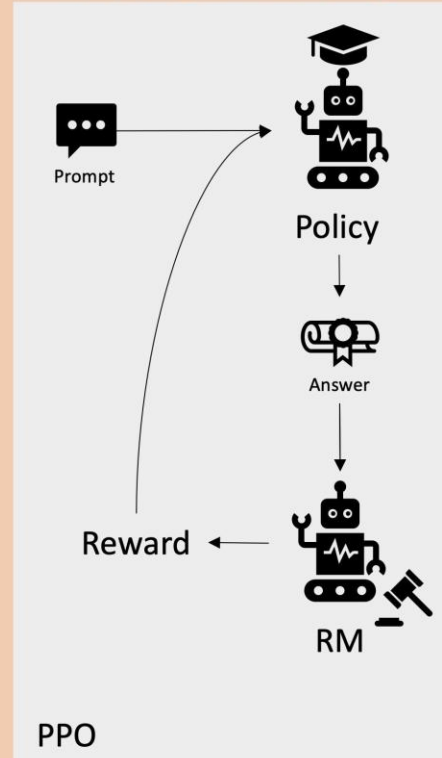
Step 1 Supervised Fine-Tuning



Step 2 Training a Reward Model



Step 3 Optimize Policy

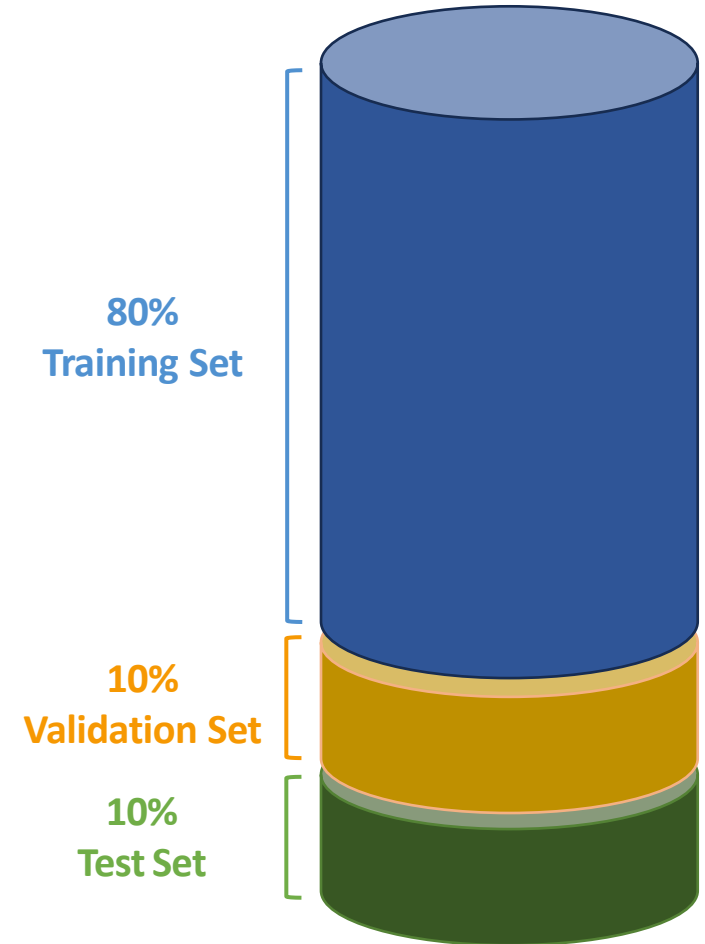


<https://aws.amazon.com/what-is/reinforcement-learning-from-human-feedback/>



Training vs Validation vs Test Set

- Training Set
 - Used to train the model
 - Percentage: typically, 60-80% of the dataset
 - Example: 800 labeled images from a dataset of 1000 images
- Validation Set
 - Used to tune model parameters and validate performance
 - Percentage: typically, 10-20% of the dataset
 - Example: 100 labeled images for hyperparameter tuning (tune the settings of the algorithm to make it more efficient)
- Test Set
 - Used to evaluate the final model performance
 - Percentage: typically, 10-20% of the dataset
 - Example: 100 labeled images to test the model's accuracy



Feature Engineering



- The process of using domain knowledge to select and transform raw data into meaningful features
- Helps enhancing the performance of machine learning models
- **Techniques**
 - Feature Extraction – extracting useful information from raw data, such as deriving age from date of birth
 - Feature Selection – selecting a subset of relevant features, like choosing important predictors in a regression model
 - Feature Transformation – transforming data for better model performance, such as normalizing numerical data
- Particularly meaningful for Supervised Learning

Before

Feature Engineering

Customer_ID	Name	BirthDate	Purchase_Amount
1	Alice	15-05-1993	\$200
2	Bob	22-08-1978	\$300



After

Feature Engineering

Customer_ID	Name	Age	Purchase_Amount
1	Alice	30	\$200
2	Bob	45	\$300



Feature Engineering on Structured Data

- **Structured Data (Tabular Data)**
- Example: Predicting house prices based on features like size, location, and number of rooms
- **Feature Engineering Tasks**
 - Feature Creation – deriving new features like “price per square foot”
 - Feature Selection – identifying and retaining important features such as location or number of bedrooms
 - Feature Transformation – normalizing features to ensure they are on a similar scale, which helps algorithms like gradient descent converge faster

Feature Engineering on UnStructured Data



- **Unstructured Data (Text, Images)**
- Example: sentiment analysis of customer reviews
- **Feature Engineering Tasks**
 - Text Data – converting text into numerical features using techniques like TF-IDF or word embeddings
 - Image Data – extracting features such as edges or textures using techniques like convolutional neural networks (CNNs)

Training Data

Training Data



- To train our model we must have good data
- Garbage in => Garbage out
- Most critical stage to build a good model
- Several options to model our data, which will impact the types of algorithms we can use to train our models
- Labeled vs. Unlabeled Data
- Structured vs. Unstructured Data

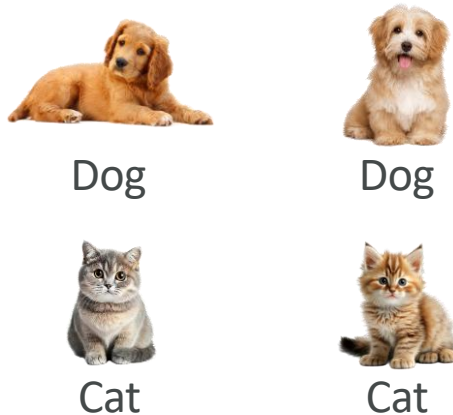


Labeled vs Unlabeled Data



Labeled Data

- Data includes both input features and corresponding output labels
- Example: dataset with images of animals where each image is labeled with the corresponding animal type (e.g., cat, dog)
- Use case: Supervised Learning, where the model is trained to map inputs to known outputs



Unlabeled Data

- Data includes only input features without any output labels
- Example: a collection of images without any associated labels
- Use case: Unsupervised Learning, where the model tries to find patterns or structures in the data



Structured Data



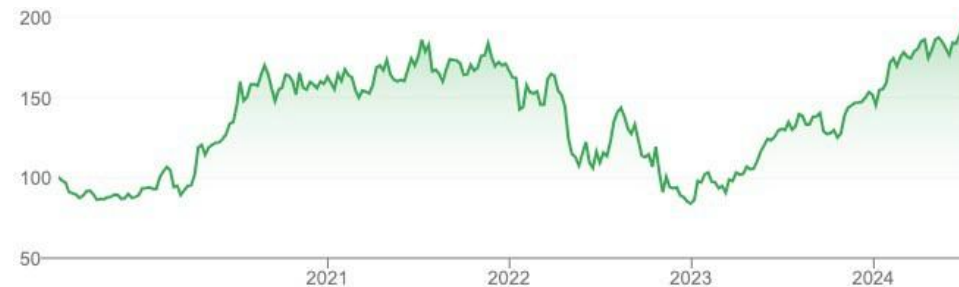
- Data is organized in a structured format, often in rows and columns (like Excel)
- **Tabular Data**
 - Data is arranged in a table with rows representing records and columns representing features
 - Example: customers database with fields such as name, age, and total purchase amount

Customer_ID	Name	Age	Purchase_Amount
1	Alice	30	\$200
2	Bob	45	\$300

Structured Data



- Data is organized in a structured format, often in rows and columns (like Excel)
- **Time Series Data**
 - Data points collected or recorded at successive points in time
 - Example: Stock prices recorded daily over a year



Date	Stock Price
01-07-2024	\$197.20
02-07-2024	\$200

Unstructured Data



- Data that doesn't follow a specific structure and is often text-heavy or multimedia content
- Text Data
 - Unstructured text such as articles, social media posts, or customer reviews
 - Example: a collection of product reviews from an e-commerce site
- Image Data
 - Data in the form of images, which can vary widely in format and content
 - Example: images used for object recognition tasks



Review: Attended a yoga class at the new studio. The instructor was excellent, and the facility was well-maintained. Loved the variety of classes offered. Only downside was the parking situation.

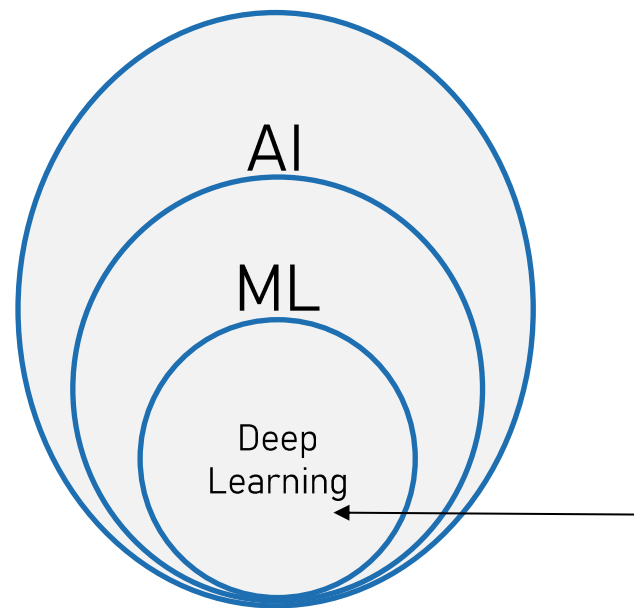


Deep Learning

Deep Learning



Deep learning is a subset of ML



Machine Learning

Supervised Learning

Unsupervised Learning

Reinforcement Learning

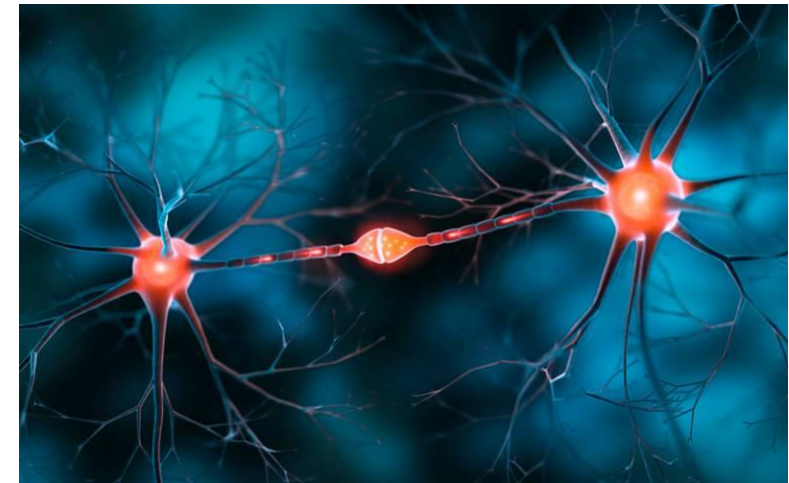
Deep Learning

Deep Learning

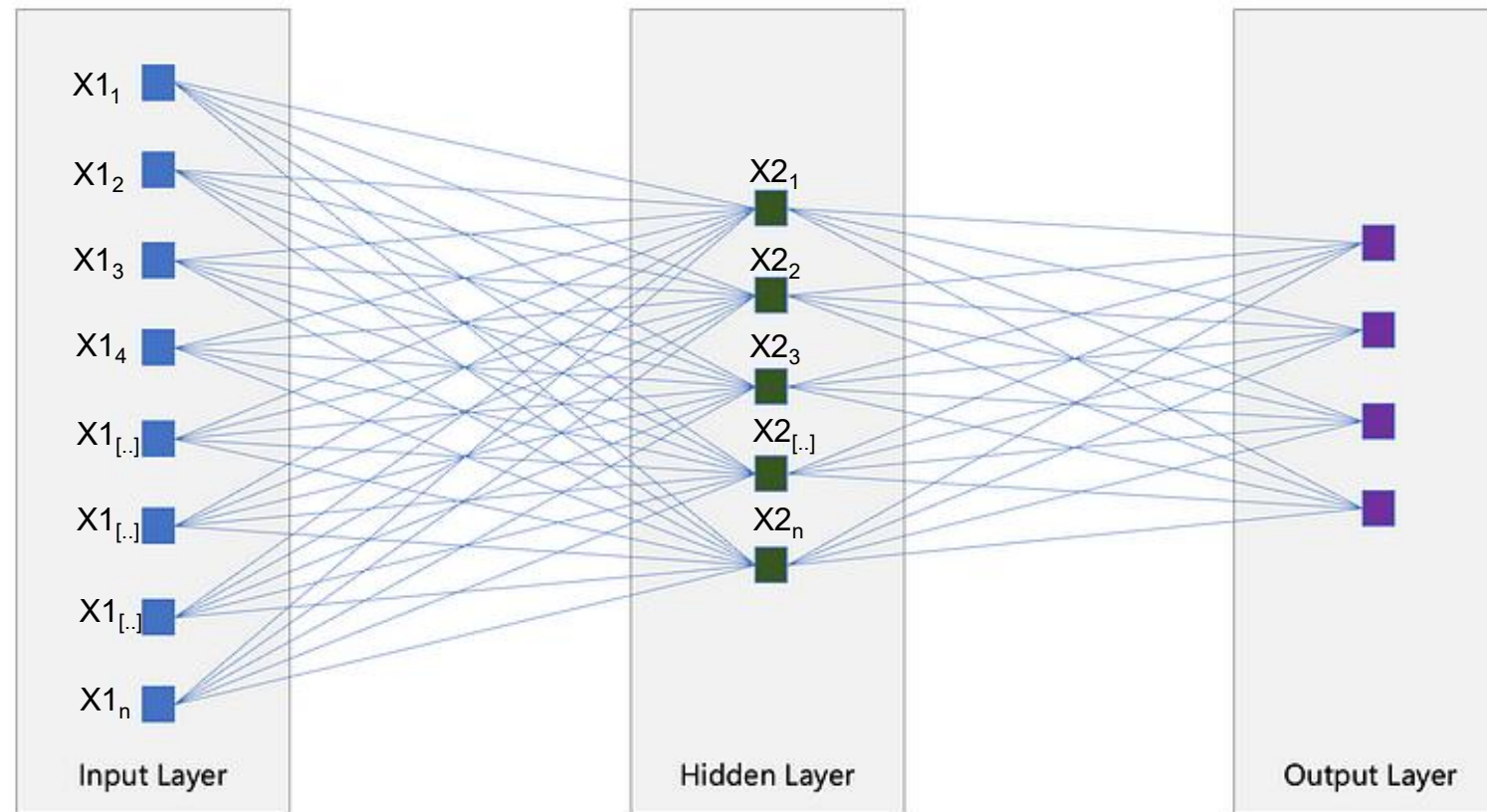


Deep learning uses Artificial Neural Networks - allowing them to process more complex patterns than traditional machine learning

- Uses neurons and synapses (like our brain) to train a model
 - Deep Learning because there's more than one layer of learning
 - Ex: Computer Vision – image classification, object detection, image segmentation
 - Ex: Natural Language Processing (NLP) – text classification, sentiment analysis, machine translation, language generation
- Large amount of input data Requires GPU (Graphical Processing Unit)



Artificial Neural Networks



Deep Learning Model Types



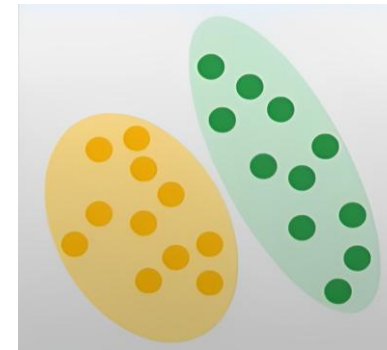
Discriminative

- Used to classify or predict
- Typically trained on a dataset of labeled data
- Learns the relationship between the features of the data points and the labels



Generative

- Used to generate
- Typically trained on a dataset of unlabeled data
- Learns the relationship between the features of the data points and generate



Deep Learning Model Types



Discriminative
Technique



Classify

Discriminative Technique
(classify as a dog or a cat)



Discriminative
Technique



Classify

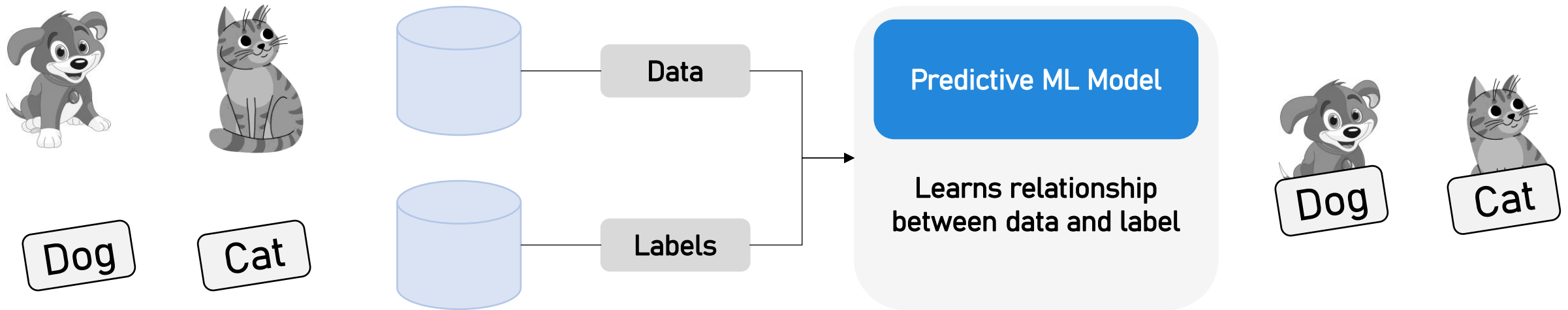
Discriminative Technique
(classify as a dog or a cat)



Deep Learning Model Types



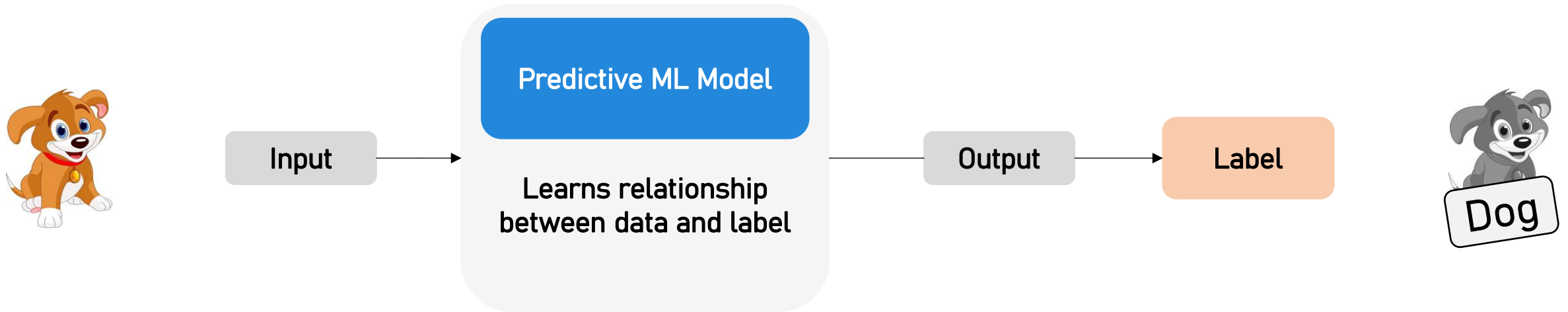
Training Phase



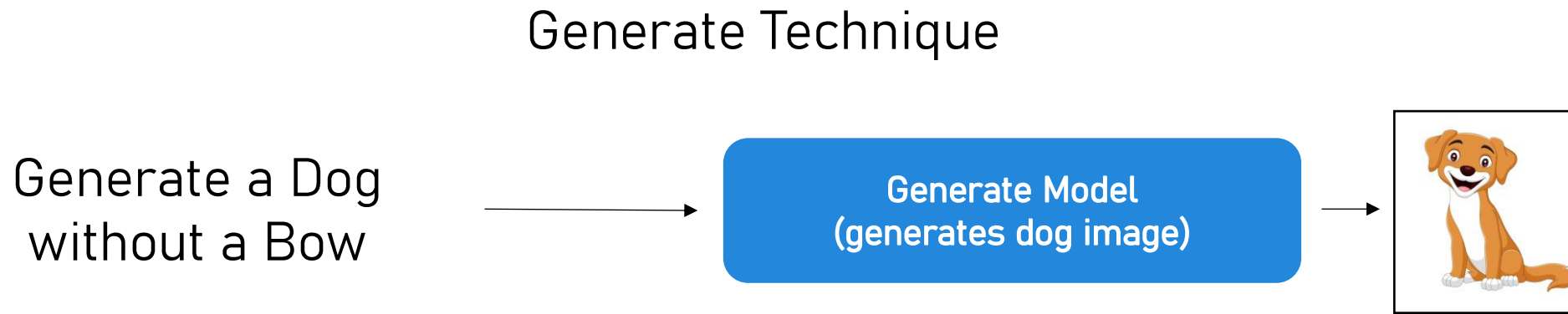
Deep Learning Model Types



Execution Phase



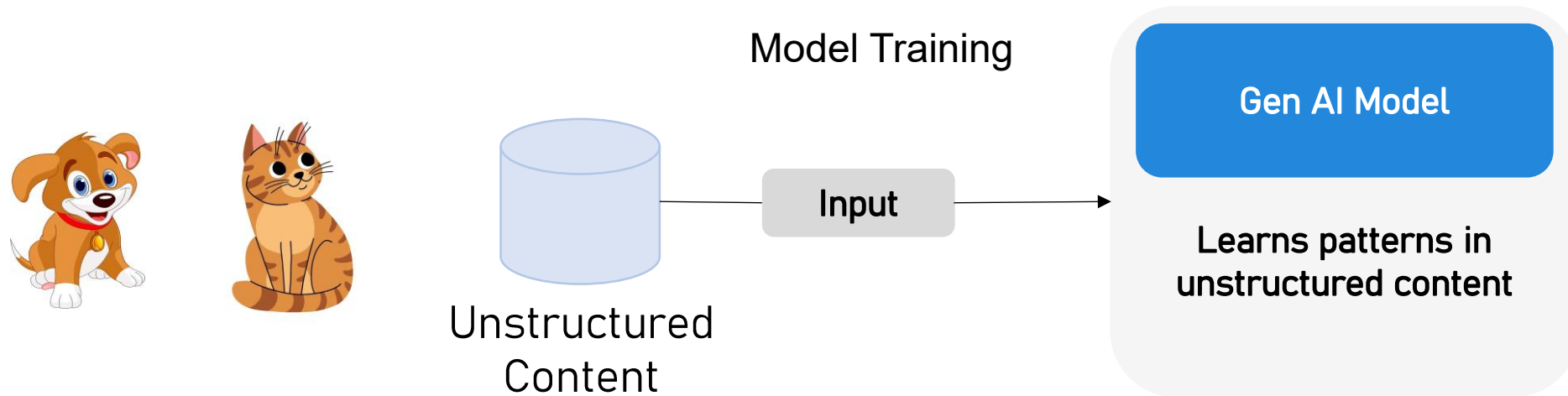
Deep Learning Model Types



Deep Learning Model Types



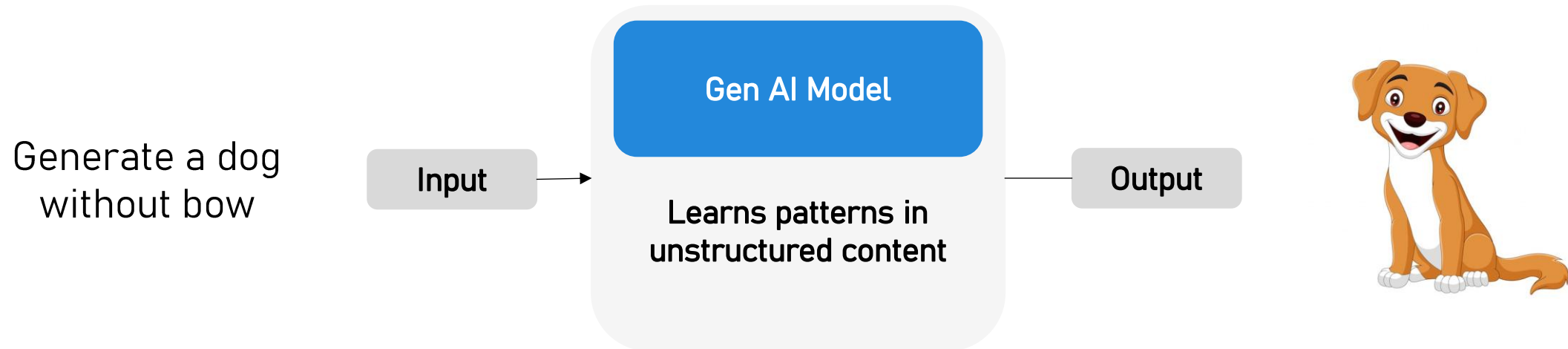
Training Phase



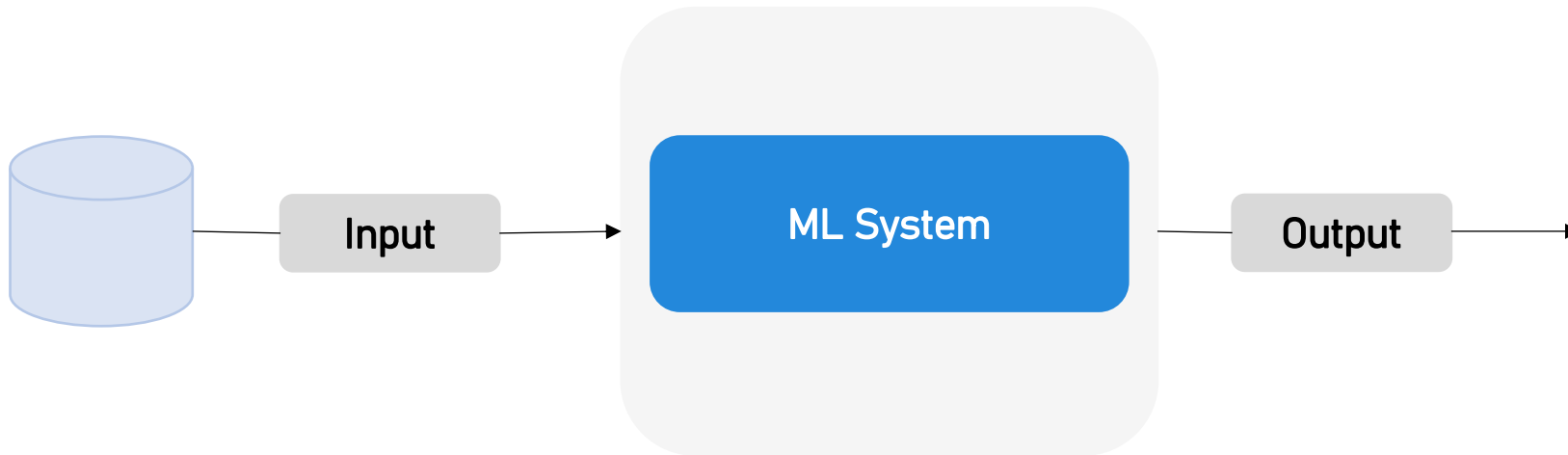
Deep Learning Model Types



Execution Phase



ML System Processing



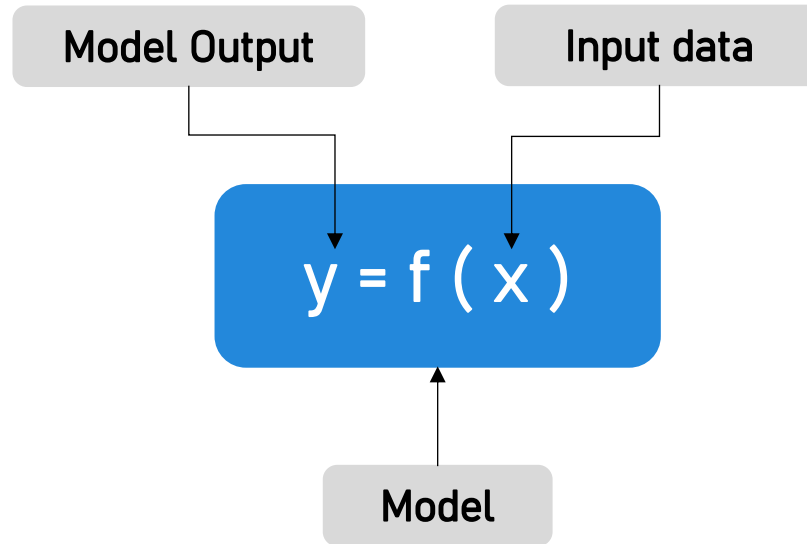
Not GenAI when y is a:

- Number
- Discrete
- Class
- Probability

Is GenAI when y is:

- Natural language
- Image
- Audio

How to Identify



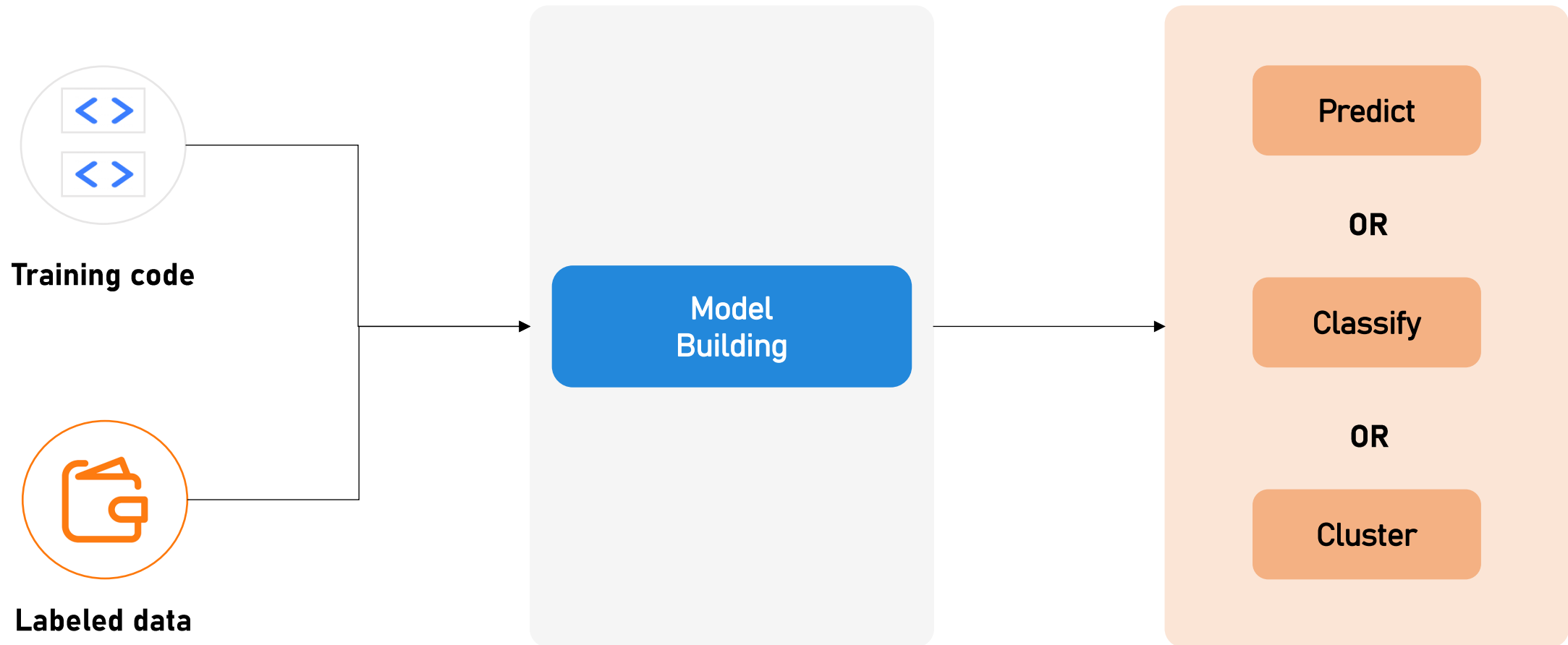
Not GenAI when y is a :

- Number
- Discrete
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- Probability

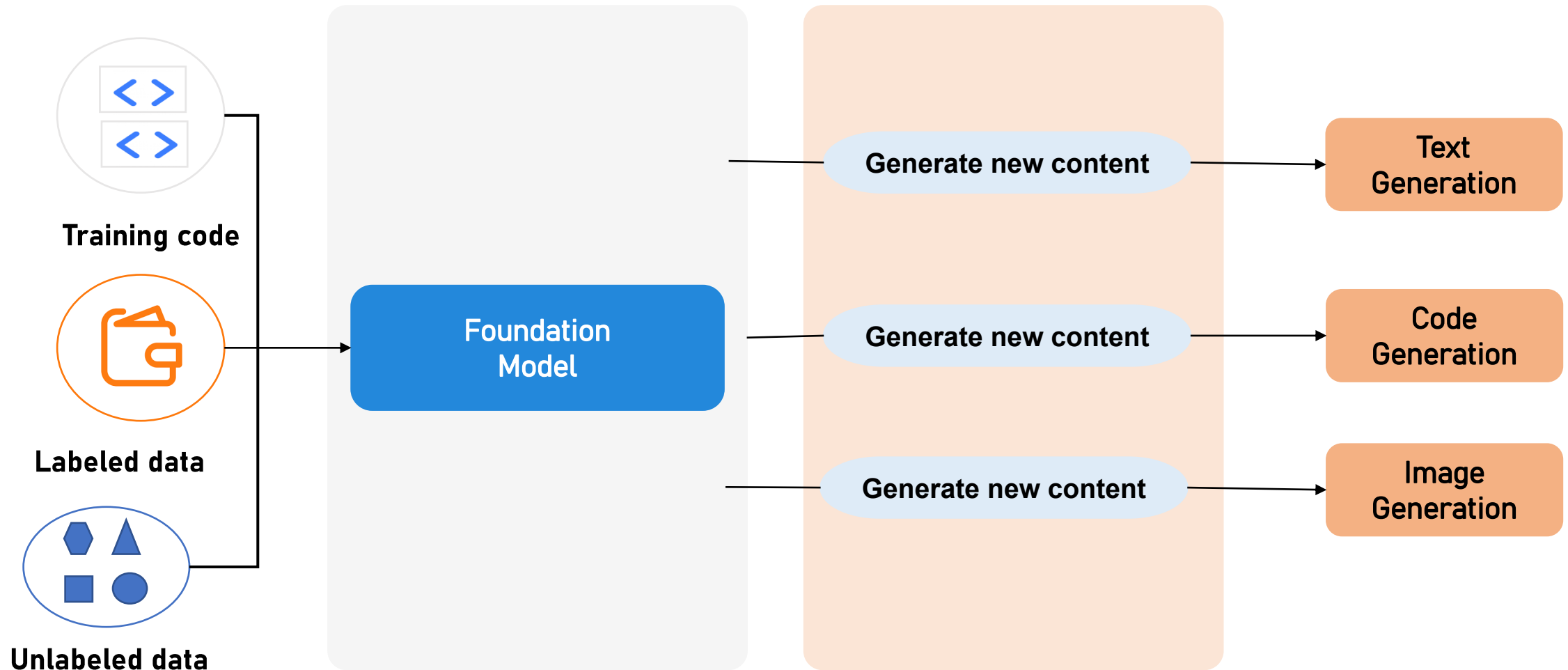
Is GenAI when y is:

- Natural language
- Image
- Audio

Classical Supervised & Unsupervised Learning



Gen AI Supervised, Semi-Supervised & Unsupervised Learning



GAI – Use Case

Real Estate Trained Model

Data Set



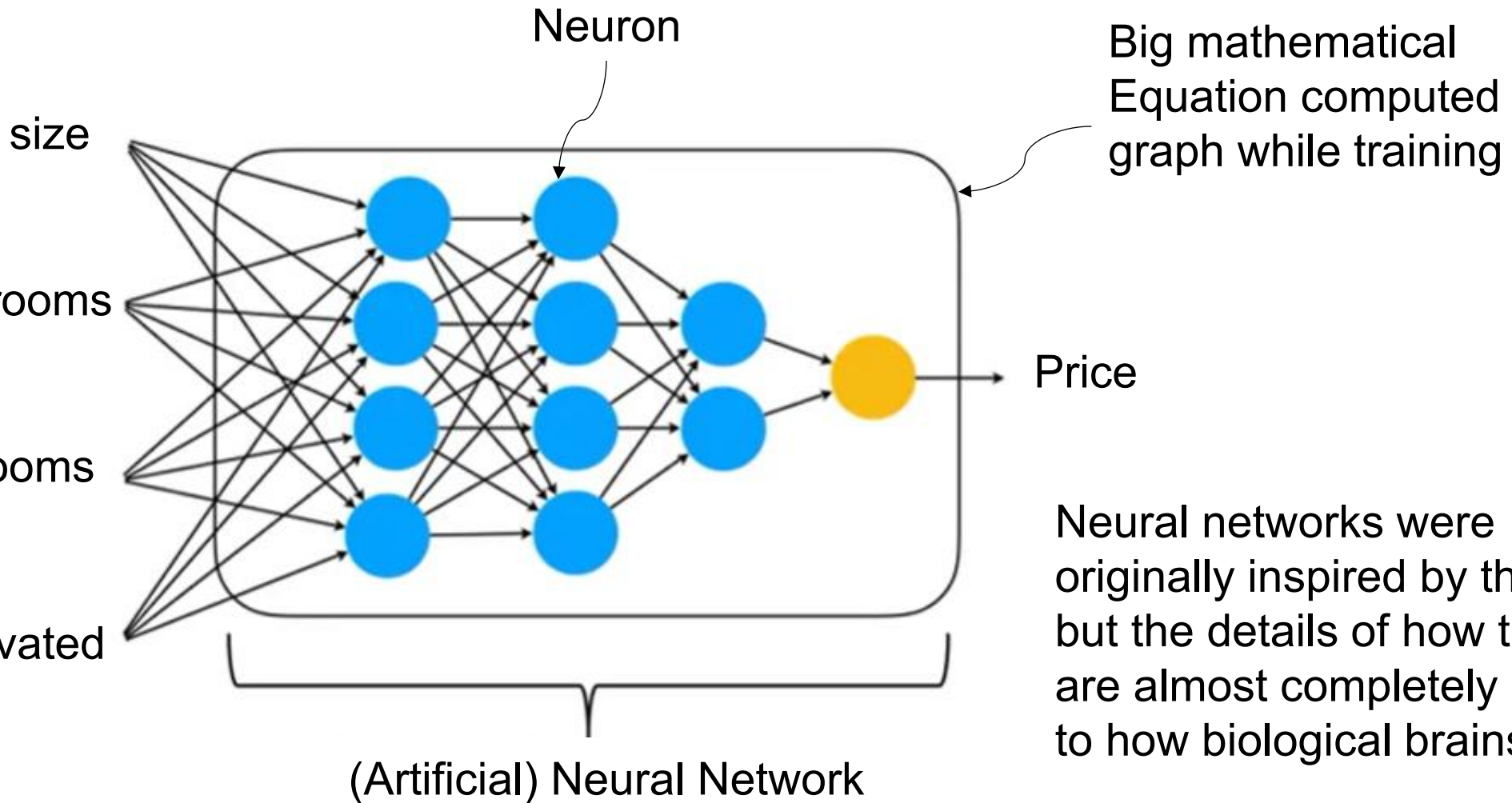
Home prices

size of house (square feet)	# of bedrooms	# of bathrooms	newly renovated	price (1000\$)
523	1	2	N	115
645	1	3	N	150
708	2	1	N	210
1034	3	3	Y	280
2290	4	4	N	355
2545	4	5	Y	440

Deep learning Model Training



Features:



Neural networks were originally inspired by the brain, but the details of how they work are almost completely unrelated to how biological brains work.

Question (Prompt)



I am looking to buy a 2-bedroom house. how much can I expect to pay for a newly renovated 2-bedroom, 1-bathroom house?

Model Response



Based on the provided Dataset, for a newly renovated 2-bedroom, 1-bathroom house, you can expect to pay approximately \$210,000.

What is Generative AI?



- GenAI is a type of Artificial Intelligence that creates new content based on what it has learned from existing content.
- The process of learning from existing content is called training and results in the creation of a statistical model (LLM).
- When given a prompt, GenAI uses this statistical model (LLM) to predict what an expected response might be-and this generates new content.



Architecture

Types of Architecture



- CNN
- RNN
- LSTM
- Transformer
- Etc...

CNN - Convolutional Neural Networks



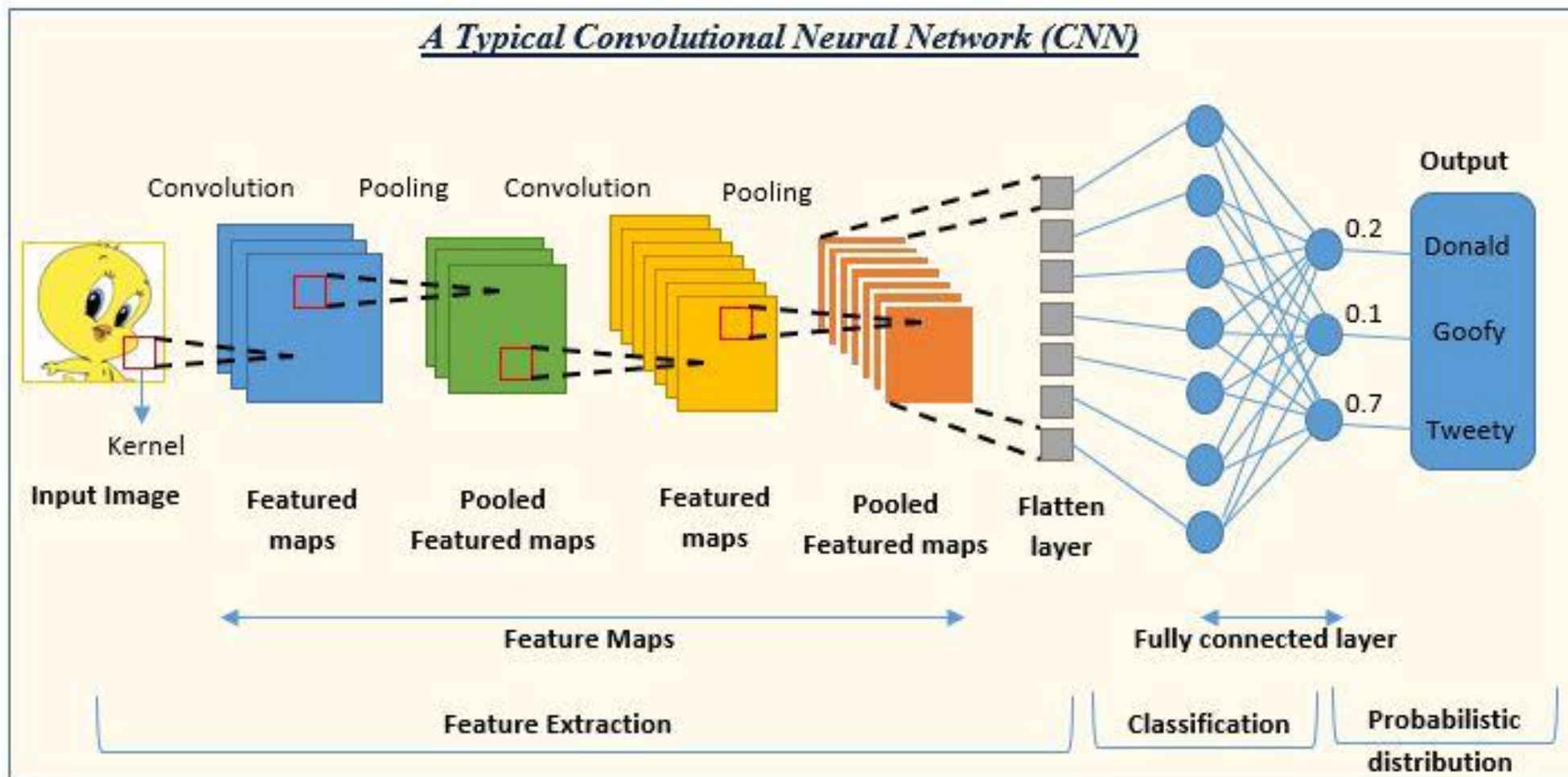
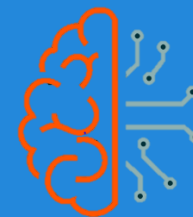
- A Convolutional Neural Network (CNN) is a type of artificial neural network primarily used for analyzing visual imagery.
- It's designed to automatically and adaptively learn spatial hierarchies of features from input data, making it particularly effective for tasks like image recognition, image classification, and object detection.

CNN - Convolutional Neural Networks



- **Focus:** Primarily used for image processing and feature extraction.
- **Strengths:** Excellent at identifying local patterns and spatial relationships within data.
- **Limitations:** Can struggle with long-range dependencies and sequential data due to their reliance on local receptive fields.

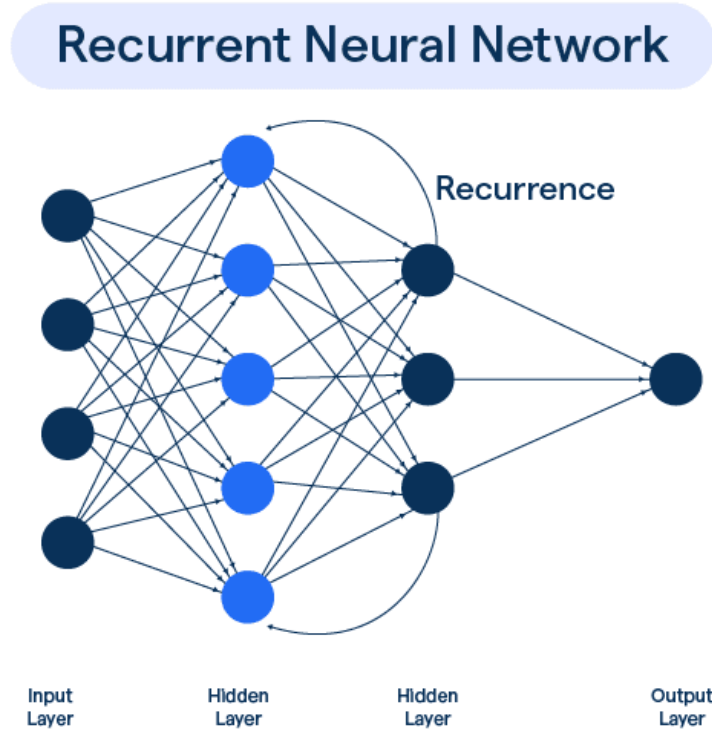
CNN - Convolutional Neural Networks



RNNs - Recurrent Neural Networks



- A Recurrent Neural Network (RNN) is a type of neural network designed to process sequential data by retaining information from previous inputs.



RNNs - Recurrent Neural Networks



- **Focus:** Designed for sequential data like text, time series, and speech.
- **Strengths:** Can maintain a "memory" of past inputs, making them suitable for tasks involving temporal dependencies.
- **Limitations:** Can suffer from vanishing or exploding gradients when processing long sequences, making it difficult to learn long-range dependencies.

LSTM

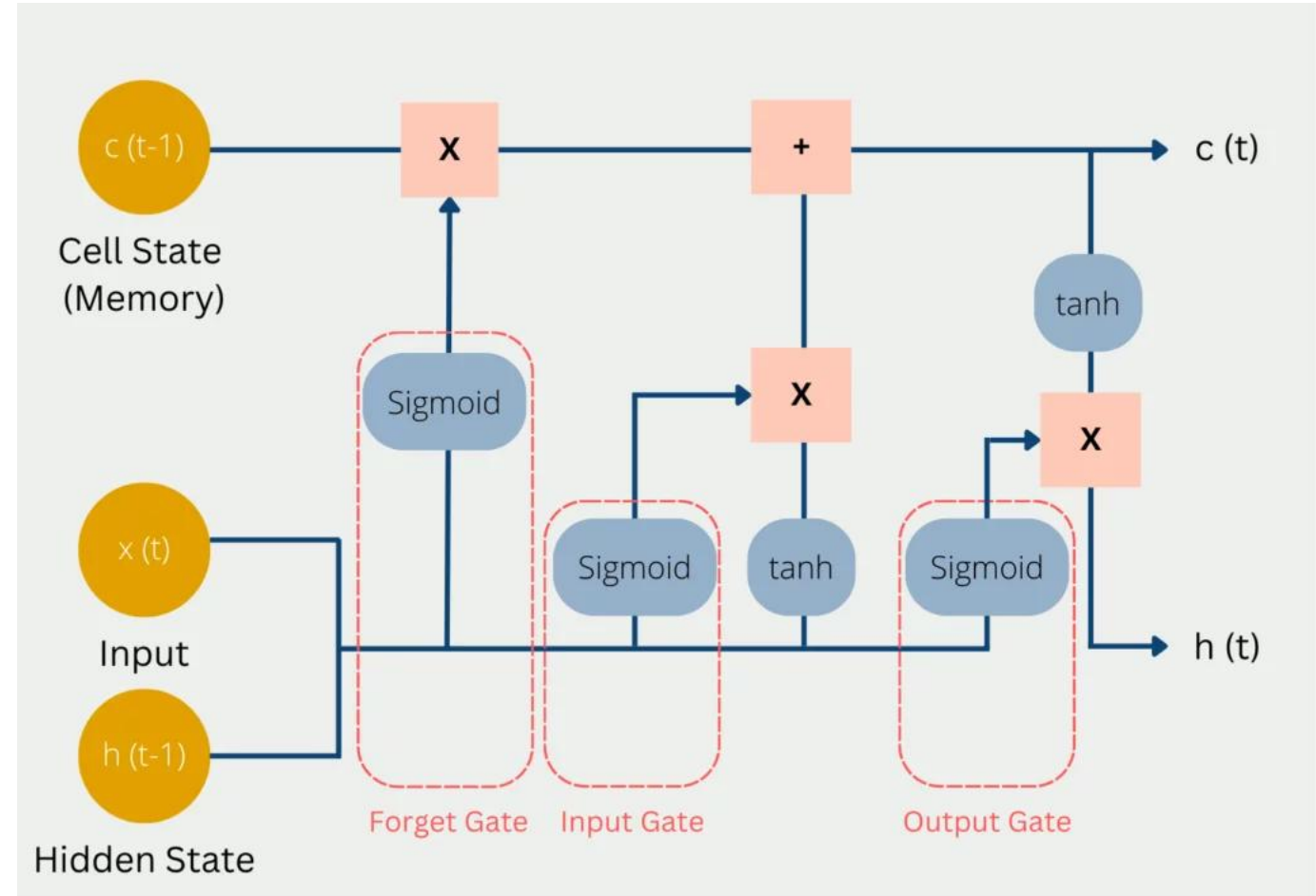


- LSTM, or Long Short-Term Memory, is a type of recurrent neural network (RNN) architecture particularly effective for processing sequential or time-series data.
- It addresses the limitations of standard RNNs by introducing a memory cell and gating mechanisms to handle long-range dependencies and mitigate the vanishing gradient problem.
- This makes LSTMs well-suited for tasks like language modeling, speech recognition, and time series forecasting.

LSTM



- Input
- Hidden State
- Memory Cell [Remember]
- Forget Gate
- Input Gate
- Output Gate



Transformers



- A transformer is a deep learning architecture developed by Google and based on the multi-head attention mechanism, proposed in a 2017 paper "Attention Is All You Need".[1] It has no recurrent units, and thus requires less training time than previous recurrent neural architectures, such as long short-term memory (LSTM)

Transformers



- **Focus:** Initially popularized for natural language processing but now used in various domains.
- **Strengths:** Utilize self-attention mechanisms to weigh the importance of different parts of the input sequence, enabling them to capture long-range dependencies effectively.
- **Advantages:** Parallelizable, meaning they can be trained more efficiently than RNNs.
- **Disadvantages:** Can be computationally intensive and require large amounts of data for training.

Attention Is All You Need [2017]



Transformers

Attention Is All You Need

Ashish Vaswani*
Google Brain
avaswani@google.com

Noam Shazeer*
Google Brain
noam@google.com

Niki Parmar*
Google Research
nikip@google.com

Jakob Uszkoreit*
Google Research
usz@google.com

Llion Jones*
Google Research
llion@google.com

Aidan N. Gomez*[‡]
University of Toronto
aidan@cs.toronto.edu

Lukasz Kaiser*
Google Brain
lukaszkaiser@google.com

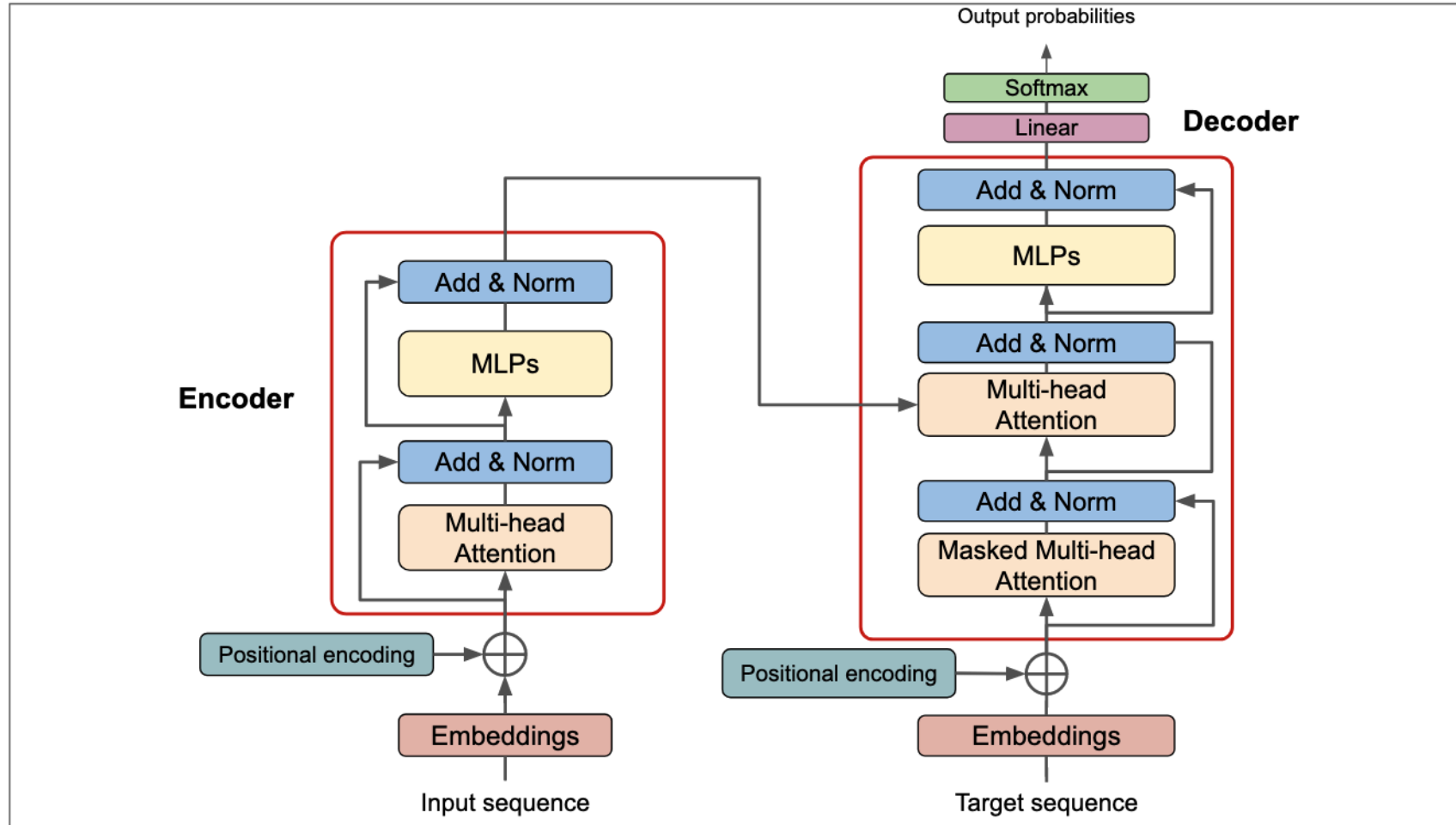
Illia Polosukhin*[‡]
illia.polosukhin@gmail.com

Abstract

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks that include an encoder and a decoder. The best performing models also connect the encoder and decoder through an attention mechanism. We propose a new simple network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolutions entirely. Experiments on two machine translation tasks show these models to

- Scale efficiently
- Parallel process
- Attention to input meaning

Transformers Architecture

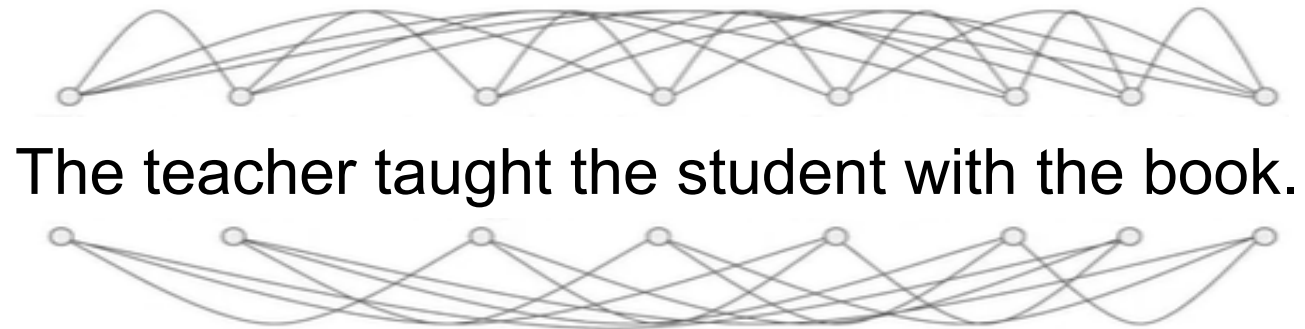


Transformers

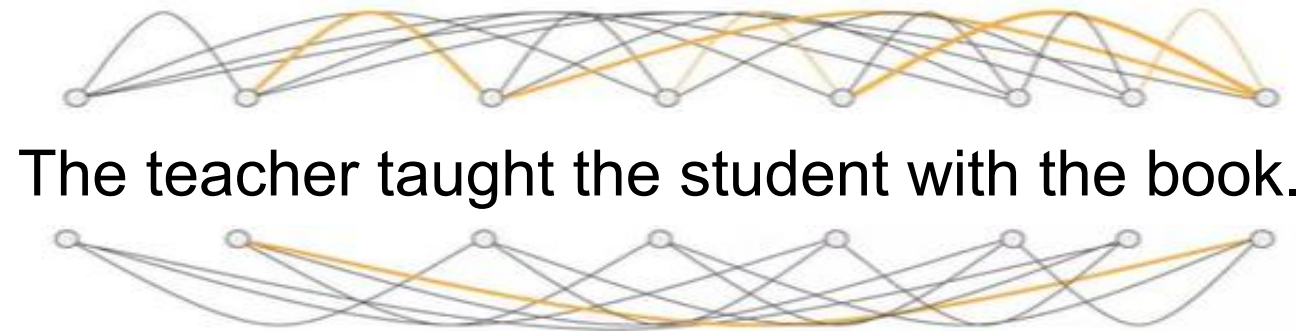


The teacher taught the student with the book.

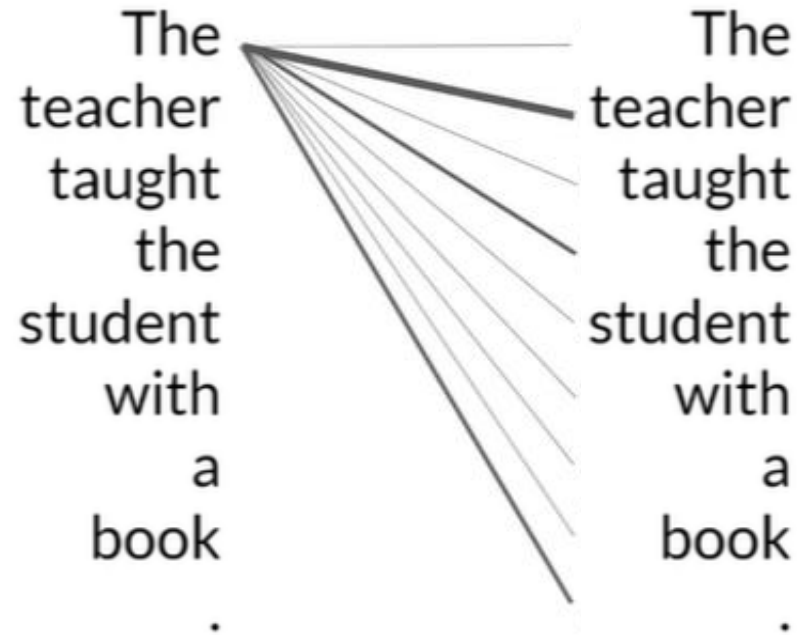
Transformers



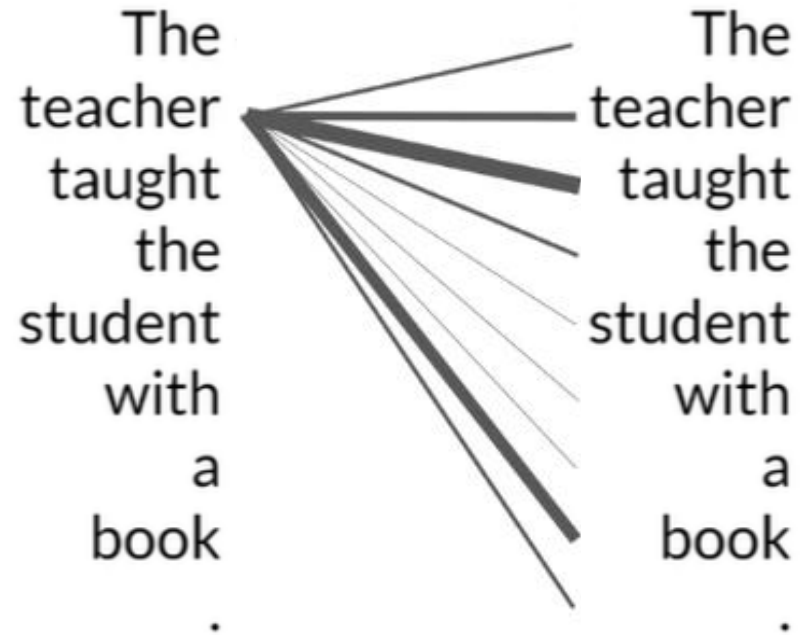
Transformers



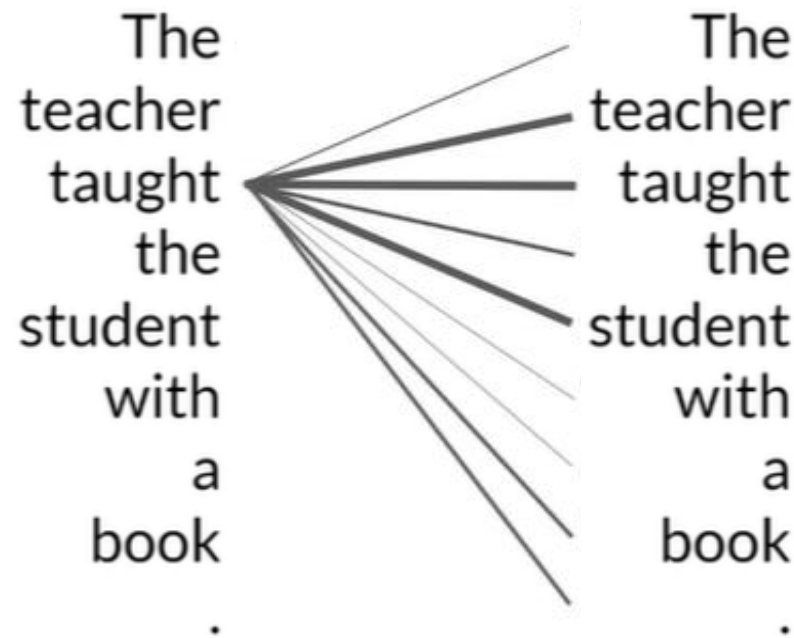
Self-attention



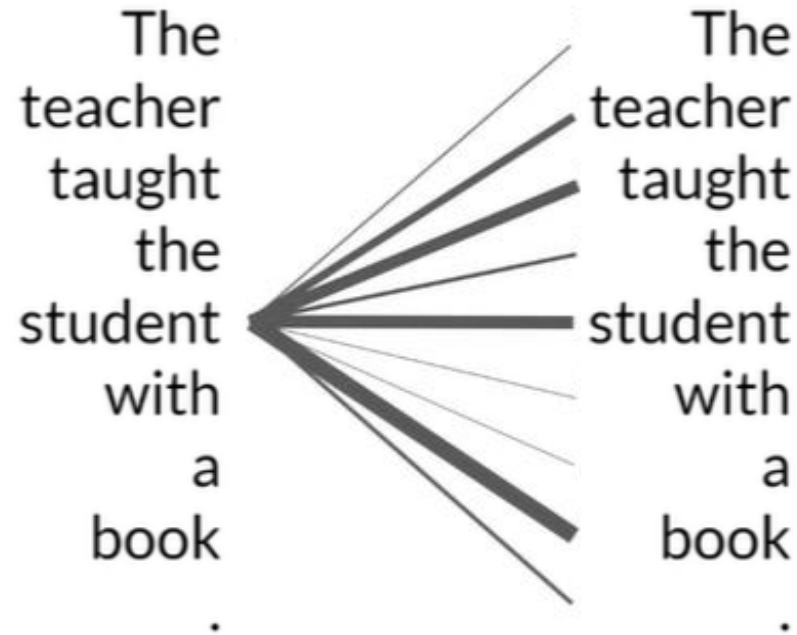
Self-attention



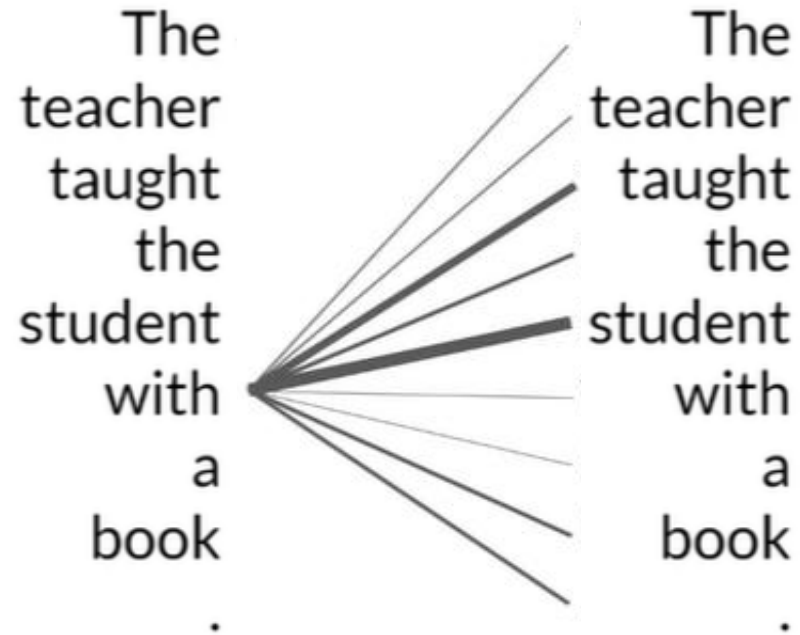
Self-attention



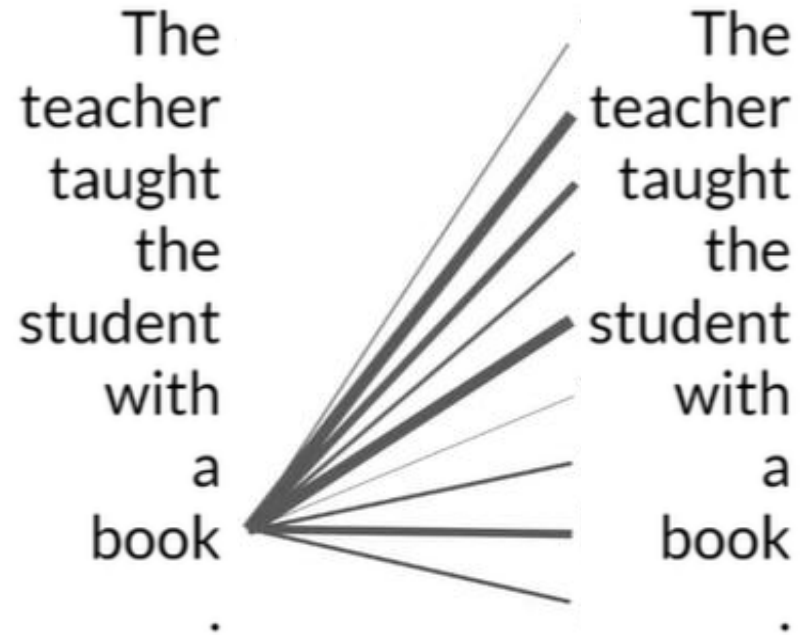
Self-attention



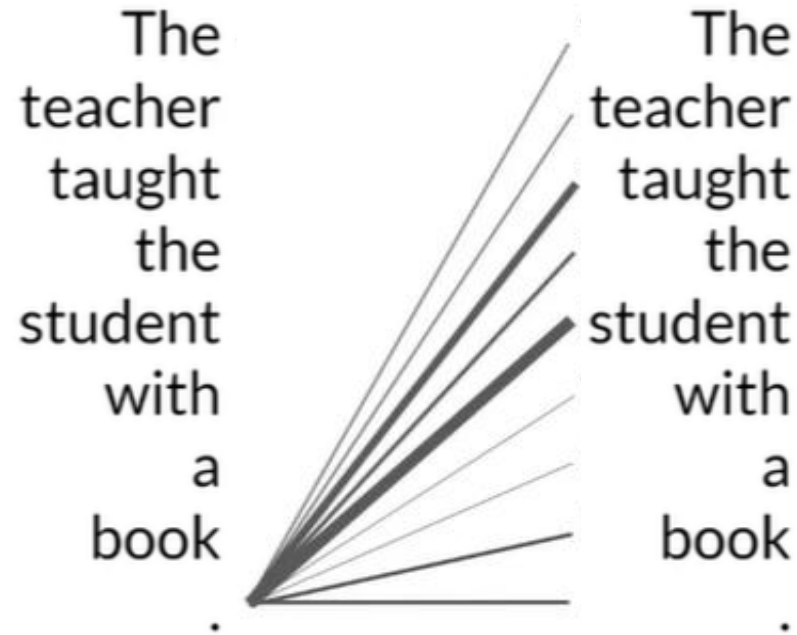
Self-attention



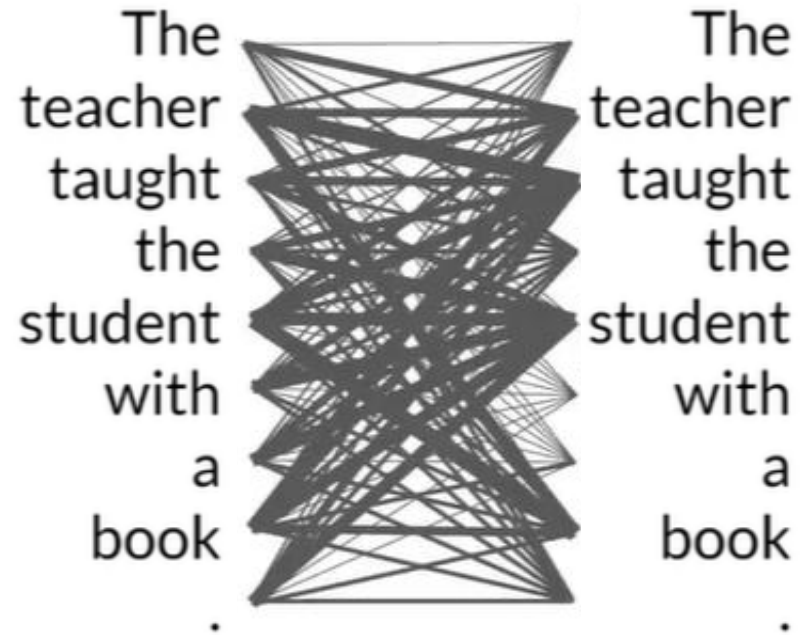
Self-attention



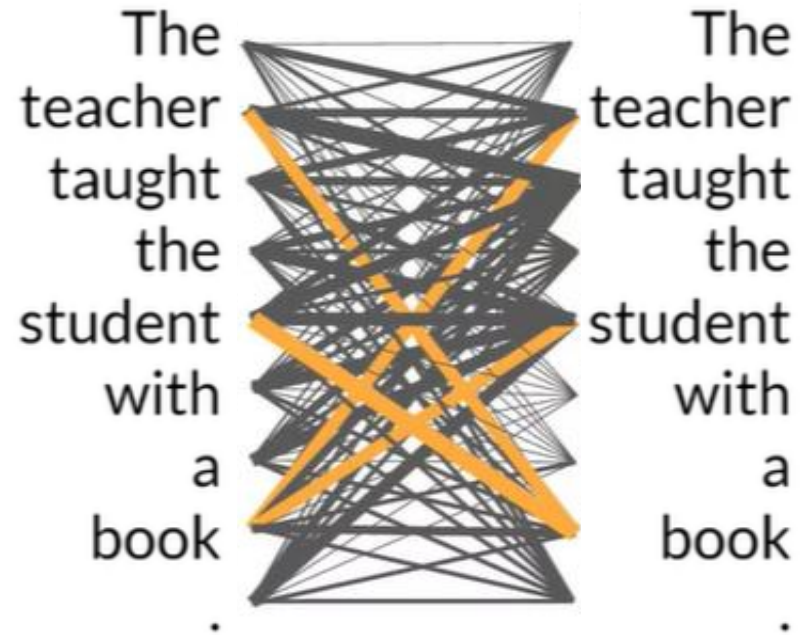
Self-attention



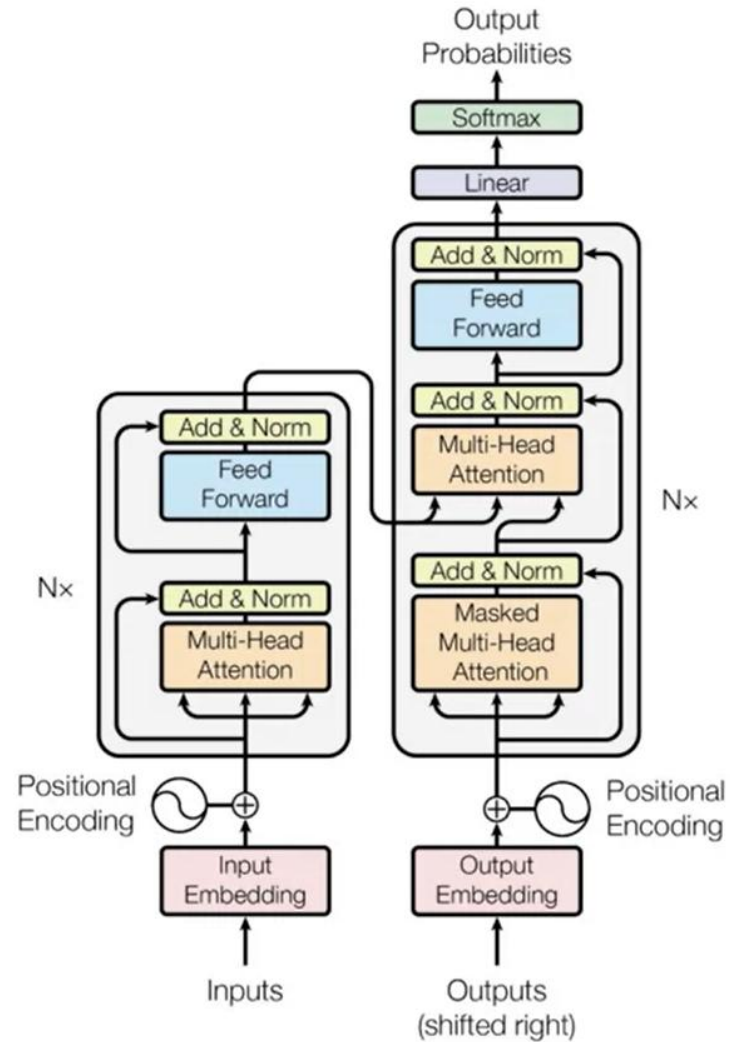
Self-attention



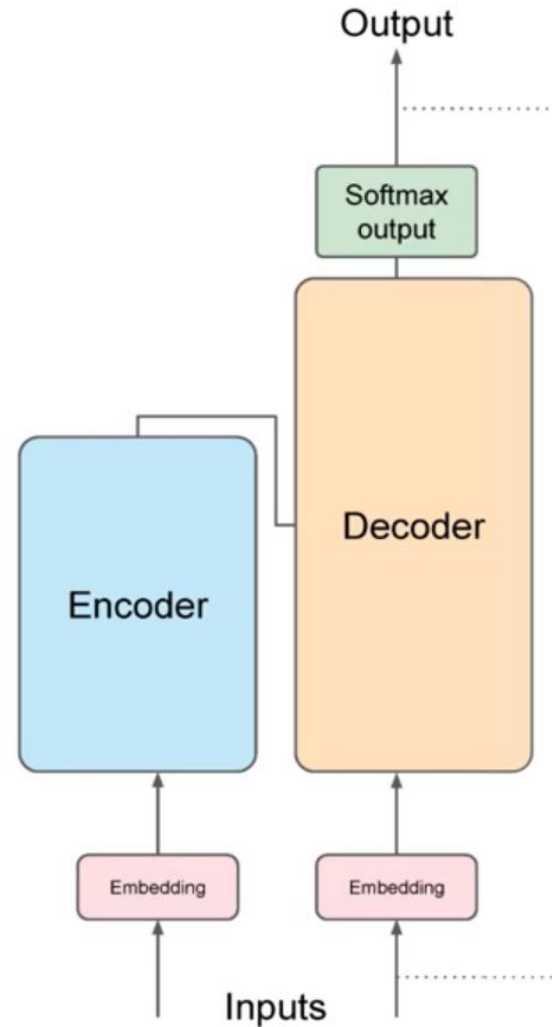
Self-attention



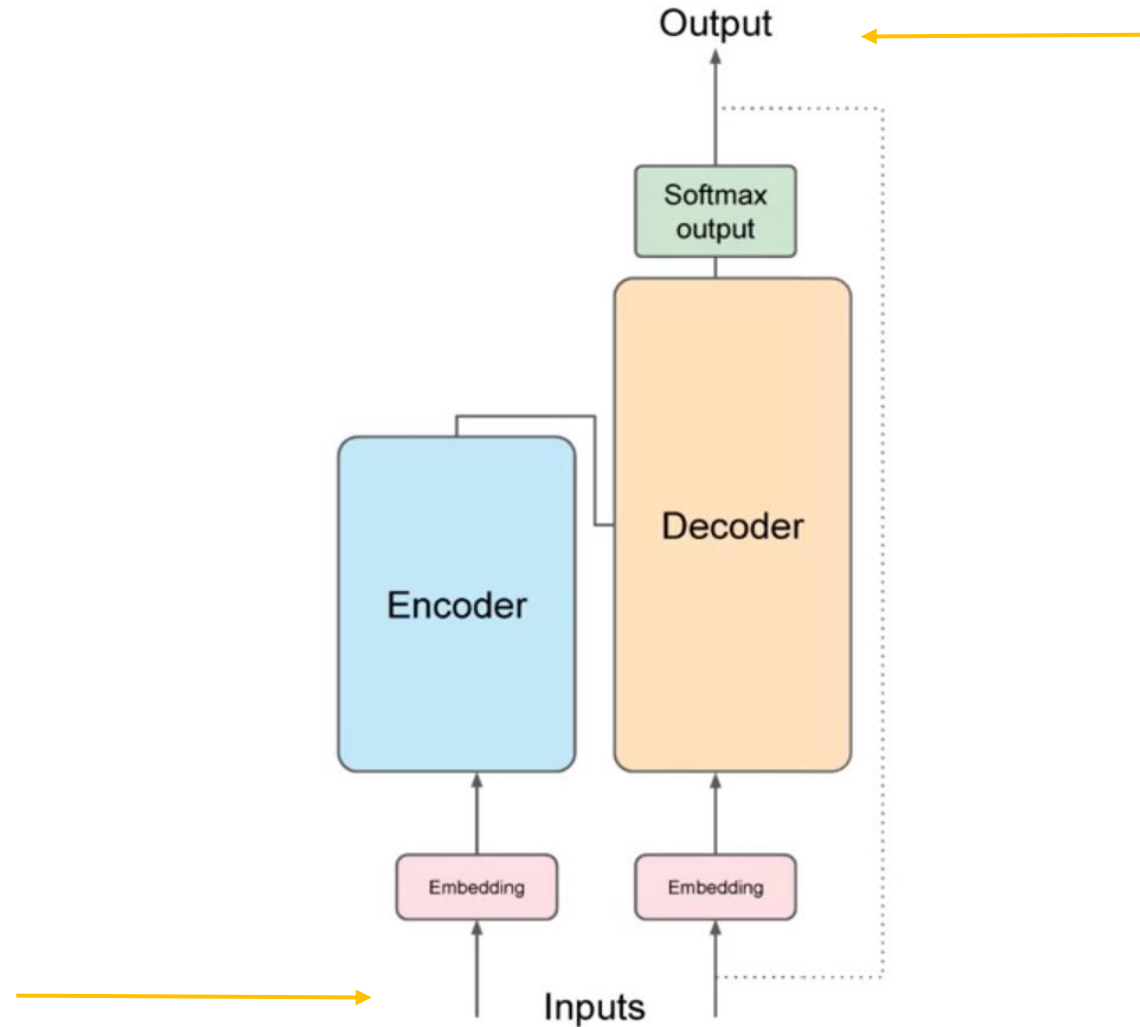
Transformers Architecture



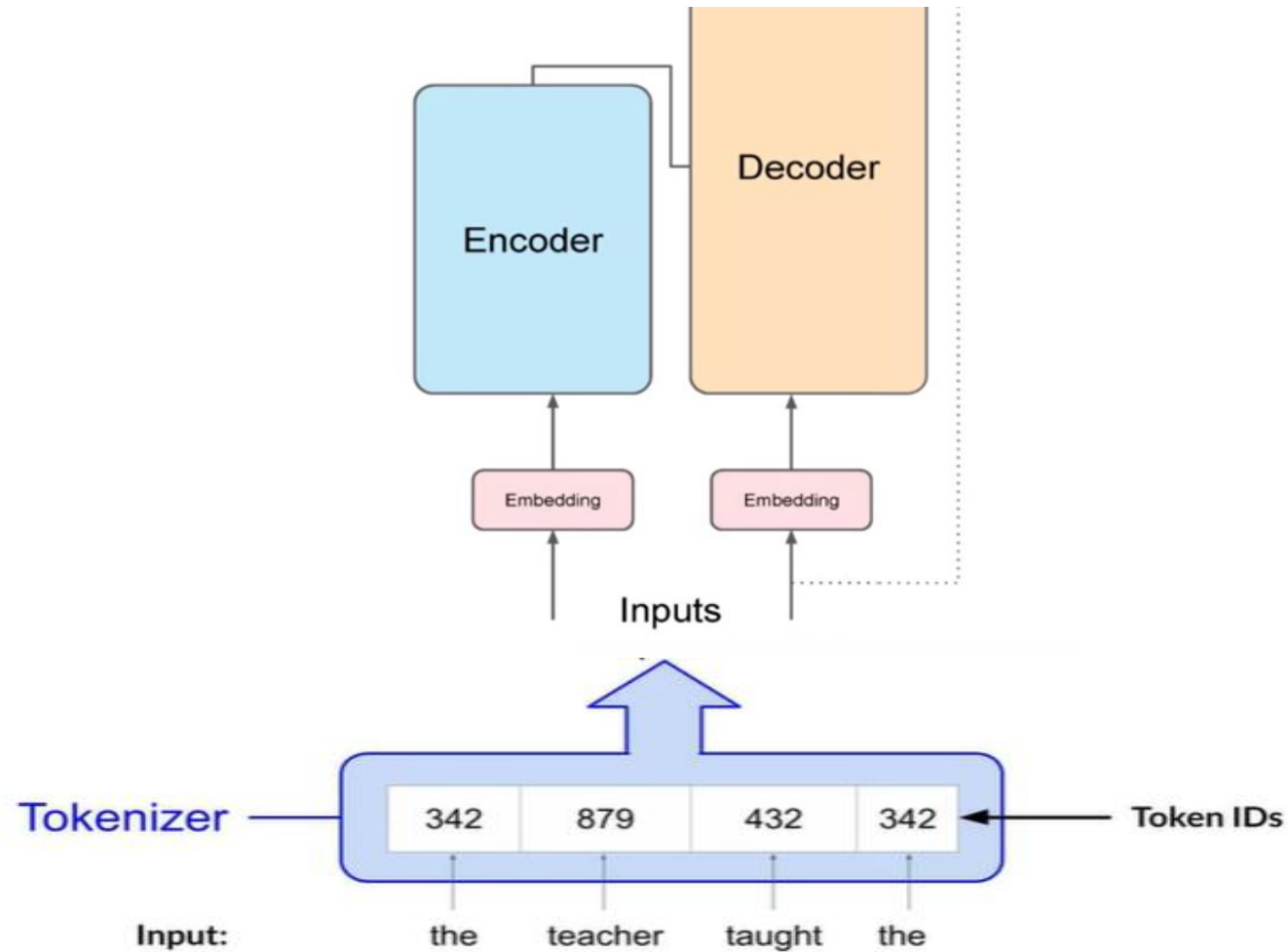
Transformers Architecture



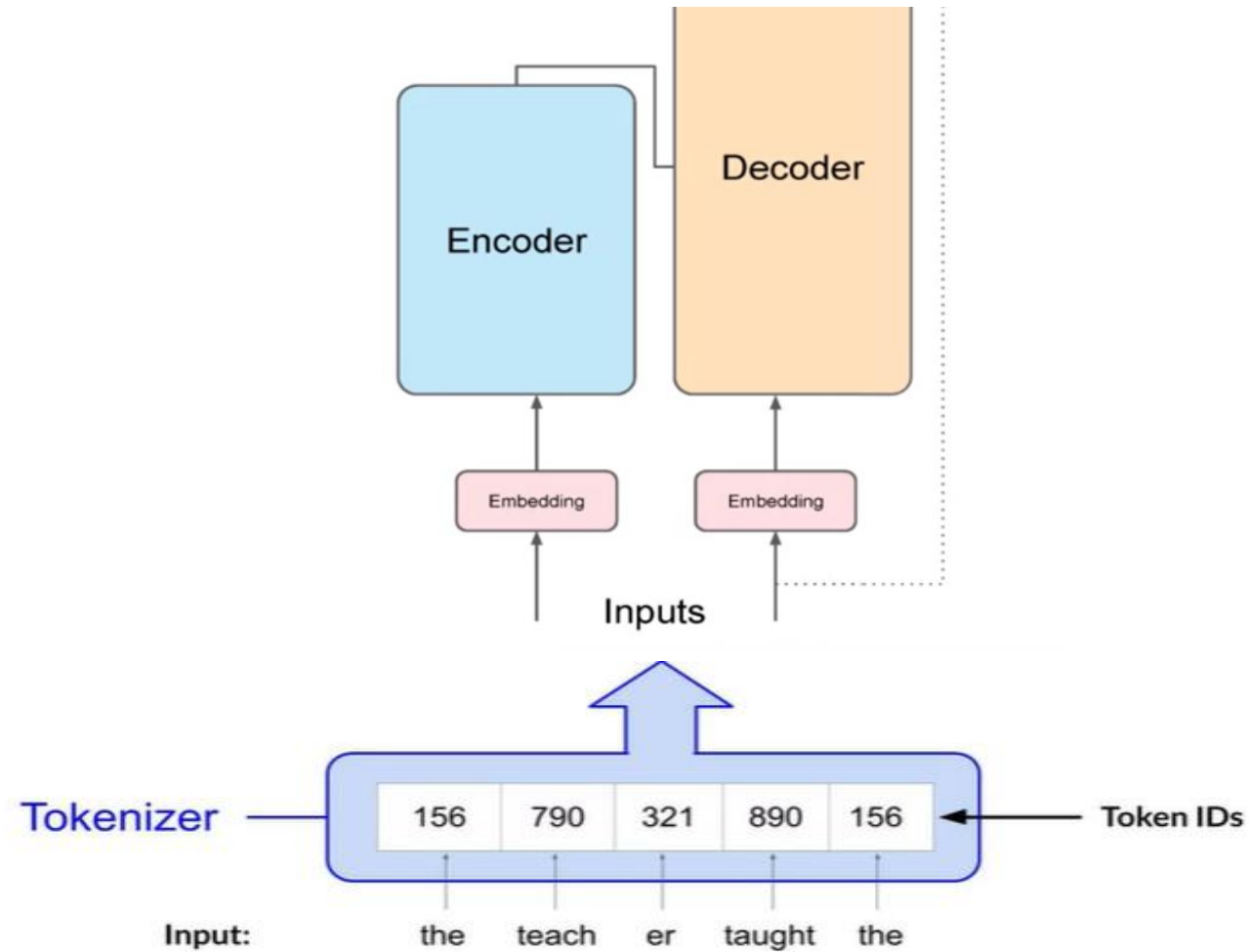
Transformers Architecture



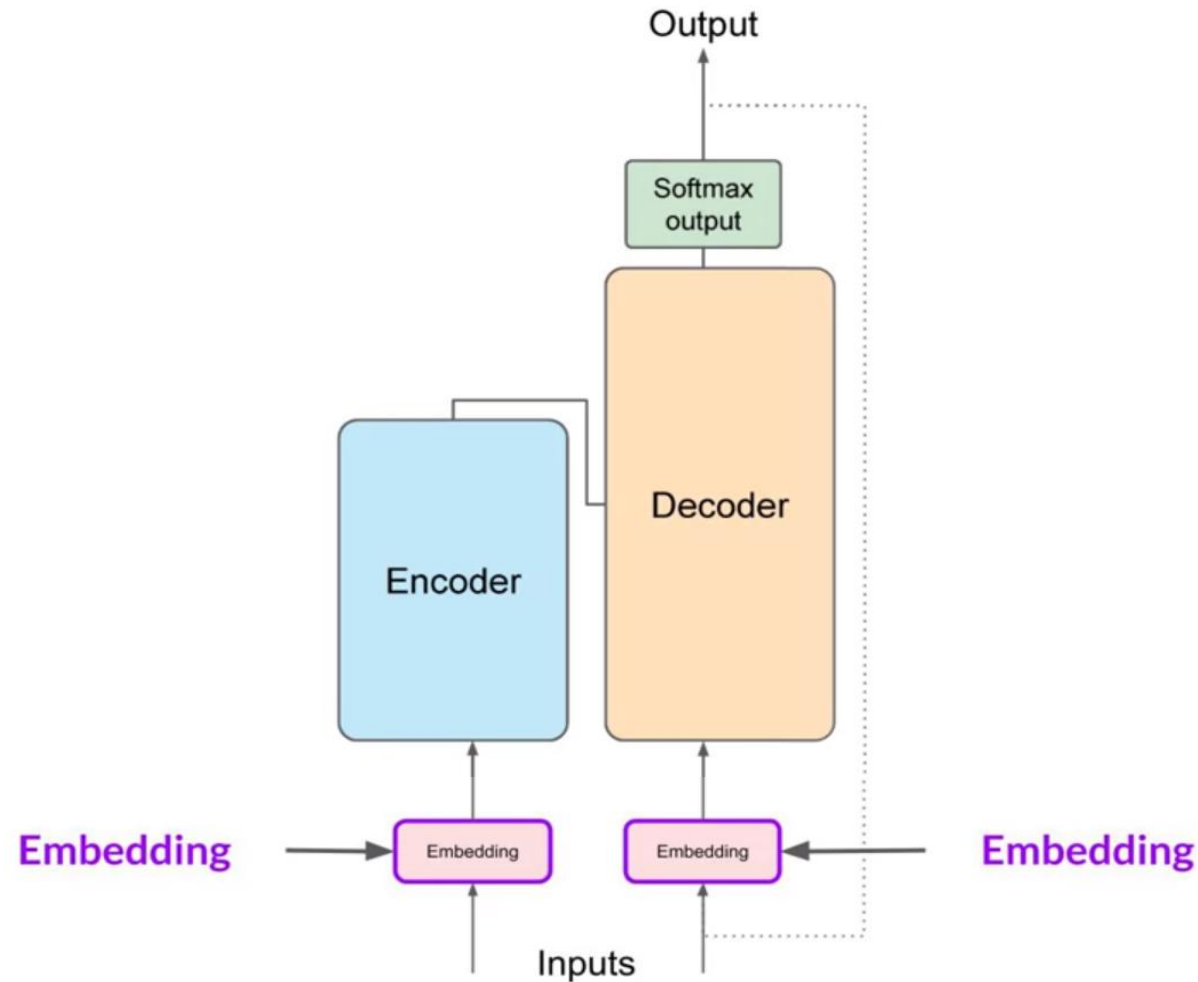
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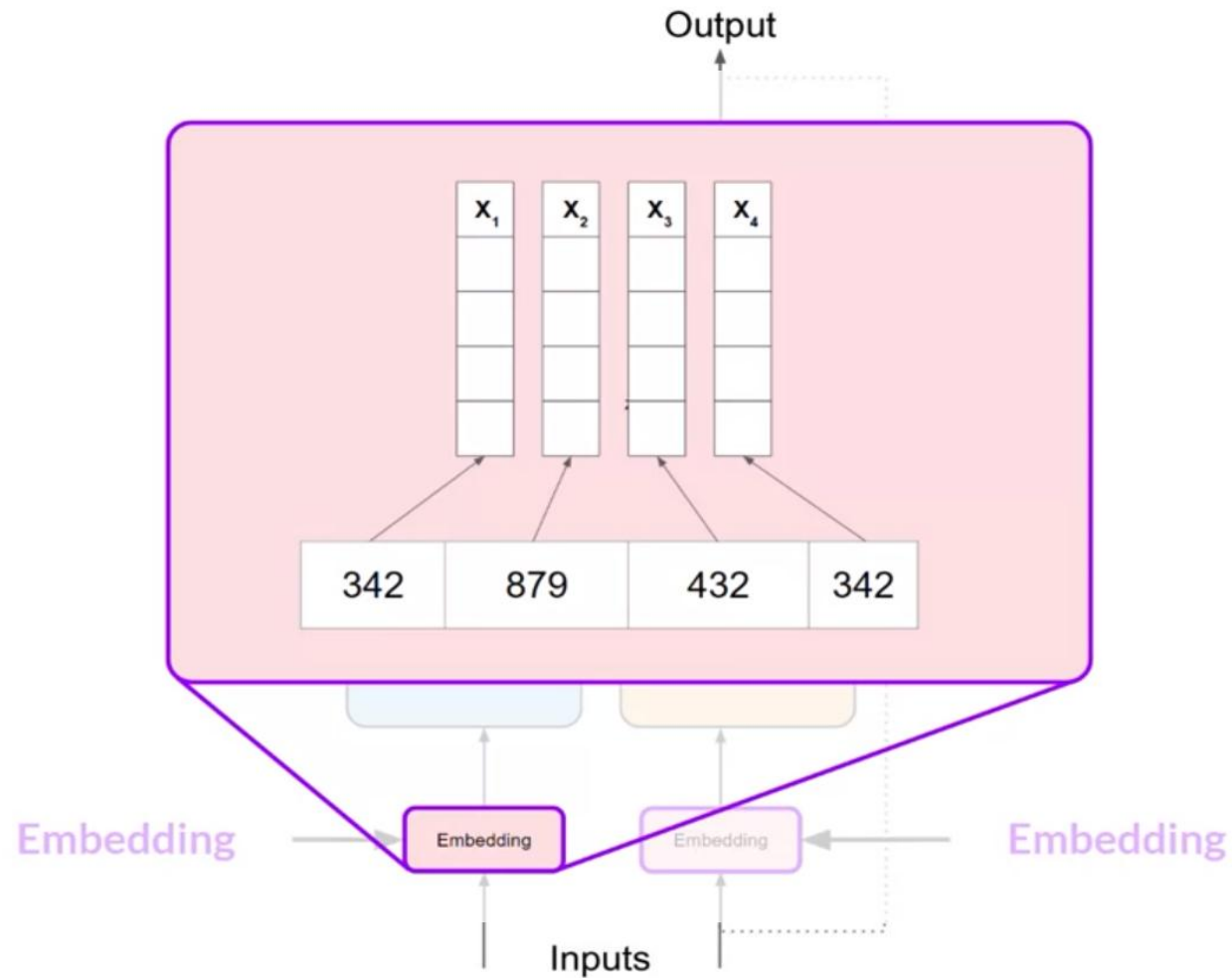
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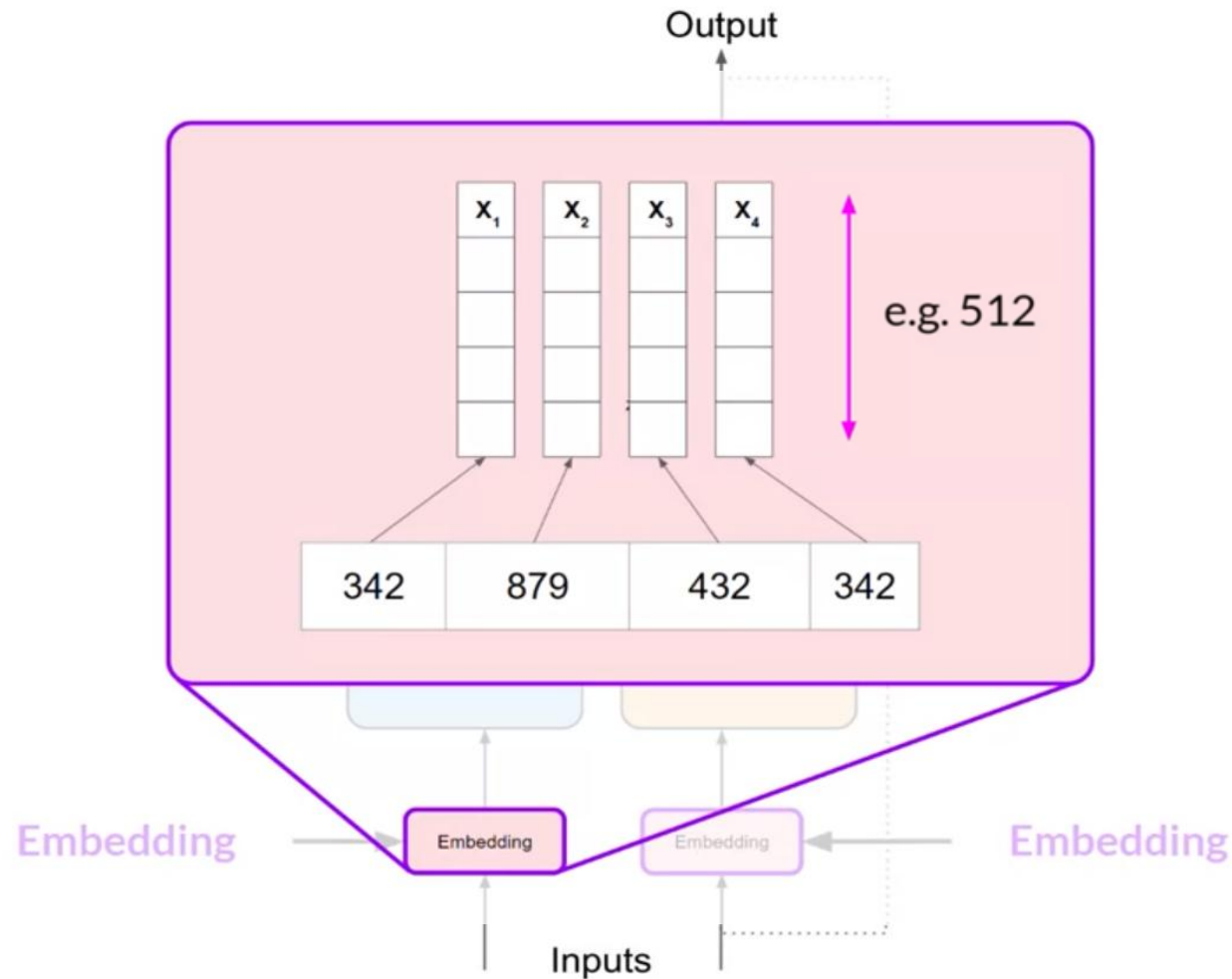
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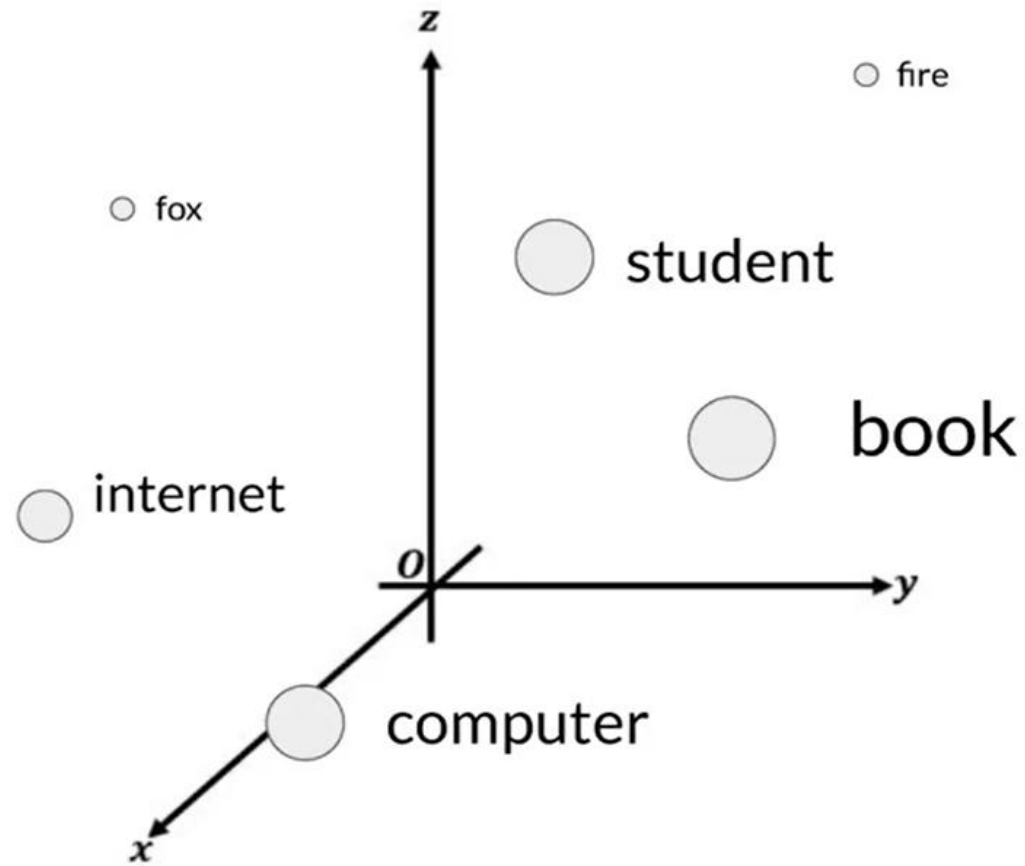
Transformers Architecture



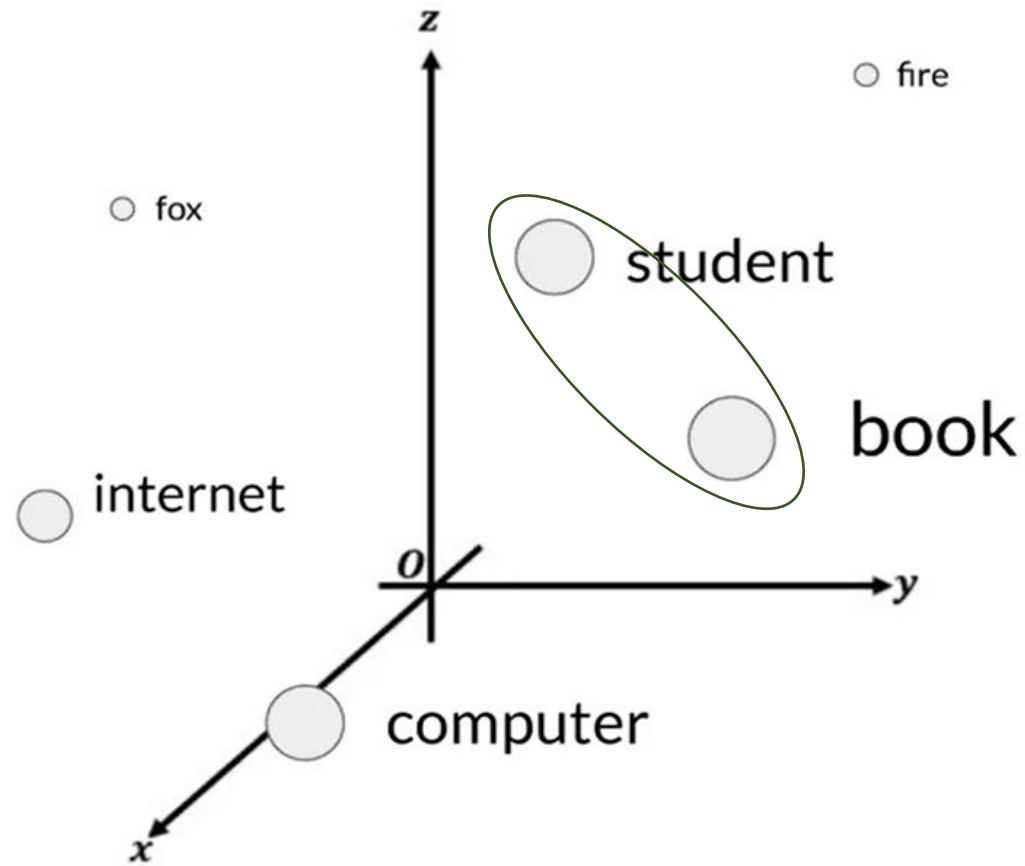
Transformers Architecture



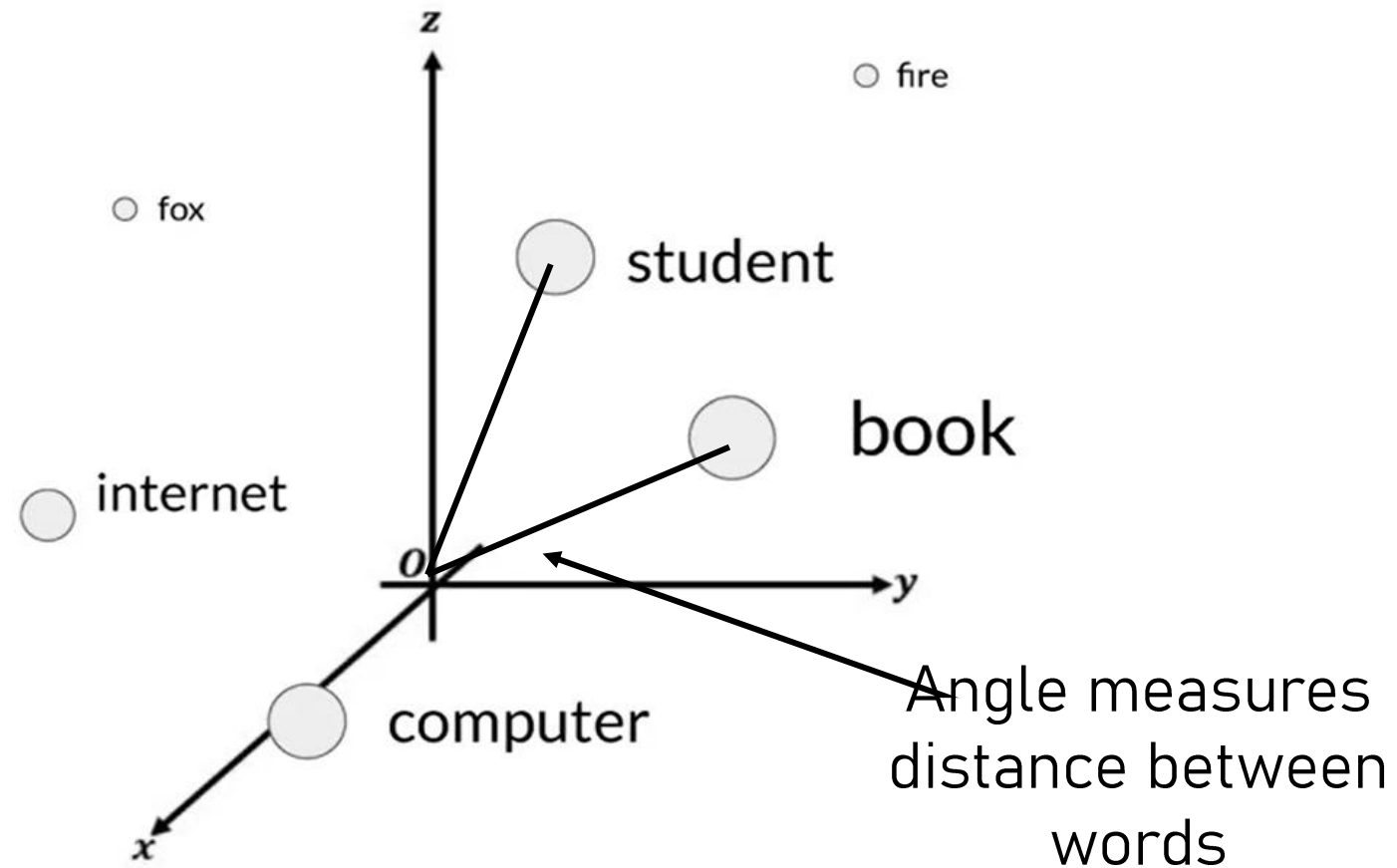
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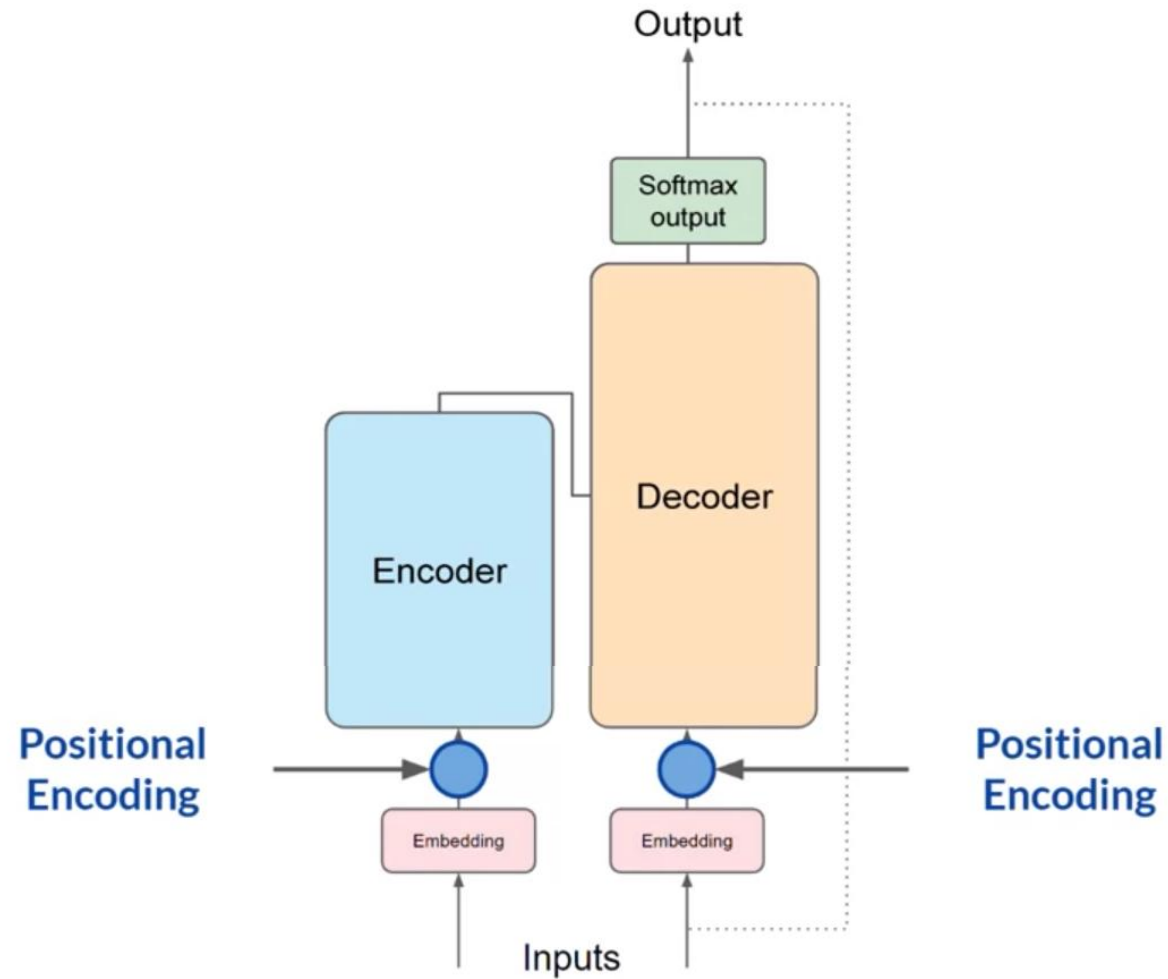
Transformers Architecture



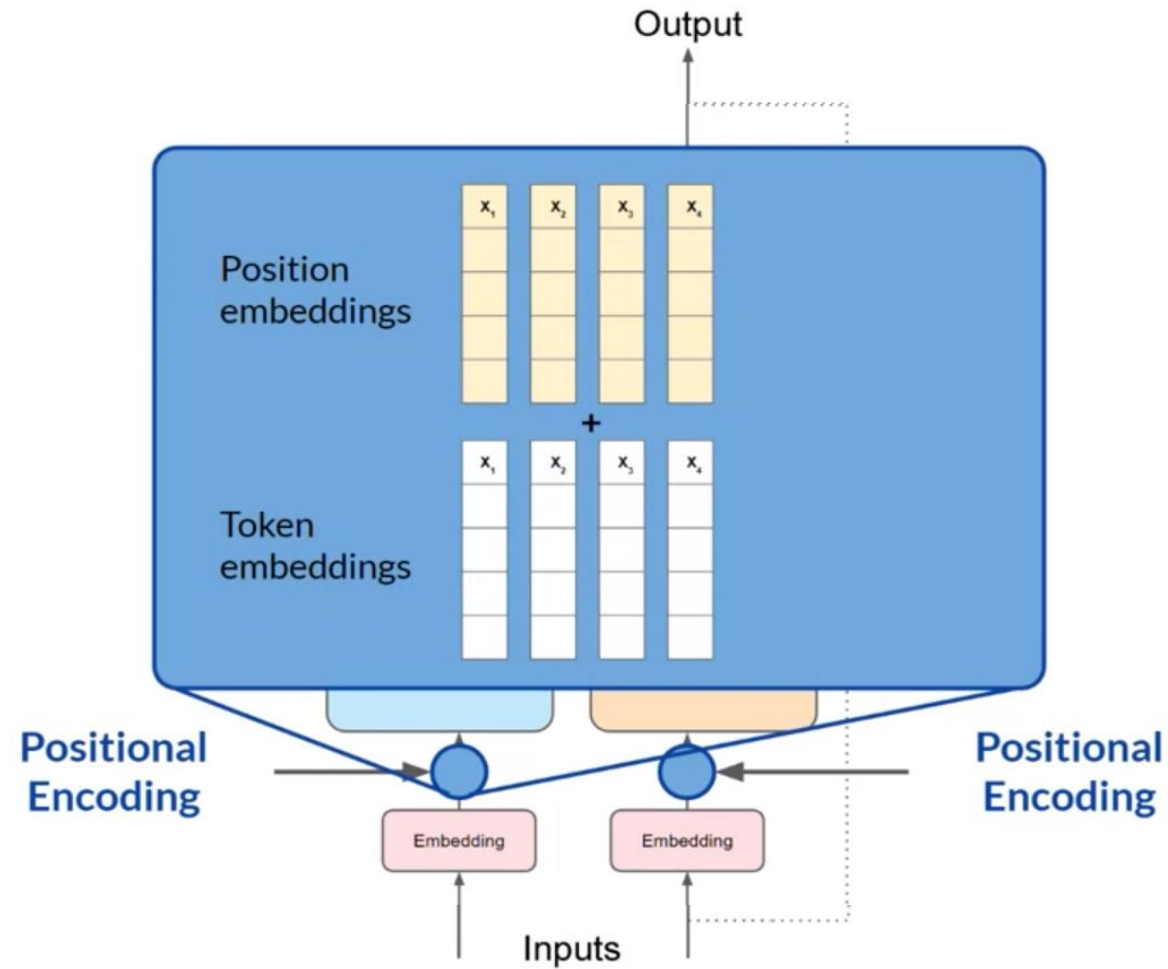
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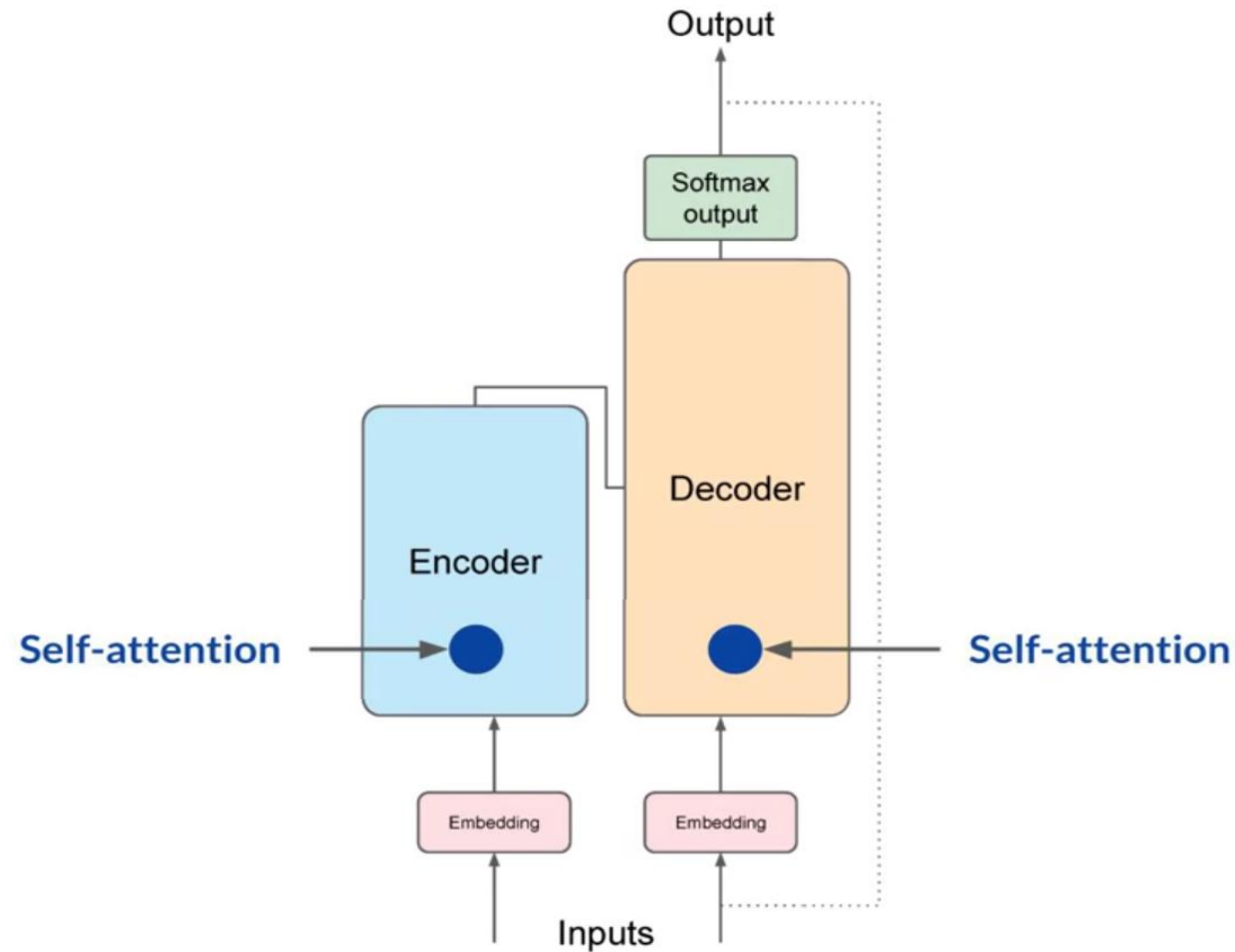
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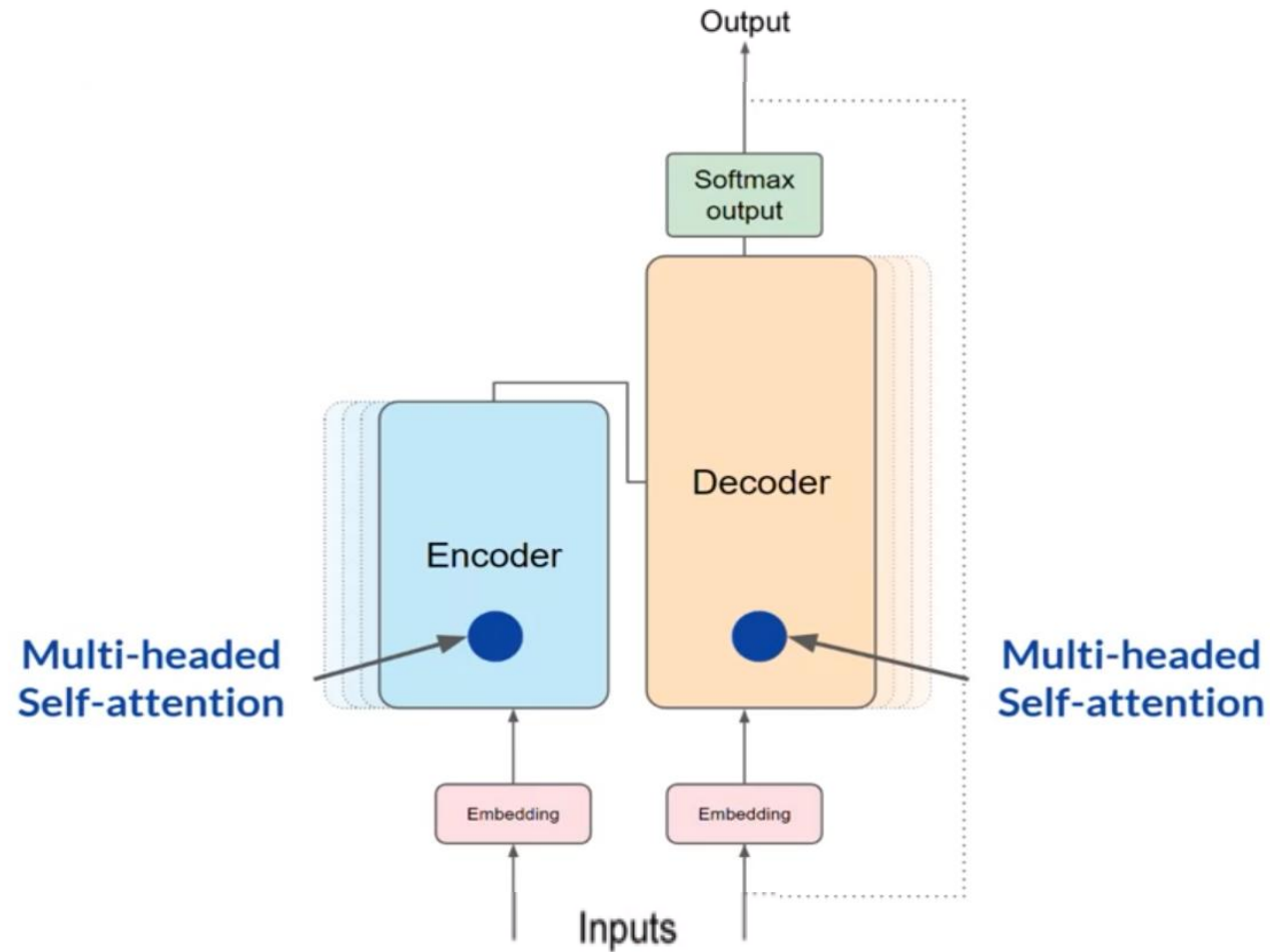
Transformers Architecture



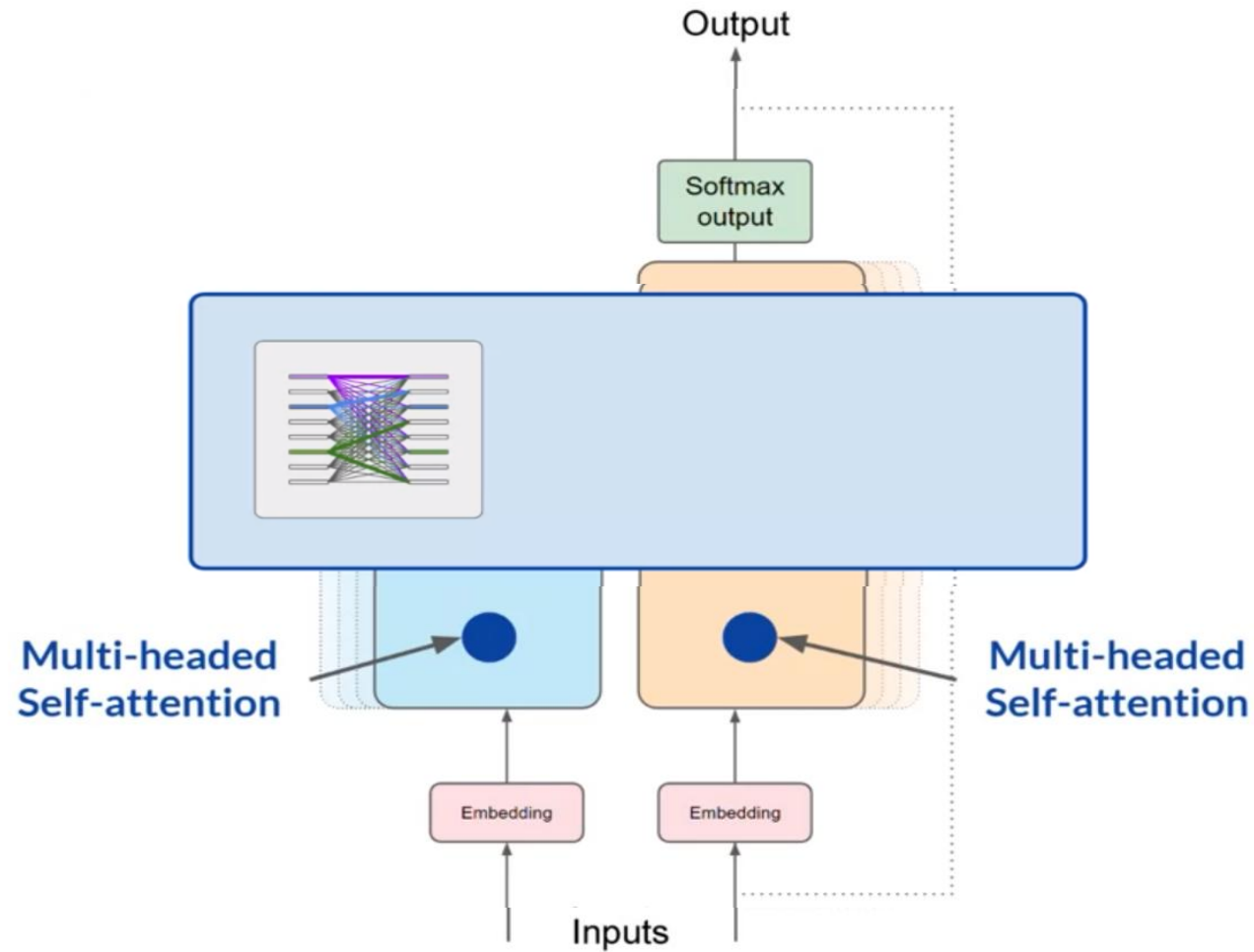
Transformers Architecture



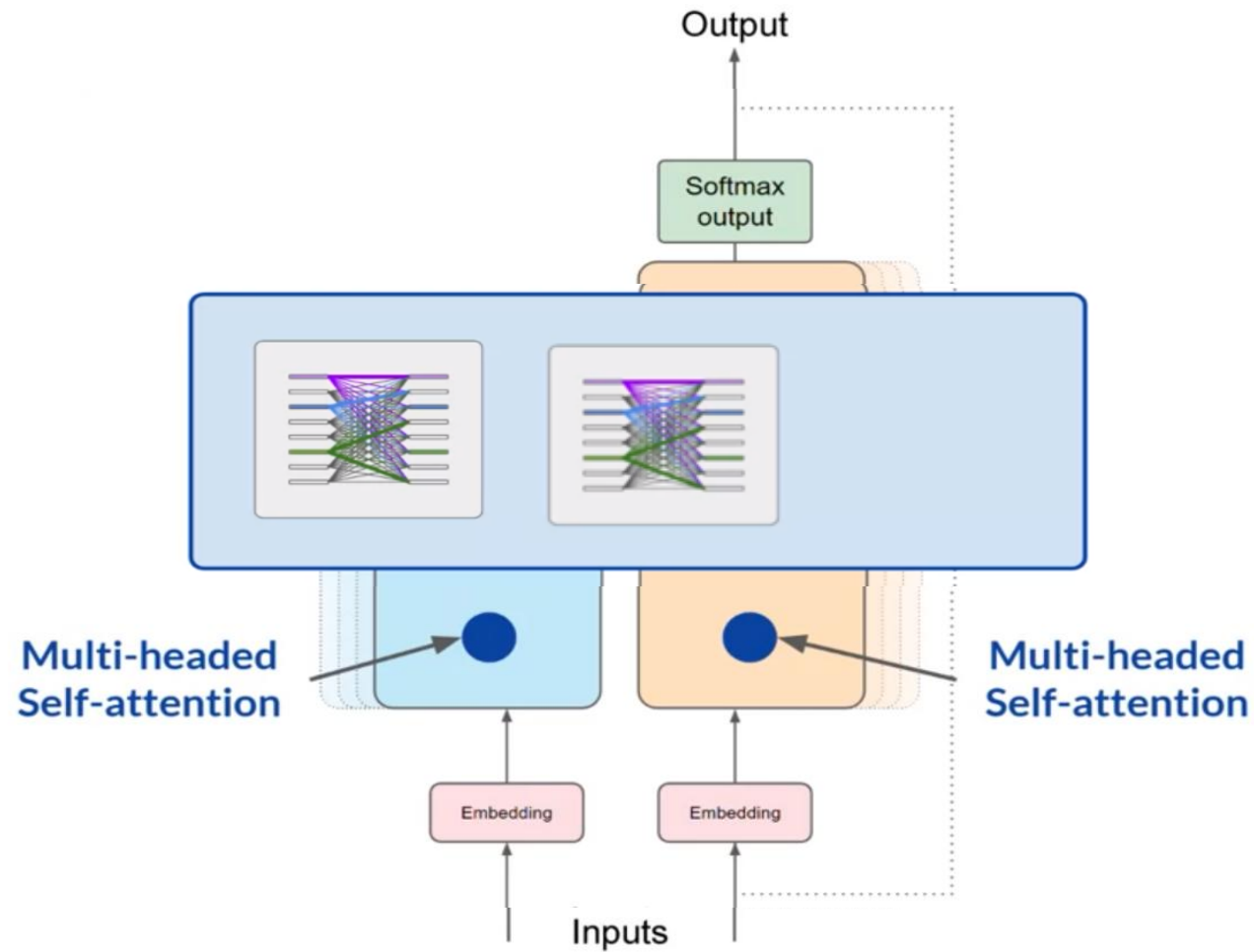
Transformers Architecture



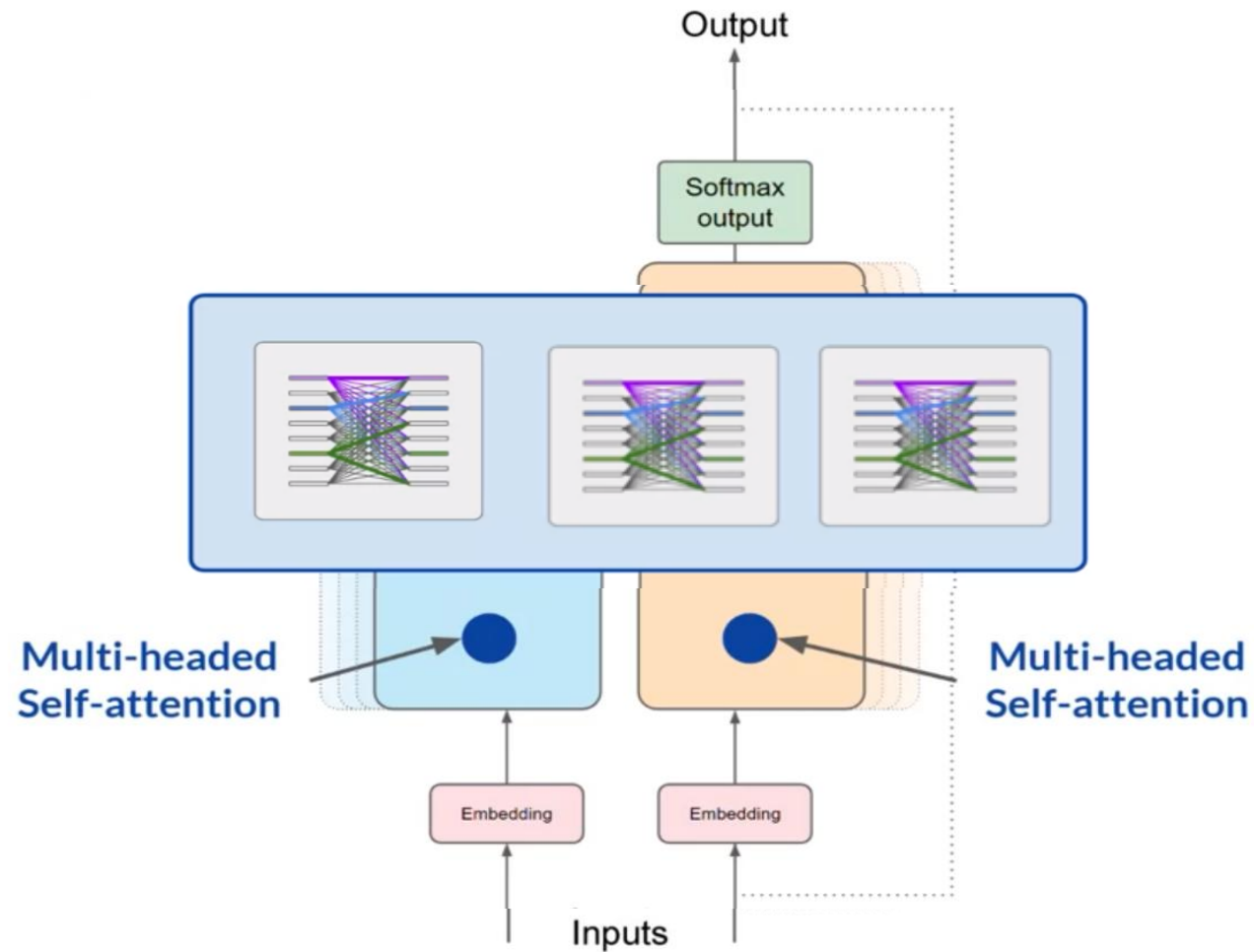
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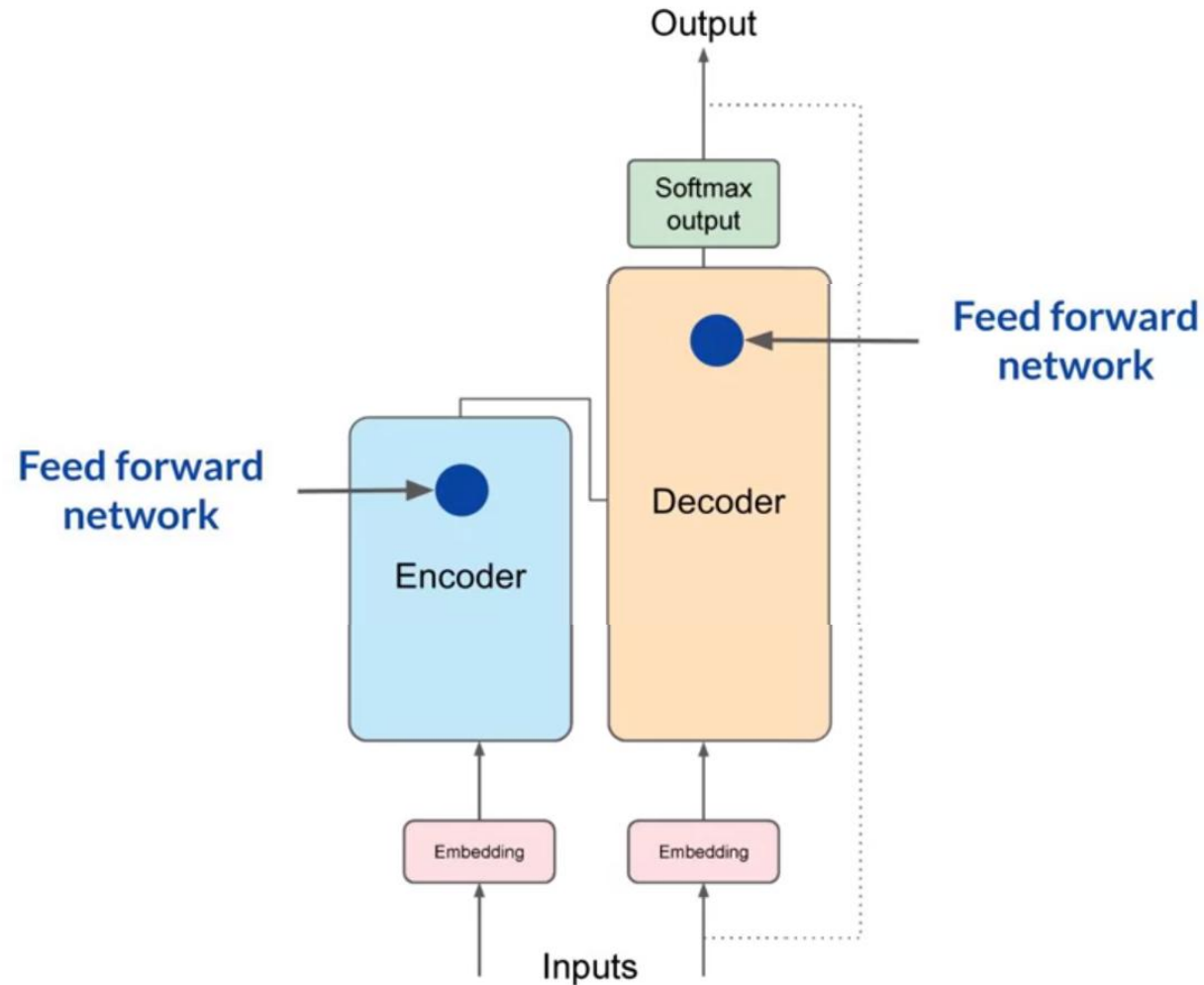
Transformers Architecture



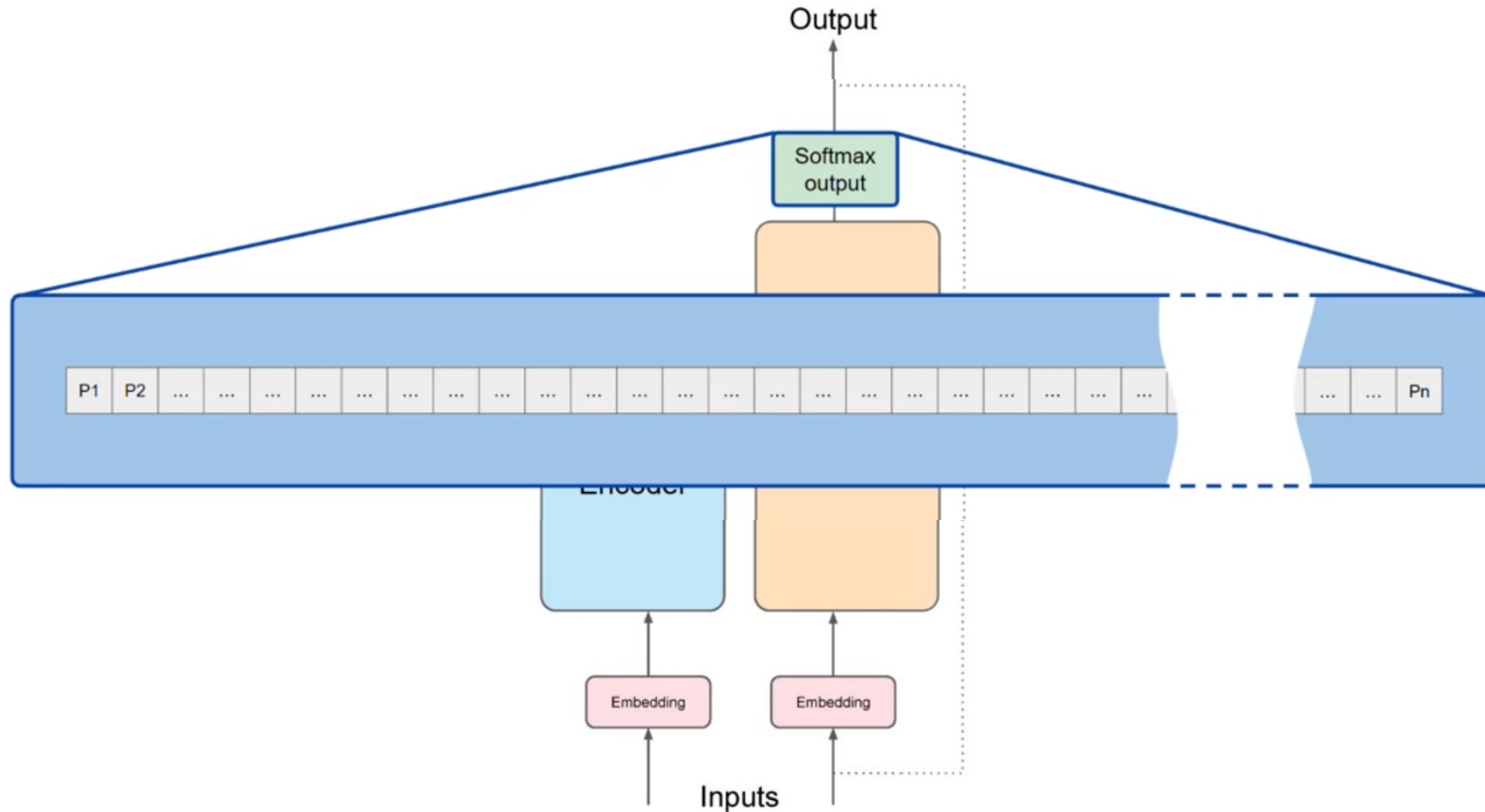
Transformers Architecture



Transformers Architecture



Transformers Architecture





Vector stores

- One of the most common ways to store and search over unstructured data is to embed it and store the resulting embedding vectors, and then at query time to embed the unstructured query and retrieve the embedding vectors that are 'most similar' to the embedded query.
- A vector store takes care of storing embedded data and performing vector search for you.

Vector Stores

1. Load Source Data



Load, Transform, Embed

Vector Store



2. Query Vector Store

Embed

5.5, -0.3...
2.1, 0.1

XXXXXXXXXXXXXX
XXXXXXXXXXXXXX

XXXXXXXXXXXXXX
XXXXXXXXXXXXXX

3. Retrieve 'most similar'

Embedding



Context:

Revenue of Apple

Apple

Embedding



Context:

Revenue of Apple

Parameters

related_to_phones

Is_location

has_stock

revenue

is_fruit

calories

Apple

Embedding



Context:
Revenue of Apple

Parameters

related_to_phones

1

Is_location

0

has_stock

1

revenue

82

is_fruit

0

calories

0

Embedding



Context:
Revenue of Apple

Parameters

related_to_phones

1

Is_location

0

has_stock

1

revenue

82

is_fruit

0

calories

0

$$\begin{bmatrix} 1 \\ 0 \\ 1 \\ 82 \\ 0 \\ 0 \end{bmatrix}$$

Embedding



Context:
Calories in Apple

Parameters

related_to_phones

0

Is_location

0

has_stock

0

revenue

0

is_fruit

1

calories

95

[0
0
0
0
1
95]

Embedding



Parameters	
related_to_phones	
Is_location	
has_stock	
revenue	
is_fruit	
calories	

Context:
Revenue of Apple

Apple

1
0
1
82
0
0

Context:
Calories in Apple

Apple

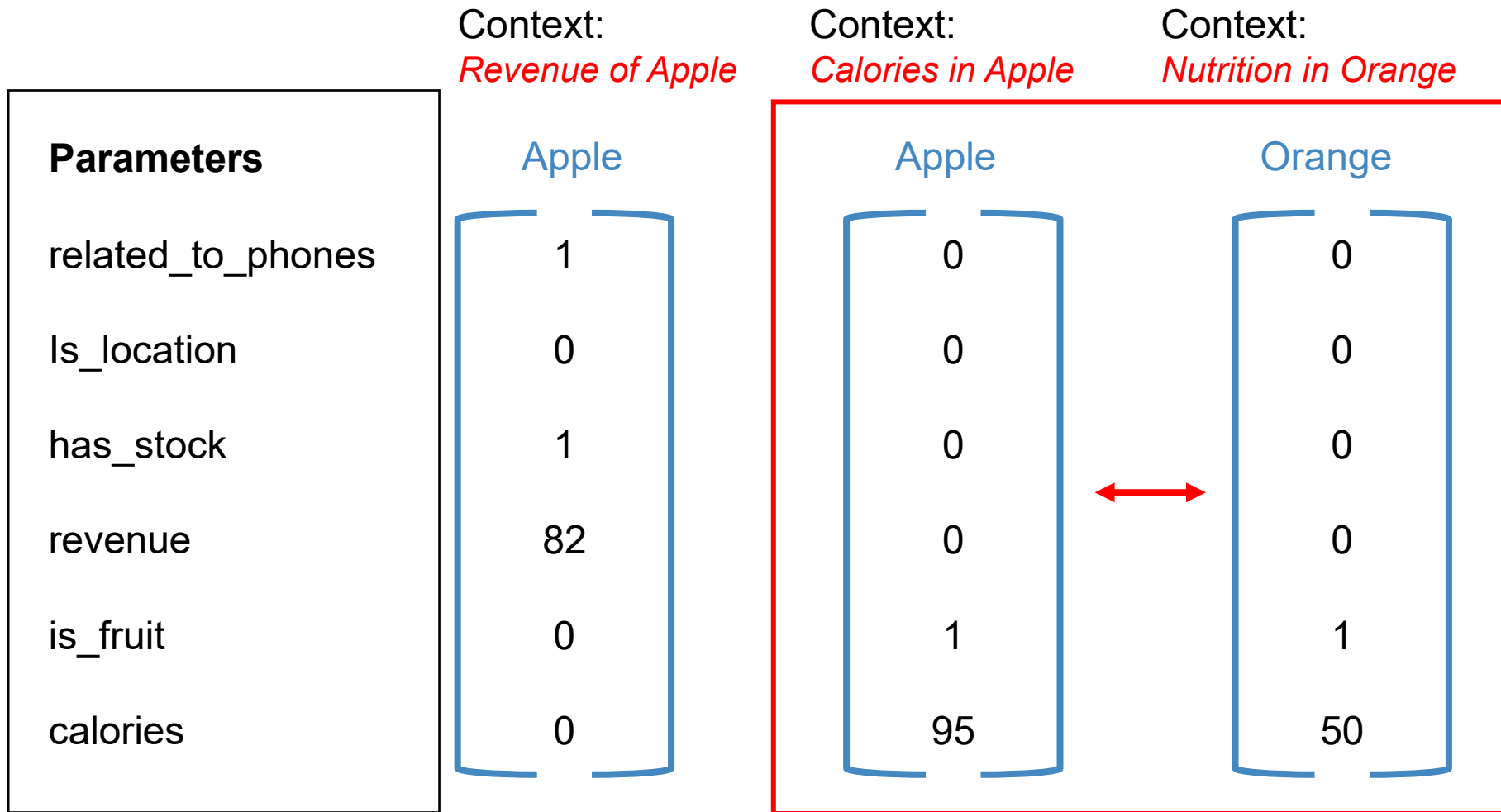
0
0
0
0
1
95

Context:
Nutrition in Orange

Orange

0
0
0
0
1
50

Embedding



Embedding



	Context: <i>Revenue of Apple</i>	Context: <i>Calories in Apple</i>	Context: <i>Nutrition in Orange</i>	Context: <i>Employees in Samsung</i>
Parameters	Apple	Apple	Orange	Samsung
related_to_phones	1	0	0	1
Is_location	0	0	0	0
has_stock	1	0	0	1
revenue	82	0	0	76
is_fruit	0	1	1	0
calories	0	95	50	0

Embedding



Context:
Revenue of Apple

Context:
Calories in Apple

Context:
Nutrition in Orange

Context:
Employees in Samsung

Parameters

related_to_phones

Is_location

has_stock

revenue

is_fruit

calories

Apple

1

0

1

82

0

0

Apple

0

0

0

0

1

95

Orange

0

0

0

0

1

50

Samsung

1

0

1

76

0

0

Word embedding techniques



CBOW, Skip gram

Word2vec

GloVe

fastText

Based on transformer
architecture

BERT

GPT

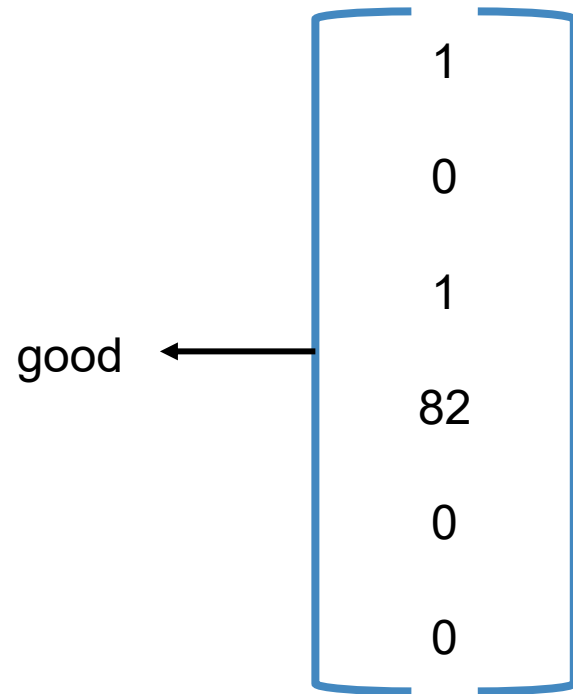
Based on LSTM

ELMo

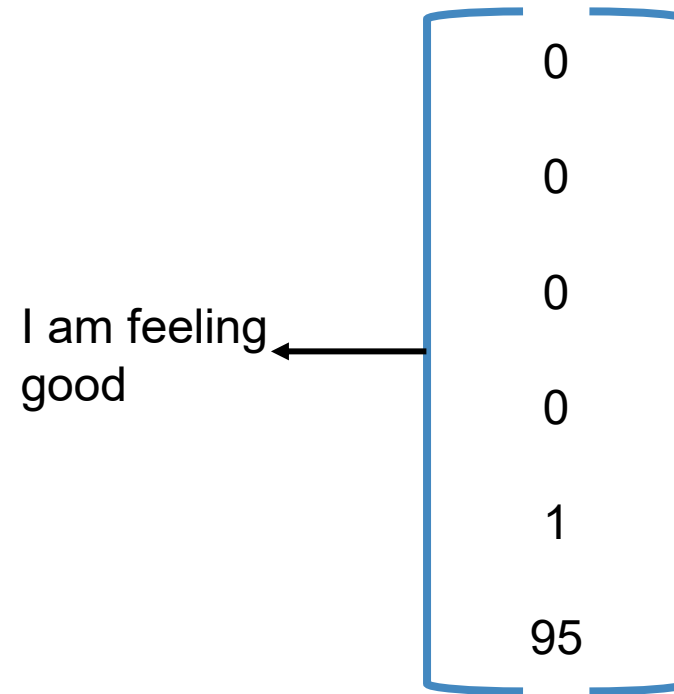
Embedding techniques



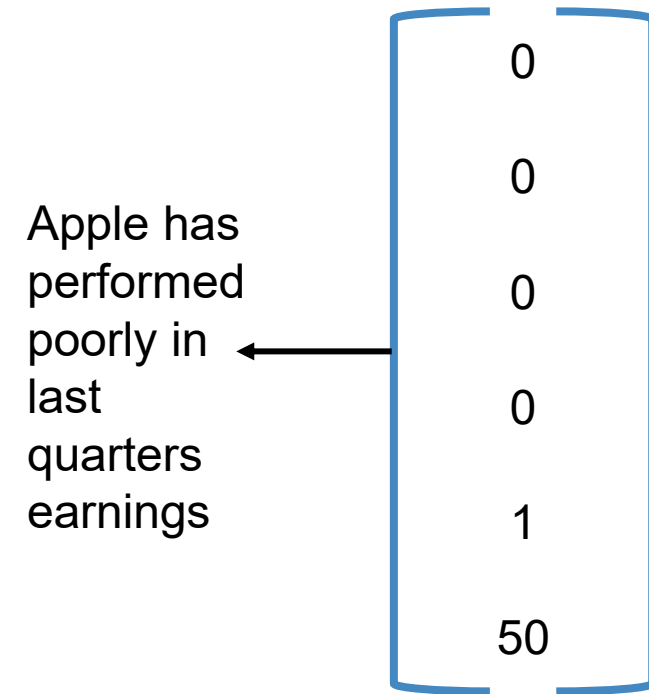
Word Embedding



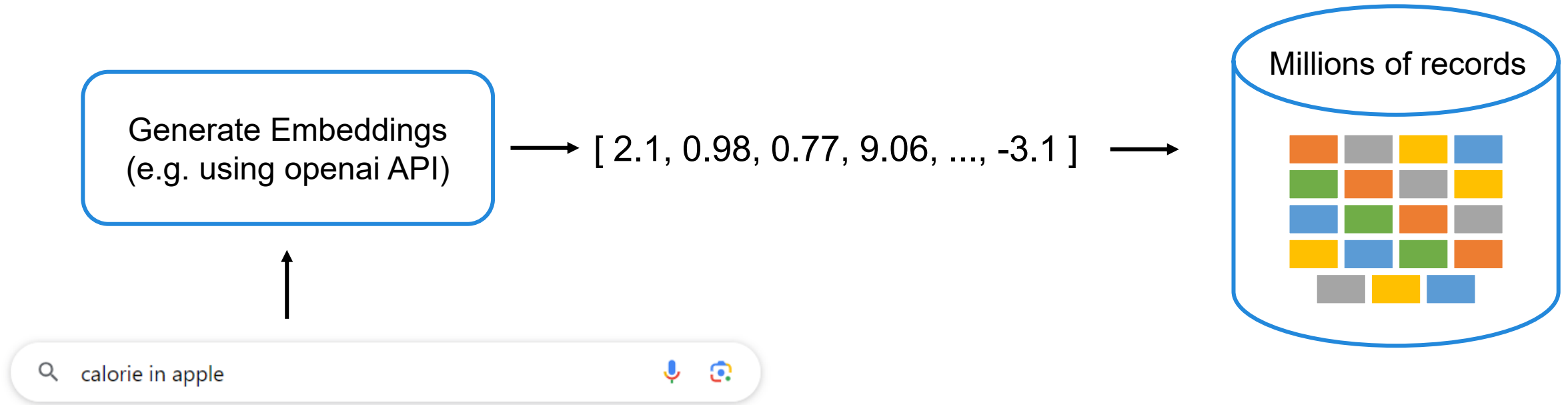
Sentence Embedding



Doc Embedding



Traditional Database



Traditional Database



Query Vector

[2.1, 0.98, 0.77, 9.06,...,-3.1]

[2.0, 0.96, 0.67, 9-4, .., -3-7]

[3.45, 6.29, 0.53, 8.47 ..., 4.86]

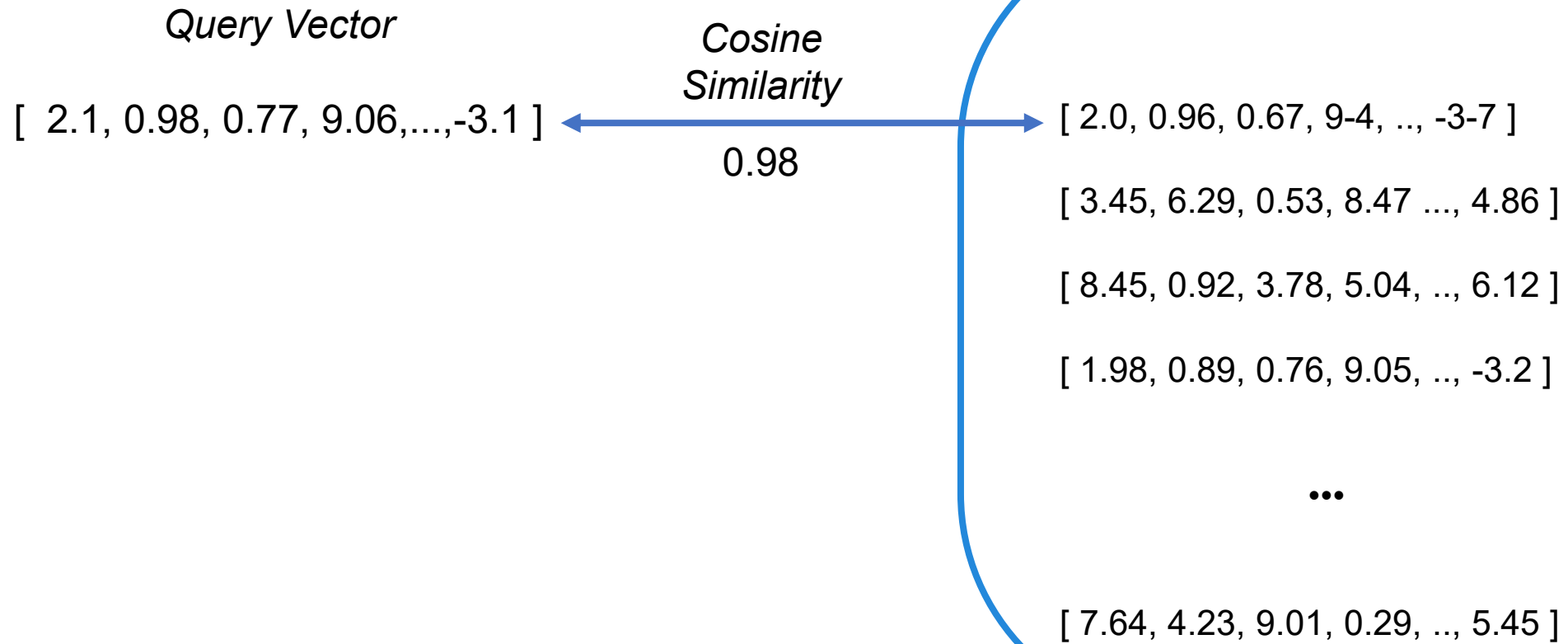
[8.45, 0.92, 3.78, 5.04, .., 6.12]

[1.98, 0.89, 0.76, 9.05, .., -3.2]

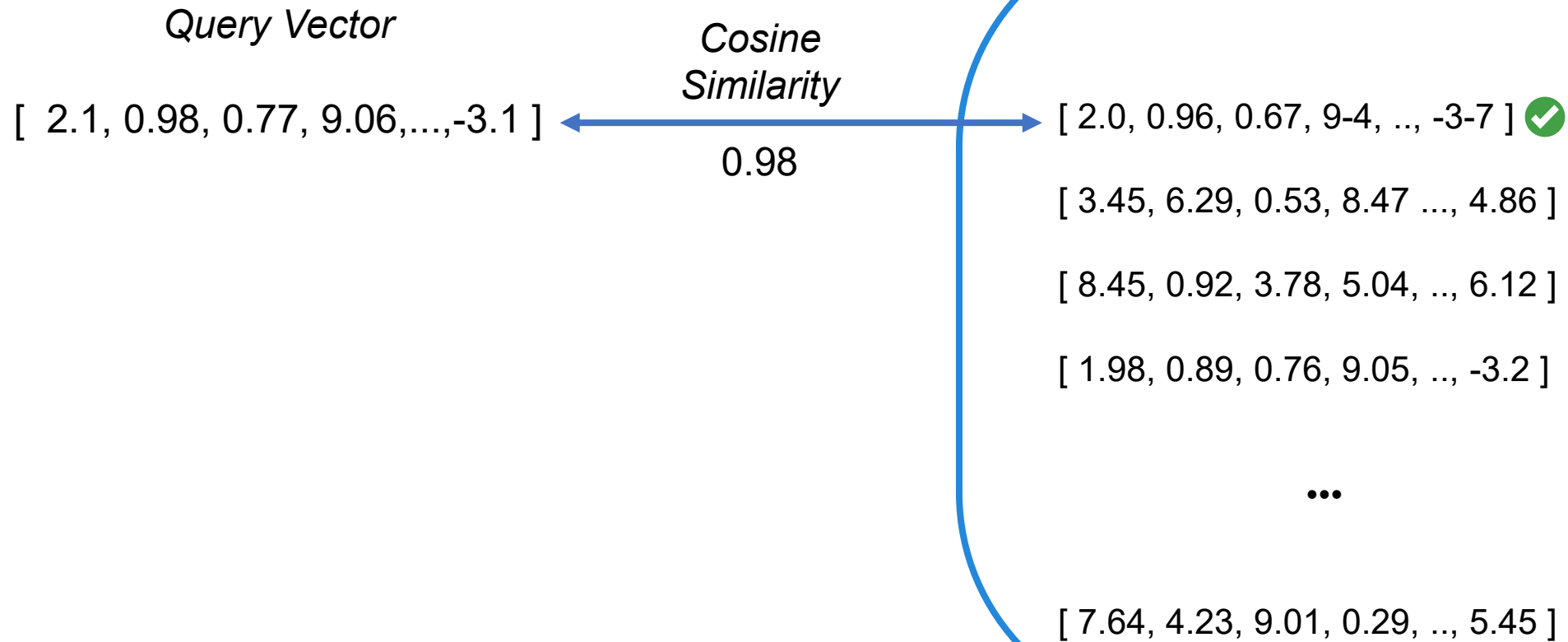
...

[7.64, 4.23, 9.01, 0.29, .., 5.45]

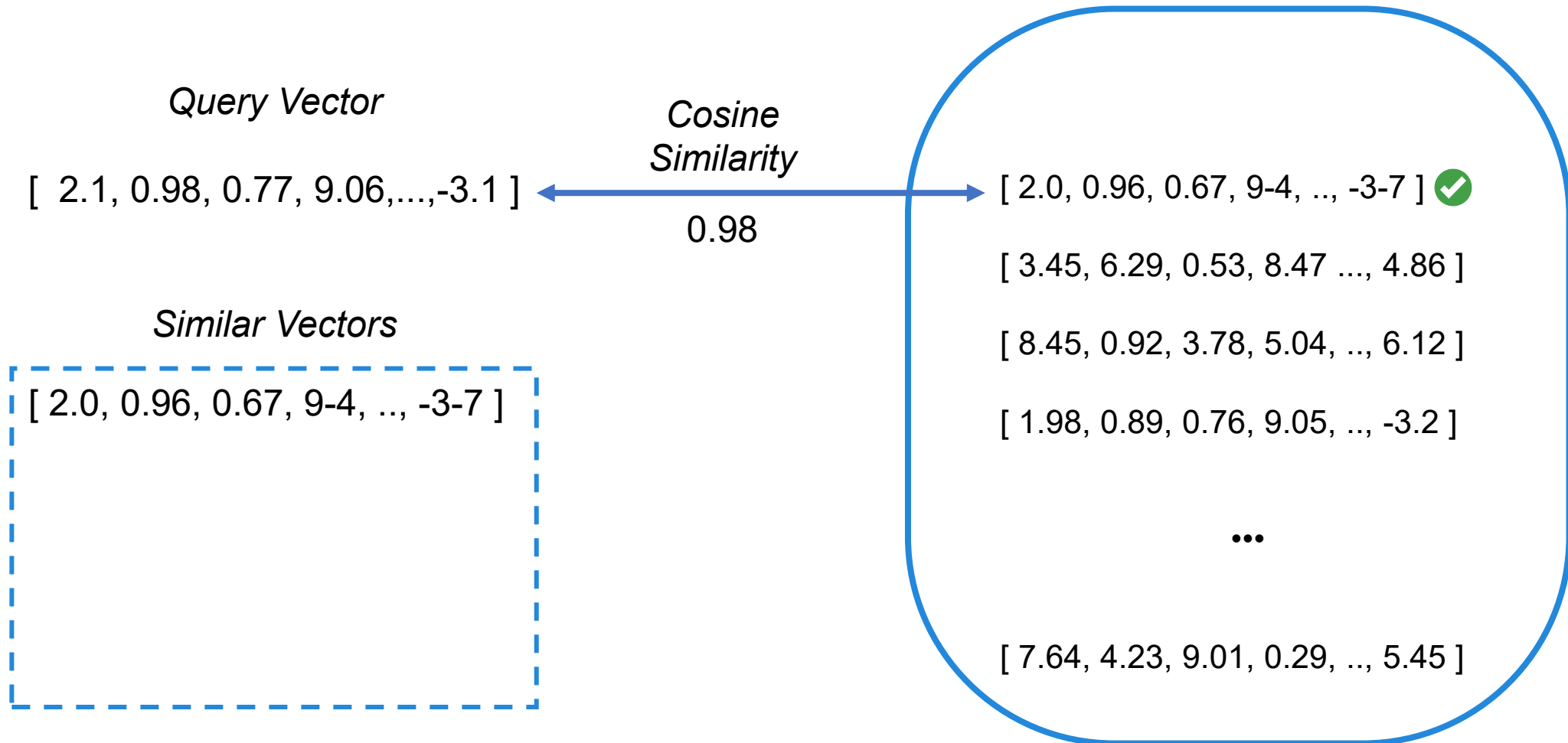
Traditional Database



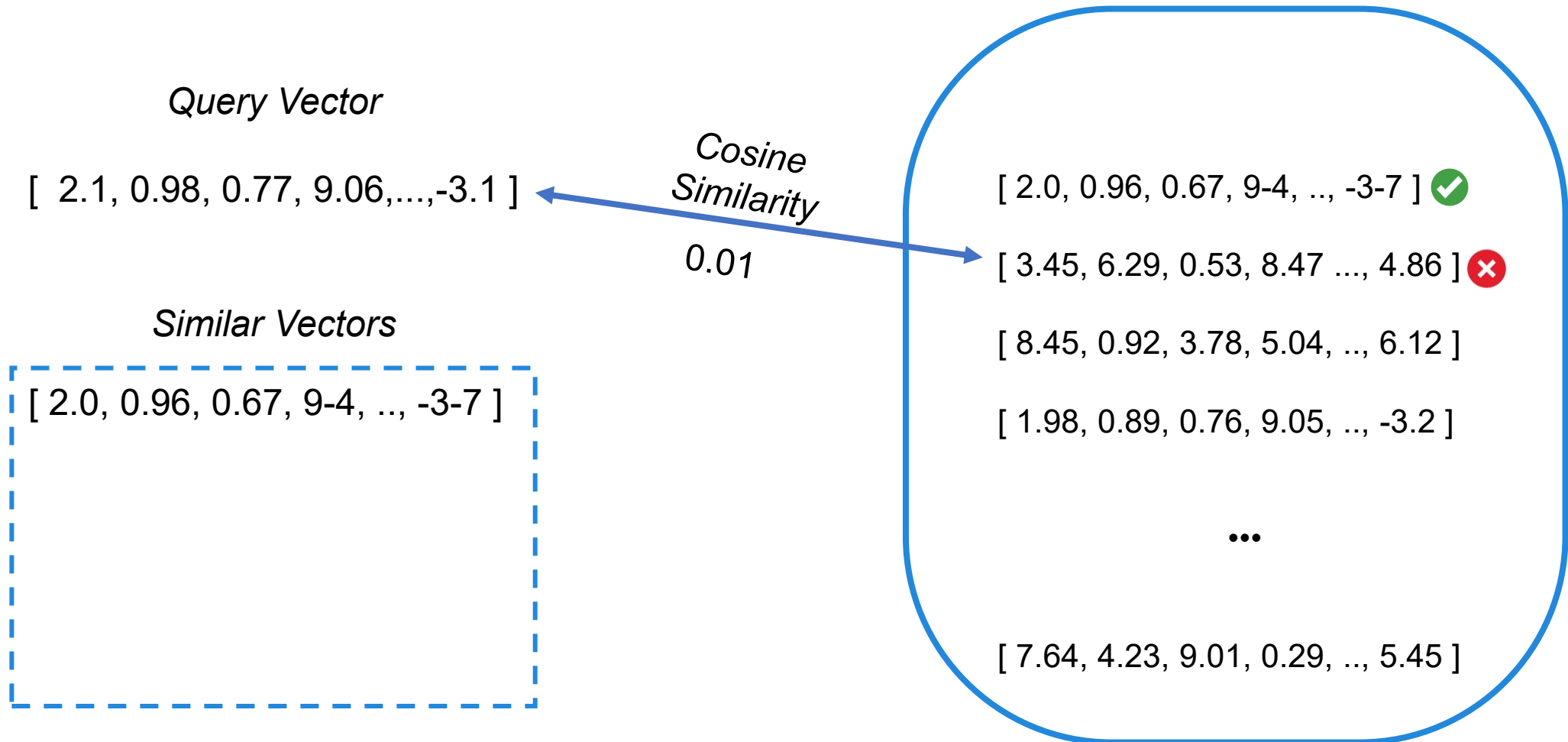
Traditional Database



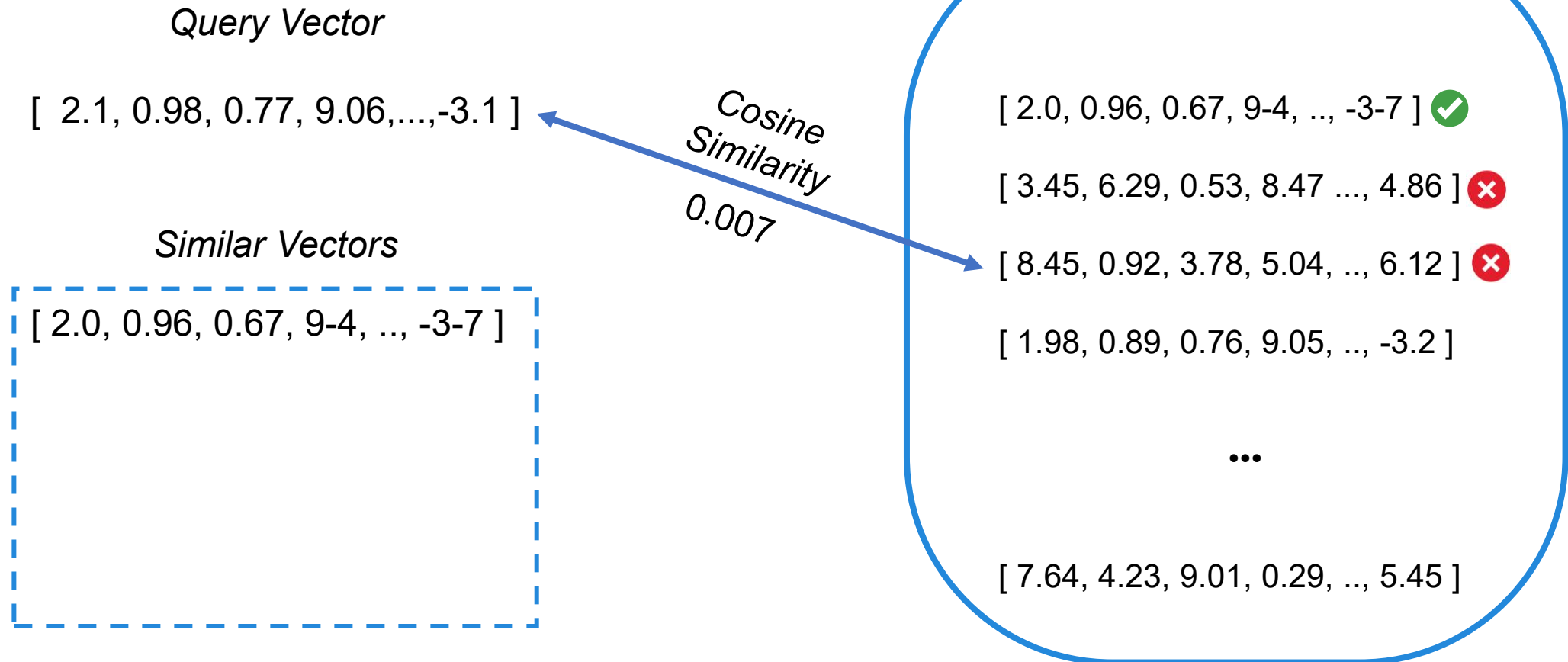
Traditional Database



Traditional Database



Traditional Database



Traditional Database



Query Vector

[2.1, 0.98, 0.77, 9.06,...,-3.1]

Similar Vectors

[2.0, 0.96, 0.67, 9-4, ..., -3-7]

[1.98, 0.89, 0.76, 9.05, ..., -3.2]

*Cosine
Similarity
0.92*

[2.0, 0.96, 0.67, 9-4, ..., -3-7] ✓

[3.45, 6.29, 0.53, 8.47 ..., 4.86] ✗

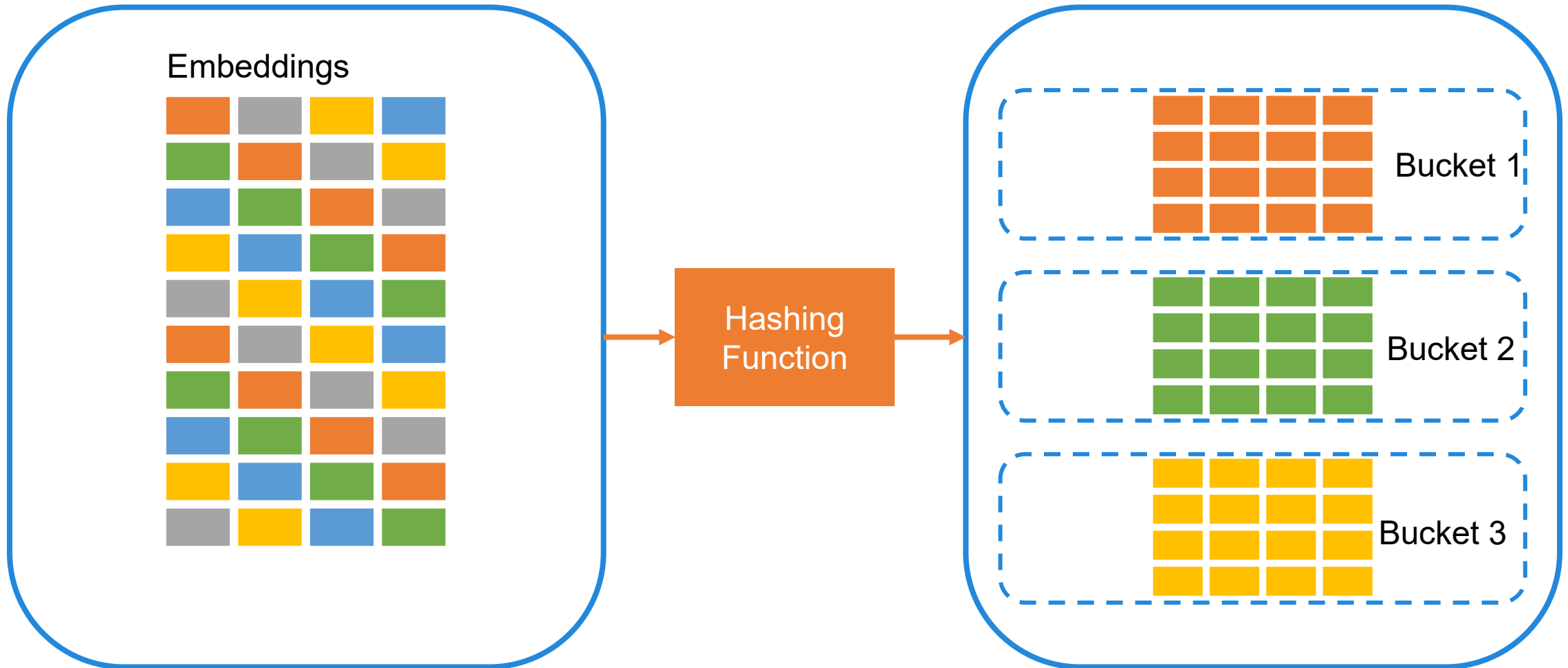
[8.45, 0.92, 3.78, 5.04, ..., 6.12] ✗

[1.98, 0.89, 0.76, 9.05, ..., -3.2] ✓

...

[7.64, 4.23, 9.01, 0.29, ..., 5.45]

Traditional Database



Output of these Architectures are Called as Models

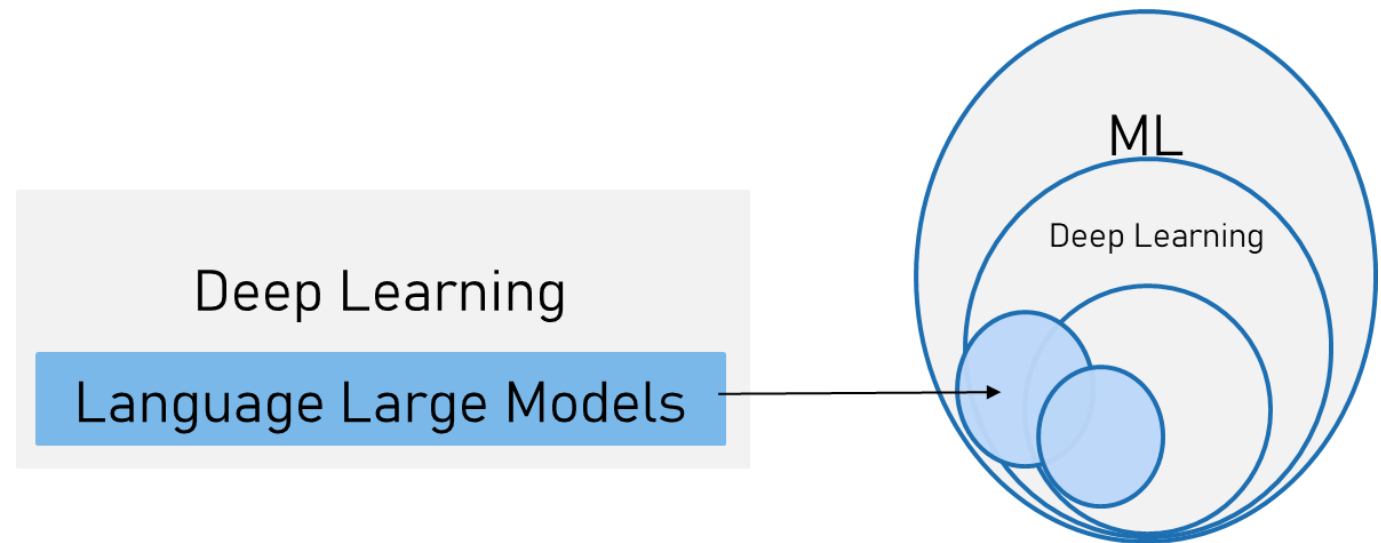
Small Language Model

Large Language Model

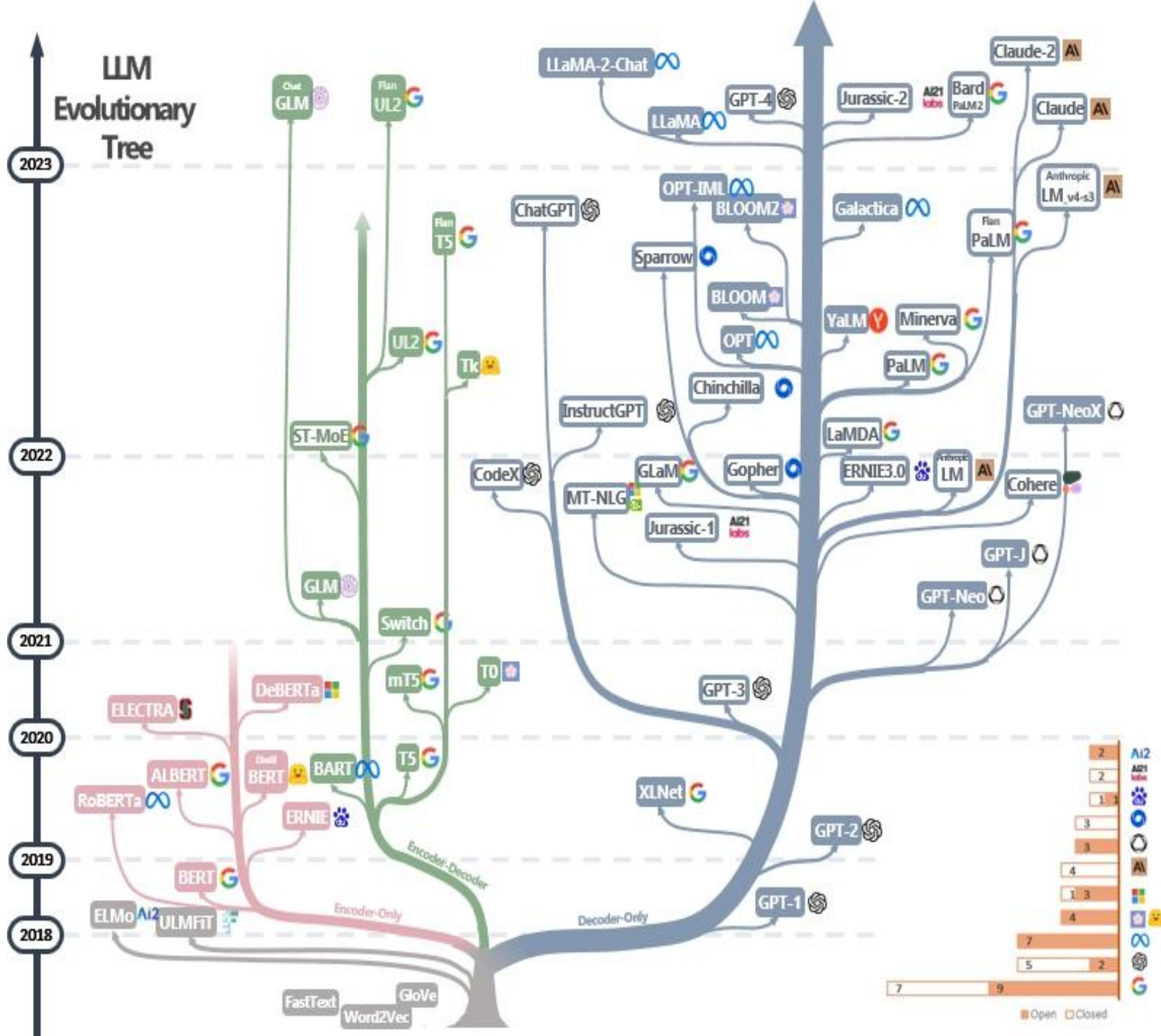
Large Language Models



- Large language models are advanced AI systems with **billions of parameters** trained on **extensive datasets**, capable of **understanding and generating human-like** text across various languages and applications.



LLM Evolution



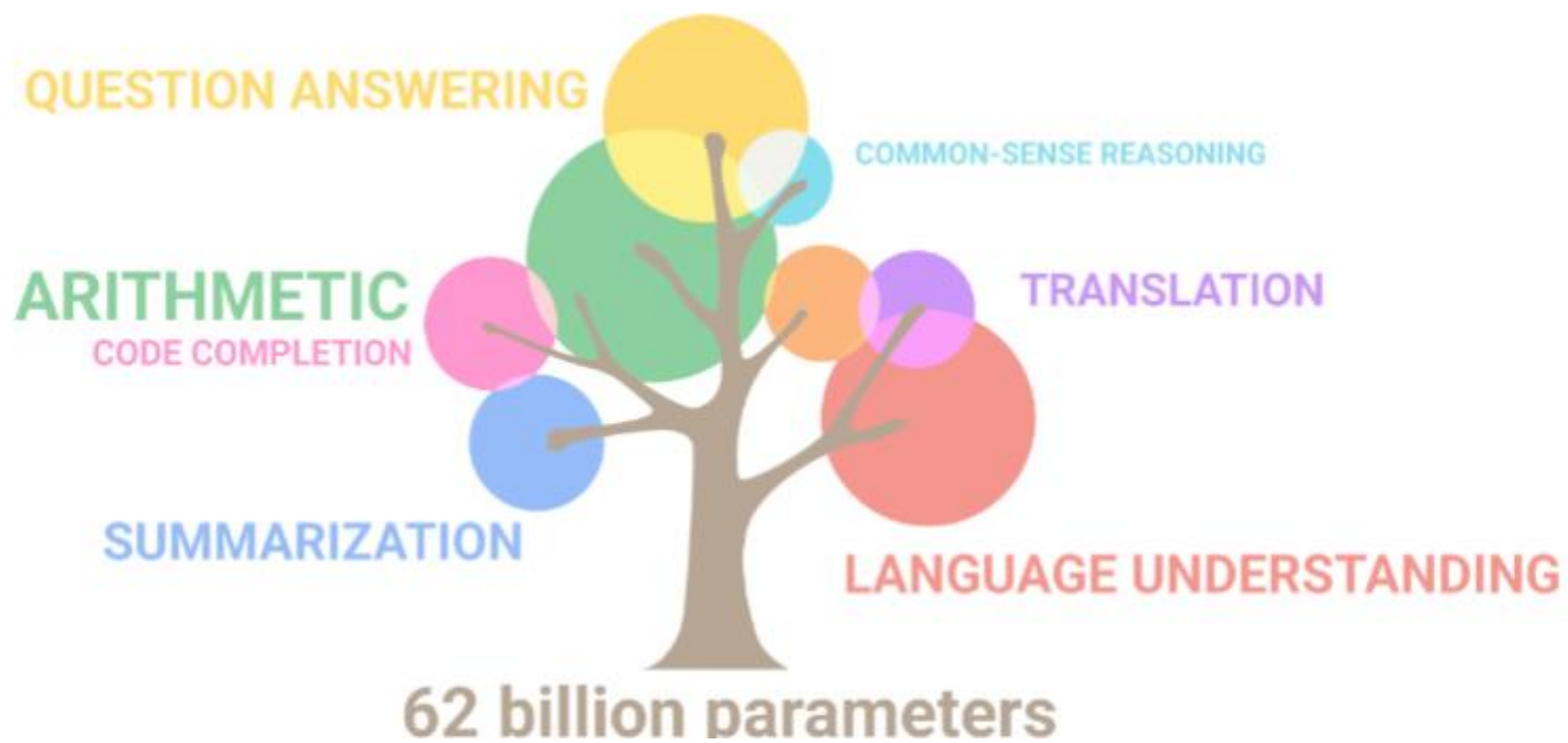
QUESTION ANSWERING

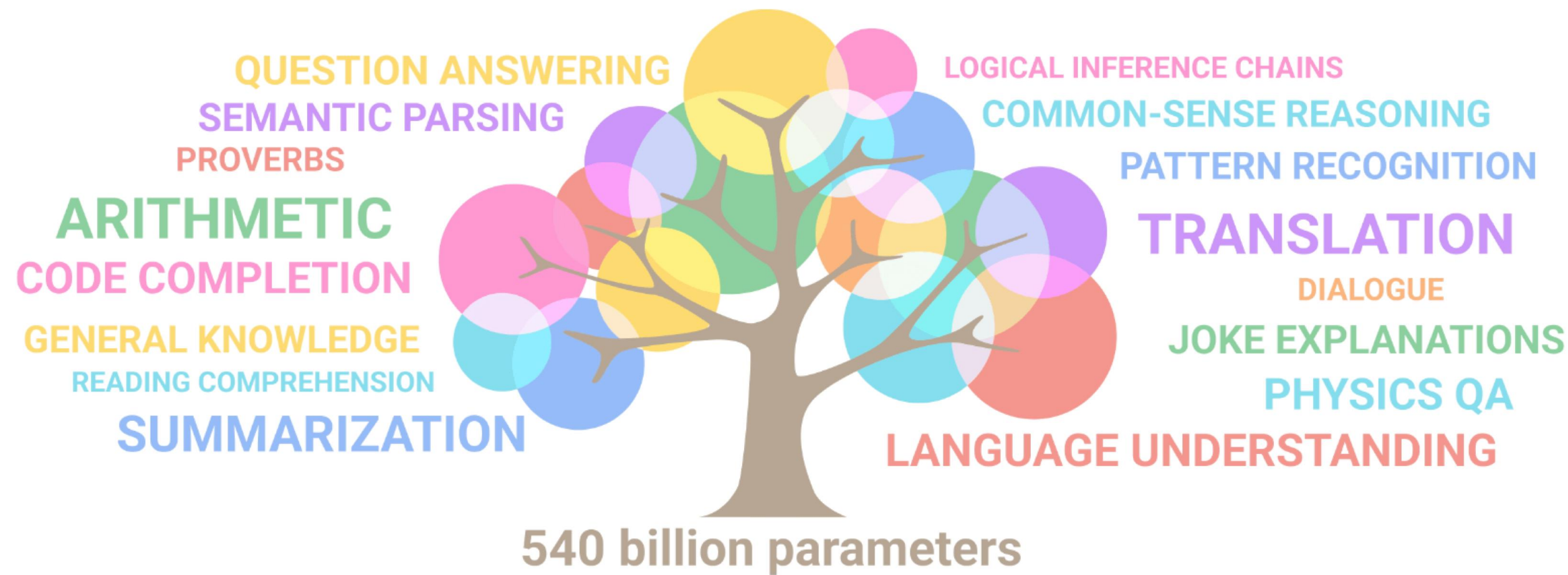
ARITHMETIC

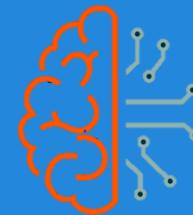


LANGUAGE UNDERSTANDING

8 billion parameters

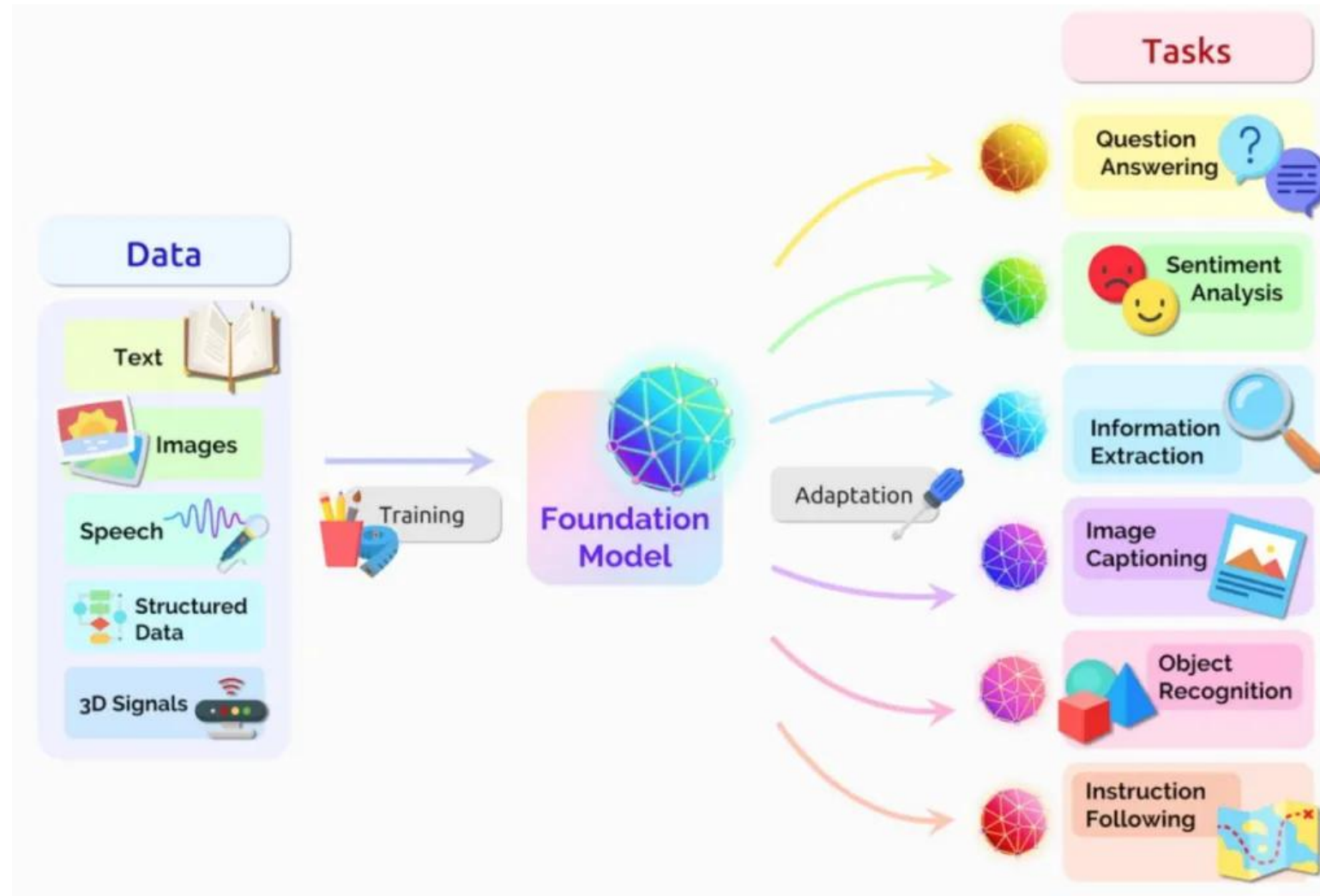






- LLM is a type of AI algorithm that uses deep learning techniques and massively large data sets to understand, summarize, generate and predict new content
- LLM are commonly used in NLP applications where a user inputs a query in natural language to generate a result.
- Modern LLMs emerged in 2017, used transformer models Some LLMs are referred to as foundation models
- Different architecture in ML history

What is Foundational Models

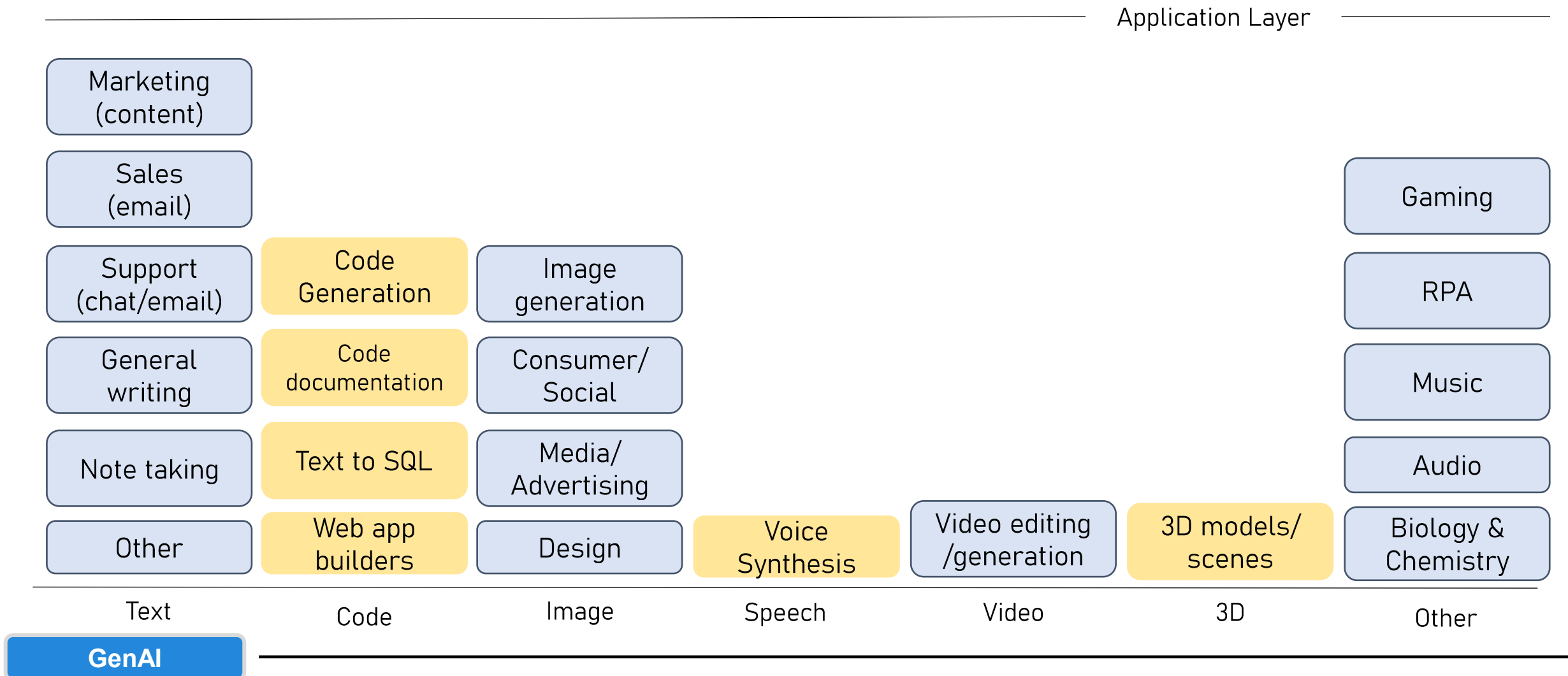


Example Foundation Models

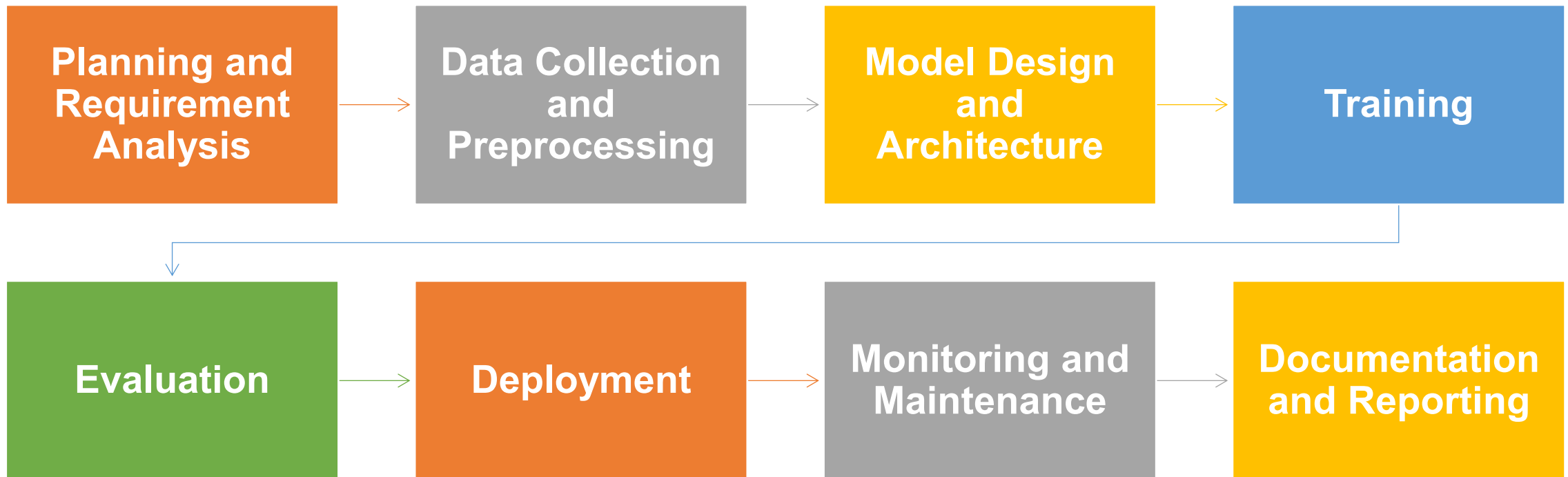


Model	Description
GPT (Generative Pre-trained Transformer)	Autoregressive model by OpenAI for text generation, summarization, and conversation. Forms the basis of ChatGPT.
BERT (Bidirectional Encoder Representations from Transformers)	Google's encoder-only model focused on deep language understanding for tasks like classification and QA.
T5 (Text-to-Text Transfer Transformer)	Google's model that frames all NLP tasks as a text to text problem flexible and powerful for both understanding and generation.
RoBERTa (Robustly Optimized BERT Pretraining Approach)	Facebook's optimized version of BERT with better performance due to improved training strategies and data.
Claude	Autoregressive model by Anthropic focused on safety, alignment, and highquality dialogue.
LLaMA (Large Language Model Meta AI)	Meta's open-source LLM series designed for research and customization popular in fine-tuning and local deployment.
Jurassic	AI21 Labs' large-scale language model available via API optimized for creative generation and enterprise use.
Cohere	Transformer-based model built for enterprise NLP offers strong multilingual support, embeddings, and retrieval-augmented generation.

The generative AI Application Landscape



LLM Development Lifecycle





To develop a chatbot that can assist hotel guests with common inquiries, such as booking information, amenities, check-in/check-out times, and local recommendations.

Why LLM models are big?



Llama 13B Model Card

- CCNet [67%]
- C4 [15%]
- GitHub [4.5%]
- Wikipedia [4.5%]
- Books [4.5%]
- ArXiv [2.5%]
- Stack
- Exchange[2%]
- 20 languages
- 2 Trillion Tokens



- Open source since 2007, available till June 2023
- 3 - 5 billion new pages added every month
- Every LLM training this corpus
- It contains 3.1 billion web pages or 370 TiB
- Page captures are from 44 million hosts or 35 million registered domains
- 82% raw tokens used in GPT-3

Foundation Model of GPT-4o



GPT-4o 'TRAINING' COST:
≈ 100+ MILLION USD

Includes: Supercomputing Clusters (GPUS), Petabytes of Data, Human Annotators, Algorithmic Development, Power Consumption.



CHANDAYAN MISSION COST:
≈ 75 MILLION USD

Includes: Rocket, Lander, Rover, Scientific Instruments, Launch, Ground Support, 4-day Journey.

Large language models by parameter count



Model	Organization	Parameter Count (approx)
Bharat Gen	IIT Bombay (IndiaAI Mission)	1 trillion
GPT-5	OpenAI	1~2 trillion
GPT-3	OpenAI	175 billion
PaLM	Google	540 billion
BLOOM	BigScience / Hugging Face	176 billion
LLaMA (max in original release)	Meta	65 billion
Wu Dao 2.0	BAAI (Beijing Academy)	1.75 trillion

Types of LLM Models



API

- No or limited visibility into the code and data, but easier to deploy



Open Source

- Offer the most transparency and flexibility, but require time and expertise to setup

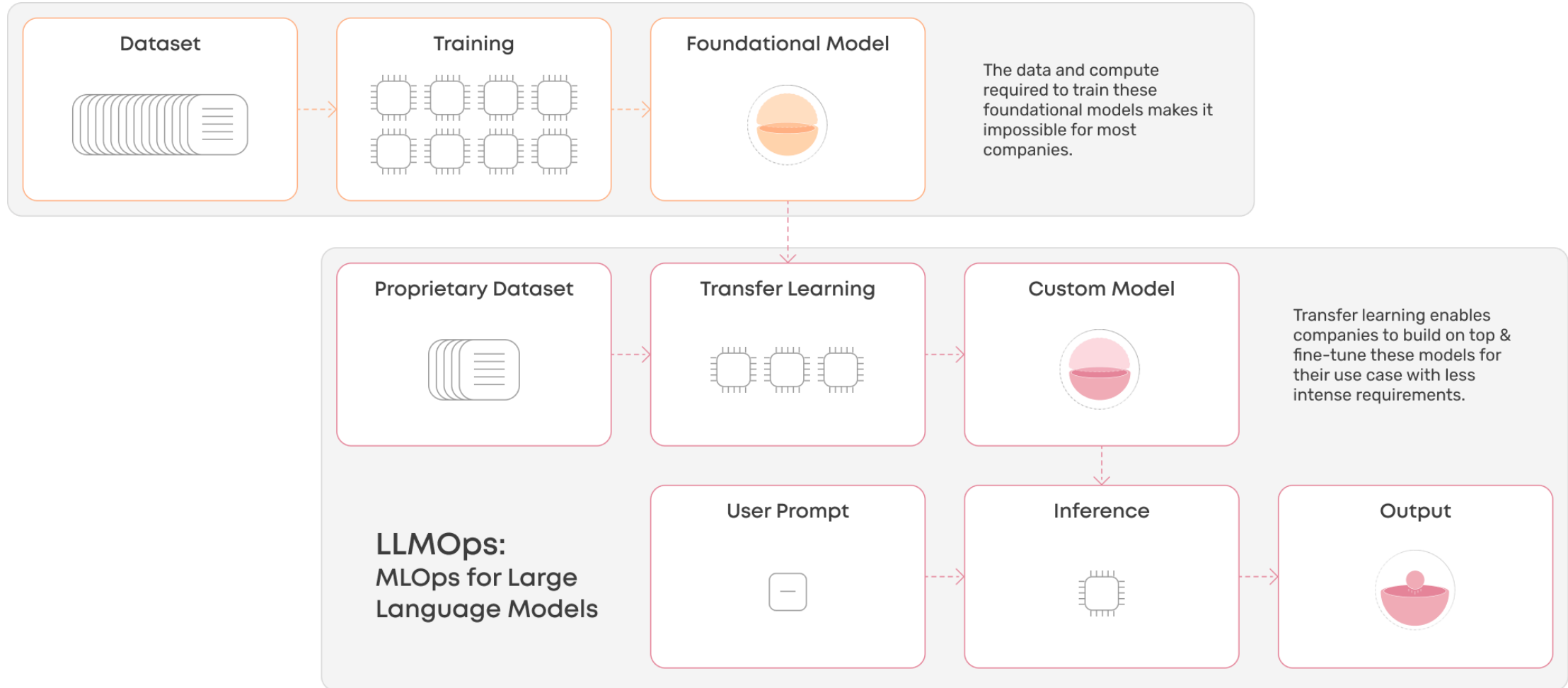


Proprietary

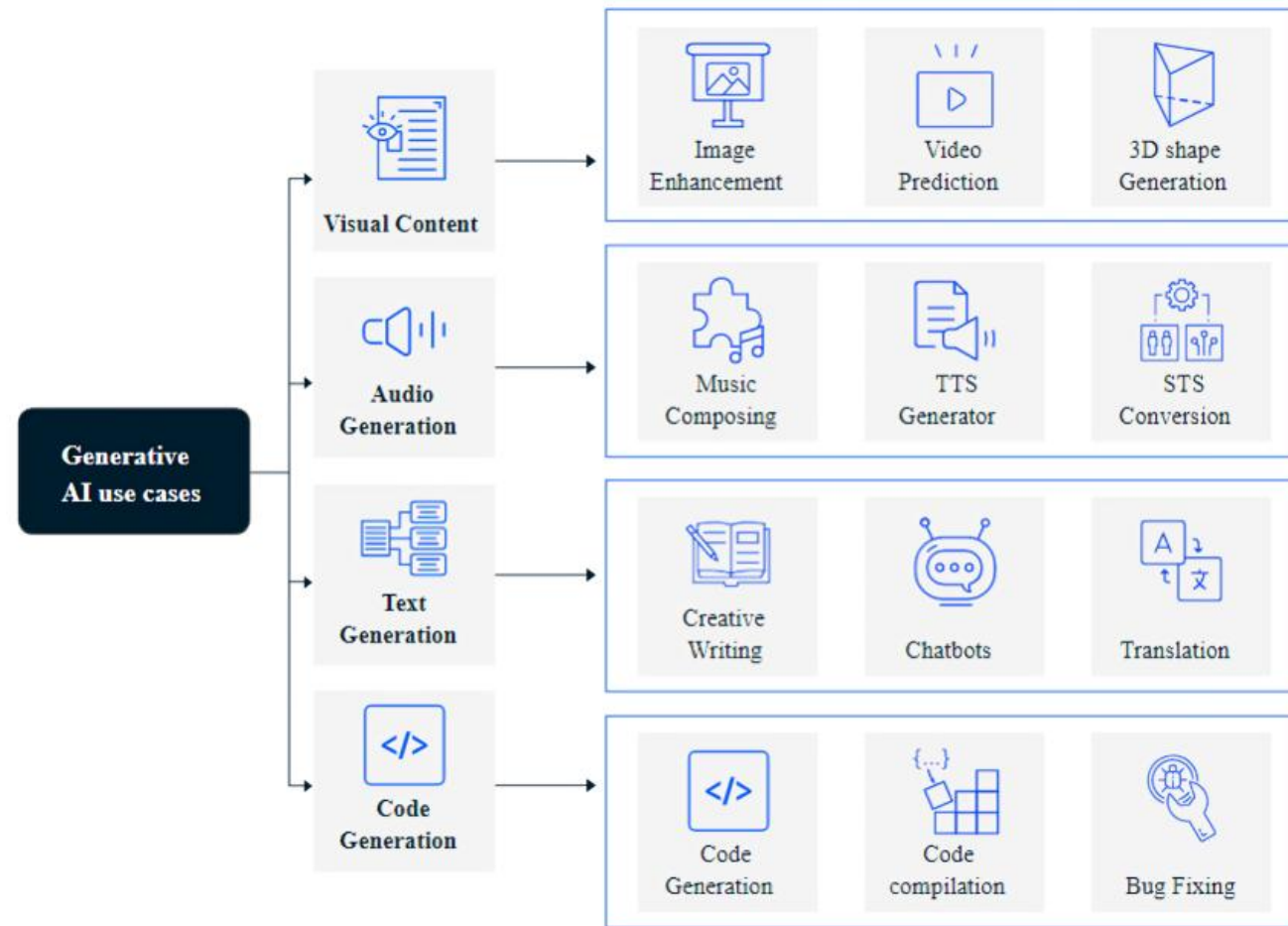
- Built in-house from scratch for the organization

Bloomberg

How LLM created & Fine-tuned?



Generative AI Applications





Gen AI Developments Options

- LangChain
- Hugging Face Hub
- OpenAI
- AWS Bedrock
- Azure AI Foundry
- Google Cloud (Vertex AI)

Thank You